

The logarithmic deficit of two inversion sequence classes (OEIS A279551 and A279556)

Commitment trees: the stretched exponential order, the sharp constant
and the next logarithmic correction

Three research reports of one external research session
(bundle Reports 126, 127 and 129), merged into one ProveIt report

Parts I–III: 2 October 2026 merged: 5 October 2026

Abstract

An inversion sequence of length n is an integer word $e_1 \cdots e_n$ with $0 \leq e_i < i$. Britt and Beaton’s classes 247 and 759, the inversion sequences avoiding the patterns (000, 010, 110, 120) and (010, 110, 120), are counted by OEIS A279551 and A279556; their exponential growth constants are $\mu = 8$ and $\mu = 9$. This report prints three manuscripts on the logarithmic deficit $\mathcal{D}_j(n) = n \log \mu_j - \log a_j(n)$ as Parts I–III, in the order in which each depends on the one before. Part I proves that $\mathcal{D}_j(n)$ has order $n^{1/3}(\log n)^{2/3}$, that $a_j(n) \leq \mu_j^n$ for every n and a_j is strictly increasing for $n \geq 1$, that the inverse threshold carries a positive correction of order $(\log T)^{1/3}(\log \log T)^{2/3}$, that the boundary function has sharp Gevrey order 3, that $1/\mu_j$ is a genuine singularity, and that neither generating function is D-finite. Part II proves $\mathcal{D}_j(n) = \sigma_D S_n + o(S_n)$ with $S_n = n^{1/3}(\log n)^{2/3}$ and $\sigma_D = (3\pi^2/(2D))^{1/3}$, where $D = 1$ for A279556 and $D = 1/2$ for A279551, together with the matching inverse and derivative-growth constants. Part III proves $\mathcal{D}_j(n) = \sigma_D S + \frac{7\sigma_D}{3} S \log \log n / \log n + O(S/\log n)$, with refined inverse and derivative consequences. Britt and Beaton’s numerical forms with a stretched term $n^{3/8}$, which they call non-rigorous, are therefore not asymptotic equivalents. The combinatorial input is their exact commitment-tree construction. The coefficient of $S/\log n$, a coefficient equivalent, ratio asymptotics and effective constants remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The three Parts are three manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 612787fb4 (batch 104). They form one chain on one spine, the commitment trees of Britt and Beaton [1]:

Part	Bundle report	Delivered title	Labels
I	126	Commitment trees and the logarithmic stretched exponential scale	<code>ivd:sc:</code>
II	127	Sharp logarithmic deficit constants for two inversion sequence classes	<code>ivd:sh:</code>
III	129 (base)	The next logarithmic correction for two inversion sequence classes	<code>ivd:ll:</code>

Arrangement. The base of the merge is Report 129, the strongest of the three: its `report129.tex` was staged as this `article.tex` by the placement commit. It is printed last, as Part III, because it is not self-contained: its Proposition 30.1 and several later steps cite proofs that exist only in Report 127, which Report 129’s own package carries unchanged under `earlier-inputs/report127/`. Report 127 is therefore printed in full as Part II. Report 127 in turn repeats Report 126’s Sections 2–4 word for word; that text is printed once, in Part I. Part I is printed in full because it alone proves the non-D-finiteness of the generating functions, the genuine singularity at $1/\mu_j$ and the strict monotonicity of $a_j(n)$, and because its proof of the order of the deficit is a different route from Parts II and III. The three Parts thus appear in dependency order, which is also the order of writing.

Numbering. Sections are numbered continuously through the three Parts, and statements and equations are numbered within sections. Part I keeps Report 126’s section and statement numbers (Sections 1–12). In Part II a section number is Report 127’s plus 12: its Section k is Section $k + 12$ here and its Theorem $k.j$ is Theorem $(k + 12).j$; so Report 127’s Theorem 1.1 is Theorem 13.1, and its Corollaries 14.1 and 14.2 are Corollaries 26.1 and 26.2. Sections 14–16 (Report 127’s Sections 2–4) keep only their titles and a pointer to Part I. In Part III the offset is 28: Report 129’s Theorem 1.1 is Theorem 29.1, its Proposition 2.1 is Proposition 30.1, its Lemmas 3.1 and 4.1 are Lemmas 31.1 and 32.1, and its Corollaries 8.1 and 9.1 are Corollaries 36.1 and 37.1. Section 39, which closes Part III, was added in the write. The manuscripts number their equations consecutively through the whole document; here equations are numbered within sections, so a delivered equation number (m) is not kept, but every delivered equation keeps its label name, with the Part’s prefix. Remarks added in the write (Remarks 1.2 and 29.2) are the last numbered statements of their sections, so no delivered statement number moved.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date. They give cross-references between the Parts, mark statements that a later Part supersedes, and mark results proved twice. Superseded statements are printed as delivered, with a dated note pointing forward, as in the merged reports `a202061-ascent-120-deficit` and `a336070-weak-ascents` of the same collection. No delivered statement was changed, and no delivered symbol was renamed. Each Part ends with a section or subsection “Further questions and research”, which collects, under Vladimir’s standing rule of 4 October 2026, every claim of that Part that is stated without proof, with its source, its sketch and what is missing; Section 39 gathers them for the whole report.

What is proved, and where

Here $j \in \{247, 759\}$, $a_{247} = \text{A279551}$, $a_{759} = \text{A279556}$, $(\mu_j, D_j) = (8, \frac{1}{2})$ and $(9, 1)$, $\sigma_D = (3\pi^2/(2D))^{1/3}$ (numerically 3.093667726280136 for A279551 and 2.455445701568578 for A279556), $\psi(t) = S_t = t^{1/3}(\log t)^{2/3}$, $N = \log n$, $J = \log \log n$, and $\mathcal{D}_j(n) = n \log \mu_j - \log a_j(n)$ is the deficit.

Statement	Where
Exact counting by multiplicity-weighted commitment-tree paths with terminal extraction; injectivity of completed decorated histories; exact cycle weights and their factorizations	Part I, Sections 2–4 (Britt–Beaton’s trees and word counts as input)
$a_j(n) \leq \mu_j^n$ for all n ; $c_j \psi(n) \leq \mathcal{D}_j(n) \leq C_j \psi(n)$; growth constants 8 and 9	Part I, Theorem 1.1 (staircase lower bound; barrier and time tilt upper bound); implied by Theorems 13.1 and 29.1
A weaker lower bound by an independent construction, the ladder $p_k = k^2$: $a_j(n) \geq \mu_j^n \exp\{-Cn^{1/3} \log n\}$	Part I, Section 5.4
No stretched equivalent $Cn^g \mu_j^n \exp\{-bn^\beta\}$, in particular not Britt–Beaton’s $\beta = 3/8$	Part I, Section 9.1 and [write] Remark 1.2; Part II, Section 13.1
$a_j(n+1) > a_j(n)$ for $n \geq 1$	Part I, Section 9.2
$\mathcal{D}_j(n) = \sigma_D S_n + o(S_n)$	Part II, Theorem 13.1
$\mathcal{D}_j(n) = \sigma_D S + \frac{7\sigma_D}{3} \frac{S_J}{N} + O(\frac{S}{N})$, with explicit envelopes	Part III, Theorem 29.1
Inverse threshold $N_j(T)$: correction of order $\psi(s)$; coefficient σ/λ ; second term $\frac{7\sigma}{3\lambda} \psi(s) \log \log s / \log s$ with error $O(\psi(s)/\log s)$	Part I, Corollary 9.1; Part II, Corollary 26.1; Part III, Corollary 36.1
Smooth boundary values; $\log \ f^{(k)}\ _\infty = 3k \log k - 2k \log \log k + O(k)$, then with linear term $[\log(2D/\pi^2) - 3]k + o(k)$, then with $-k \log \log k / \log k + O(k/\log k)$; sharp Gevrey order 3	Part I, Theorem 10.1; Part II, Corollary 26.2; Part III, Corollary 37.1
$1/\mu_j$ is a genuine singularity; the generating functions are not D-finite over $\mathbb{C}(z)$, hence not algebraic	Part I, Theorem 10.1 (only there)
Uniform sharp local critical root and the singular variational problem; their quantitative and anchored forms	Part II, Theorems 18.1 and 19.1; Part III, Lemmas 31.1 and 32.1
<i>Open in all three Parts</i>	
The coefficient of $S/\log n$; a coefficient equivalent (multiplier, polynomial factor, all lower exponent terms); ratio asymptotics for $a_j(n+1)/a_j(n)$; all orders, a transseries, effective constants and onsets; a numerical crossover for finite-range fits; a unique dominant singularity and continuation on the circle $ z = 1/\mu_j$. See Section 39.	

Finite computations in all three Parts (exact rational checks, mutation campaigns, certified log-arithmic floors) are regression checks of formulas and implementations; no asymptotic statement rests on them. The manuscripts make no global priority claim.

Notation across the three Parts

Common to all Parts: inversion sequences e ; labels (p, c) (initial-zero-run length, remaining commitment batches); Catalan numbers C_b and $d_b = C_b/4^b$; multiplicities $M_j(\ell, b)$; terminal sets $\mathcal{T}_j$; the tilts $(\mu, x, y) = (9, 3, \frac{1}{2})$ for class 759 and $(8, 2, \frac{1}{3})$ for class 247, $h(p, c) = x^p y^c$ and the normalized

weight q ; $a = x/\mu$, A_0 , α , $r = a + 2A_0$; the transforms $A_j(\lambda, \eta)$, $R_j(\lambda, \eta)$; the cycle weight $Q_j(b, \ell, w)$ with original duration $L = w + 1$; $\psi(t) = t^{1/3}(\log t)^{2/3}$. In the write's own text the deficit is written $\mathcal{D}_j(n) = n \log \mu_j - \log a_j(n)$, because the letter D is taken. The symbols below change meaning between the Parts or inside one; each Part's text fixes its meaning there, and each Part opens with a short table of its own readings.

Symbol	Meanings
D	Part I: $D_j(n)$ is the deficit $\mathcal{D}_j(n)$ (Theorem 1.1, Sections 9.1, 11); $D = w - \ell$ the displacement of a renewal step (Section 6); $D(t q)$ the binomial divergence (Lemma 4.1); constants D in the proofs of Corollary 9.1 and Theorem 10.1. Parts II and III: $D = D_j \in \{1, \frac{1}{2}\}$, the diffusion coefficient v/α ; Report 127 warns that it is “not a deficit”. <i>Tempting false reading</i> : σ_D is not a function of Part I's $D_j(n)$.
Δ	Parts II, III: the displacement $w - \ell$ of a cycle (Part I's D in Section 6).
M	All: multiplicities $M_j(\ell, b)$. Part I: constants M_1, M_2 (Section 5); moments $M_k = \sum n^k b_n$ (Section 10). Part II: profile height $M = (2D/(3\pi^2))^{1/3}$; the reward martingale M_k (proof of Theorem 21.1) and the moments M_k (Section 26). Part III: M_A ; the reward martingale M_k (Section 34) and the moments M_k (Section 37).
a, b	$a_j(n)$ the counts; $a = x/\mu$ the ordinary-edge weight. Part I: $a > 0$ in (10.5); b the number of commitments, b_k stair batches, $b_n = a_j(n)\mu^{-n}$, b in (1.3). Part II: $a_n^* = E_n/n$, the entrance fraction (named so “not to be confused with the counting sequence”); $0 \leq a < b \leq 1$ the time interval of Theorem 21.1; $b = 1 - \delta$ in Section 23. Part III: $a_n = E(P_n)/n$ and $b_n = 1 - \delta$ (Section 34), but $b_n = a_j(n)\mu^{-n}$ (Section 37); $a = 7/3$ and $b = c/\sigma$ in the proof of Corollary 37.1.
e	The inversion sequence e , Euler's number in every exponential, and: Part II, the slack $e > 0$ (Lemma 24.1, Section 25); Part III, the strip height $e = e_n = N^{-6}$ ((32.1)), so that $\sqrt{e} = N^{-3}$ and ceH are heights. <i>False reading</i> : in Part III $e^{3/2}$, $\sqrt{e}$ and eH involve N^{-6} , not Euler's number.
N, J	Part I: $N_j(T)$ the threshold; $N_k = k^3/(\log(k+2))^2$; N an integer to be filled (Lemma 5.1). Part II: $N(T)$; N the number of cycles in Theorem 21.1; N an exact length in Lemma 22.1; J_p the active interval (20.4). Part III: $N = \log n$ and $J = \log \log n$, but also $N(T)$ in Corollary 36.1. Part I: J_q , J logarithmic degrees in Section 10.
L, ℓ	ℓ the drop. L : Part I, the duration of a renewal step, $L = \ell_K + 1$ (Section 5) and $L = \log \mu_j$ (proof of Corollary 9.1); Part II, the duration $w + 1$ and $L = \ell_K + 1$; Part III, the duration and $L = \log k$ (proof of Corollary 37.1).
λ	The spatial tilt (all; $\lambda_p = \phi'(p)$ in Part I, Section 7); $\lambda = \log \mu$ in Part II, Section 26, and Part III, Sections 36–37.
μ	The growth constant; Part II also $\mu_i = \mathbb{E}[L_i \mid \text{past}]$ in the proof of Theorem 21.1.
κ	Part I and Part II: the barrier constant. Part III: the barrier constant in Proposition 30.1, but $\kappa_p = \eta_p(0)$ the scalar root in Lemma 31.1.
R	All: the Catalan ratio $R_j(\lambda, \eta)$. Part I: $R = \lfloor K \log(K+1) \rfloor$ the plateau window. Part II: the same window (Lemma 22.1) and the strip bound RH (Sections 24–25). Part III: the strip bound RH , and $R(t) = \psi(t)/\log t$ the remainder scale of Theorem 29.1.
G, V	Part I: $G_j(u, y)$ the boundary commitment sums; $V = e^\phi$ the barrier. Part II: $G(p) = \sqrt{(p+2)\log(p+2)}$ and V the barrier (Lemma 24.2). Part III: $G(t)$ the main term of Theorem 29.1; $V_c = G + cR$ in Corollary 37.1.
k	Part I: the stair index and the derivative order. Part II: the stair index, $k = S/n$ (Sections 24–25) and the derivative order (Section 26). Part III: $k_0 = S/n$ and the derivative order.
P	Part I: P_m the embedded chain; $P(\log w)$ a polynomial (Section 10). Part II: $P_n = \lfloor \varepsilon H \rfloor$; $\mathcal{P}_{p,\lambda}$ the tilted law. Part III: $P(n)$ a fixed power of $\log n$ (“never hides a positive power of n ”), $P_n = \lfloor eH \rfloor$, $\mathbb{P}_{p,\lambda}$ the tilted law.

Symbol	Meanings
ρ, r	$\rho = 1/\mu$ the radius (Part I, Section 10; Part II, Section 26; Part III, Section 37), but also $\rho < 1$ a row contraction (Part I, (7.4); Parts II and III, the space–time kernel). $r = r_j$ the zero-tilt boundary row mass; Part II: $[l, r]$ a strip in Lemma 20.2 and Theorem 21.1; $r_\zeta = 1 - 2\zeta$, Part III r_n .
H, S	Part I: $H \geq 2$ a height cutoff, then $H = \lfloor n^{2/3}(\log n)^{1/3} \rfloor$ (Section 8). Parts II and III: $H = n^{2/3}(\log n)^{1/3}$ and $S = n^{1/3}(\log n)^{2/3} = \psi(n)$, unrounded. S and T are also inversion sequences in Section 2.
T	The threshold in $N(T)$ (all); T_K the ladder length (Part I); $T_p(\lambda, \eta)$ the legal row (Parts II, III).
δ	Part I: the support margin of Lemma 4.1 and the time-tilt constant (8.1). Part II: $\delta = A\varepsilon^{3/2}$, $\delta_n = n^{-1/12}$. Part III: $\delta = A_{\text{rep}}e^{3/2}$; its good-cycle threshold is $d_n = n^{-1/12}$.
A	$A_j(\lambda, \eta)$, A_0 (all). Part I: the exponent in k^{-A} . Part II: a large constant A (Section 23). Part III: $A_n = \frac{1}{3} + \frac{7}{6} \frac{J}{N}$ and A in Lemma 32.1; A_{rep} .
f, F	Part I: the barrier $f(p)$ (Section 7) and the boundary function $f(\theta) = F(\rho e^{i\theta})$; $F(z)$, $F_H(z)$, F_j fixed lengths. Part II: the profile f , f_* , and the boundary function $f(t)$; $F_{b,\lambda}$ the drop CDF; $F(p)$ a repair length. Part III: f_A , f_n , the boundary function f ; $F_{b,\lambda}$; $F_t(x)$ the dual tangent.
τ	Part I: τ_H a first-hit index. Part II: $\tau = t/n$. Part III: $\tau_n = n^{-1/14}$ the tube radius.

Provenance and merge decisions

Three manuscripts of the session bundle of Reports 1–243, all dated 2 October 2026. Their title blocks read “Report 126”, “Report 127” and “Report 129”; none names an author, a tool or an addressee (each source sets an empty PDF author field), and none carries “prepared for private review” wording. None pins a ProveIt commit or names a repository path. They arrived in commit 60f54ea06 and were placed by 612787fb4 (batch 104).

- **Part I**, bundle Report 126, archive A279551_A279556_Logarithmic_Deficit_and_Inverse_Source.zip (report126.tex, 1025 lines, 17 pages). Contributes the order theorem by its own route, the exclusion of stretched exponents, strict monotonicity, the first inverse theorem, the boundary-regularity theorem with non-D-finiteness, and the counting model of Sections 2–4.
- **Part II**, bundle Report 127, archive A279551_A279556_Sharp_Deficit_Constants_and_Inversion_Source.zip (report127.tex, 1612 lines, 24 pages). It does not cite Report 126, but its Sections 2–4 are Report 126’s Sections 2–4 word for word. Contributes the sharp constant σ_D , the local root and variational theorems, the tilted laws, the strip and repair constructions, the spatial tail and the space–time supersolution, and the sharp inverse and derivative corollaries.
- **Part III**, bundle Report 129, archive A279551_A279556_Loglog_Correction_and_Refined_Inversion_Source.zip (report129.tex, 1148 lines, 17 pages). The base. Its archive carries Report 127’s whole package under earlier-inputs/report127/; all 19 files are byte copies of Report 127’s delivery (SHA-256, checked at placement). Contributes the second term $\frac{7\sigma_D}{3}SJ/N$ with remainder $O(S/N)$, the quantitative and anchored local root, the corrected potential with integrable height error, the error ledger, and the refined corollaries.

Where the merge had to choose:

1. *Order and base.* Dependency order, I–II–III, with the base last (see the Guide). Later Parts improve earlier statements; the earlier statements stay, with dated notes.
2. *Verbatim repetition.* Report 127’s Sections 2–4 equal Report 126’s Sections 2–4 character for character (checked by the merge script); they are printed once, in Part I, and Sections 14–16 keep their titles and a pointer. Report 127’s references to their displays and to Lemma 4.1 point to Part I, where the numbers are the same. Report 129’s Section 30 restates parts of that model and of Report 127’s Section 5 in its own condensed words; it is printed in full, because it fixes the statements that Part III uses. The joint transforms (6.3)–(6.4) of Part I reappear in Part II as (17.2)–(17.3) and in Part III as (30.2)–(30.3), each in a different surrounding argument; all three are printed.
3. *Second routes.* Part I’s proof of the order of $\mathcal{D}_j(n)$ (a logarithmic staircase with integer filling below; a concave barrier and a uniform time tilt above) differs from the variational proofs of Parts II and III, which imply its statement; it is kept as a second route. Part I’s spatial barrier (Section 7) and Part II’s Lemma 24.2 use the same function and the same supermartingale argument; Part II adds the coefficient clause that Part III uses.
4. *Front matter and bibliography.* The three title blocks and abstracts are printed at the heads of their Parts; one table of contents replaces three. The three bibliographies are merged into one, keeping every detail of every entry; the entry for Report 127 cited by Part III now points to Part II.
5. *Write additions.* Remark 1.2 (Britt–Beaton’s numerical forms, with a proof that they fail, extended to every stretched exponent), Remark 29.2 (the parallel with `a202061-ascent-120-deficit`), the further-questions material (Sections 11.1, 27.1 and 39), the dated notes and this front matter. Everything else is delivered text.

Relation to neighbouring reports

All neighbours are in the collection directory `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`. No other report treats A279551 or A279556 (searched 5 October 2026).

The same method for another sequence. `a202061-ascent-120-deficit` treats 120-avoiding ascent sequences (A202061) and proves results of the same shape by the same mechanism: the order $n^{1/3}(\log n)^{2/3}$, the sharp constant, the second term with coefficient 7/3, the exclusion of a numerically estimated stretched exponent 3/8, and, beyond this report, the next constant and every fixed inverse-logarithmic order. Remark 29.2 gives the details. Neither report cites the other; the proofs here are an independent instance of the method, not re-proofs of anything there.

Other Britt–Beaton classes. Batch 104 also placed `a279544-inversion-kernel-classes` (A279544, A279567, A279569, A279558: algebraic kernels and square-root singularities) and `a279571-inversion-cone-walk` (A279571: a quadrant cone walk with an irrational polynomial exponent). They share with this report only the definition of inversion sequences and the source [1], whose Section 4.2 tabulates the numerical estimates for all of these classes; their methods and asymptotic regimes are unrelated, and none of the eight Britt–Beaton manuscripts of the batch cites another. They are siblings, not hosts.

Formal status. No statement of this report is formalized: no Lean or Rocq file in the repository

mentions inversion sequences, commitment trees, A279551 or A279556 (searched 5 October 2026). Placement in the `SetTheory/Cardinals` report collection confers no formal status.

Part I

Commitment trees and the logarithmic stretched exponential scale

Bundle Report 126, archive A279551_A279556_Logarithmic_Deficit_and_Inverse_Source.zip, delivered as `report126.tex` (17 pages), dated 2 October 2026. Delivered title block “Report 126. Commitment trees and the logarithmic stretched exponential scale. A279551 and A279556. Standalone proof and exact finite verification companion”; no author line. Printed as delivered, with **[write]** notes; section and statement numbers are those of the delivery (equations are renumbered within sections). Remark 1.2 and Section 11.1 are added in the write.

Reading conventions of Part I	Elsewhere in the report
$D_j(n) = n \log \mu_j - \log a_j(n)$, the deficit	Write: $\mathcal{D}_j(n)$; Parts II, III: $D \in \{1, \frac{1}{2}\}$ is a diffusion coefficient
$D = w - \ell$, displacement (Section 6)	Parts II, III: Δ
$H = \lfloor n^{2/3}(\log n)^{1/3} \rfloor$ (Section 8)	Parts II, III: H unrounded
M_k moments; $\rho = 1/\mu$ (Section 10); $\rho < 1$ (Section 7)	Parts II, III: also martingales M_k , profile heights M , M_A
$N_k = k^3/(\log(k+2))^2$ (Section 10)	Part III: $N = \log n$

Abstract of this Part (as delivered). For the inversion-sequence classes numbered 247 and 759, we prove that the exponential growth constants are respectively 8 and 9, and that the logarithmic deficit from exponential growth has order $n^{1/3}(\log n)^{2/3}$. The lower counting bound uses completed commitment cycles along a logarithmic staircase, exact integer-length filling, and a fixed valid endpoint. The upper counting bound uses exact joint space–time renewal transforms, a global concave spatial barrier, and a uniform positive time tilt below a height cutoff. These two arguments are deductions from the exact commitment trees of Britt and Beaton. They rule out the literal negative, nonzero $n^{3/8}$ term in the numerically proposed equivalents, while preserving the exact enumeration and finite coefficient data. We also obtain a positive correction of the same order for the inverse counting threshold. We further prove smooth boundary values, sharp Gevrey order 3, and non-D-finiteness of both generating functions. No leading deficit constant, coefficient equivalent, quantitative coefficient remainder, or transseries is asserted.

1 Definitions and the main theorem

An inversion sequence of length n is an integer sequence $e = (e_1, \dots, e_n)$ with $0 \leq e_i < i$. Let $a_{247}(n)$ count such sequences with no $i < j < k$ satisfying $e_i \leq e_j$ and $e_i \geq e_k$. Let $a_{759}(n)$ count those with no $i < j < k$ satisfying these two inequalities and $e_j \neq e_k$. In the usual pattern notation, these are avoidance of (000, 010, 110, 120) and (010, 110, 120), respectively. Thus

$$a_{247}(n) = A279551(n), \quad a_{759}(n) = A279556(n), \quad a_j(0) = 1.$$

All logarithms are natural. Put $\mu_{247} = 8$, $\mu_{759} = 9$, and $\psi(t) = t^{1/3}(\log t)^{2/3}$ for $t > 1$.

Theorem 1.1. *For each $j \in \{247, 759\}$ there are constants $0 < c_j \leq C_j < \infty$ and n_j such that, for every integer $n \geq n_j$,*

$$\mu_j^n e^{-C_j \psi(n)} \leq a_j(n) \leq \mu_j^n e^{-c_j \psi(n)}. \quad (1.1)$$

Equivalently, $D_j(n) := n \log \mu_j - \log a_j(n)$ satisfies

$$c_j \psi(n) \leq D_j(n) \leq C_j \psi(n).$$

Moreover $a_j(n) \leq \mu_j^n$ for every $n \geq 0$, and

$$\lim_{n \rightarrow \infty} a_{247}(n)^{1/n} = 8, \quad \lim_{n \rightarrow \infty} a_{759}(n)^{1/n} = 9. \quad (1.2)$$

The statement is an order-of-magnitude theorem for the logarithm. It does not imply that $D_j(n)/\psi(n)$ has a limit. In particular, writing $a_j(n) = \mu_j^n \exp\{-\Theta(\psi(n))\}$ is shorthand for the two inequalities in (1.1), not for an equivalent with an unspecified constant. Fixed prefactors in intermediate bounds will be absorbed into c_j, C_j only after n is sufficiently large.

[write] 5 October 2026. Superseded in precision, not in validity. Part II, Theorem 13.1, proves that $D_j(n)/\psi(n)$ does converge, to $\sigma_{D_j} = (3\pi^2/(2D_j))^{1/3}$ with $D_{759} = 1$, $D_{247} = \frac{1}{2}$ (Part II's diffusion coefficients, not this Part's deficit), and Part III, Theorem 29.1, adds the term $\frac{7\sigma_D}{3}\psi(n) \log \log n / \log n$ with remainder $O(\psi(n)/\log n)$. Both imply (1.1) for every $c_j < \sigma_{D_j} < C_j$. The proof of this Part (Sections 5–8) is a different route to the order statement and is kept in full as a second route. In the write's own text the deficit $D_j(n)$ is written $\mathcal{D}_j(n)$.

1.1 Published inputs and comparison

The combinatorial input is Britt and Beaton's exact succession rules [1, Sections 3.8–3.9, Lemmas 5–6 and Appendix A], in version 3 dated 29 September 2026. Their Section 4.2 explicitly labels its asymptotic estimates as nonrigorous numerical analysis. Equations (4.12)–(4.13) propose forms of the type

$$An^g \mu_j^n \exp\{-bn^{3/8}\}, \quad A, b > 0. \quad (1.3)$$

Theorem 1.1 disproves (1.3) as a literal full asymptotic equivalent. It does not dispute the finite enumerations or the exact trees that make the present proof possible. No global literature priority claim is made here. The estimates below are derived from those trees and do not use fitted numerical constants.

Remark 1.2 ([**write**] Britt–Beaton's numerical forms, on record; 5 October 2026). Section 4.2 of [1], which its authors describe as non-rigorous and based only on numerical analysis, gives as best estimates, with thanks to A. J. Guttmann for assistance,

$$\begin{aligned} \text{A279551 (their (4.12)) : } I_n &\sim C n^g 8^n \mu_1^{n^{3/8}}, \quad g \approx 4.25, \log \mu_1 \approx -13, \log C \approx 8; \\ \text{A279556 (their (4.13)) : } I_n &\sim C n^g 9^n \mu_1^{n^{3/8}}, \quad g \approx 3.25, \log \mu_1 \approx -10.4, \end{aligned} \quad (1.4)$$

that is, (1.3) with $b = -\log \mu_1 > 0$. As asymptotic equivalents both forms are false, and so is every form

$$a_j(n) \sim C n^g \mu_j^n \exp\{-bn^\beta\} \quad (C > 0; g, b, \beta \in \mathbb{R}).$$

Proof. Such an equivalent gives, on taking logarithms, $\mathcal{D}_j(n) = bn^\beta - g \log n - \log C + o(1)$. By Theorem 1.1, $c_j \psi(n) \leq \mathcal{D}_j(n) \leq C_j \psi(n)$ for large n , with $\psi(n) = n^{1/3}(\log n)^{2/3}$; note $\log n = o(\psi(n))$. If $b \leq 0$ or $\beta \leq 0$, then bn^β is bounded above, so $\mathcal{D}_j(n) \leq |g| \log n + O(1) = o(\psi(n))$, contradicting the lower bound. If $b > 0$ and $0 < \beta \leq 1/3$, then $n^\beta/\psi(n) = n^{\beta-1/3}(\log n)^{-2/3} \rightarrow 0$, so again $\mathcal{D}_j(n) = o(\psi(n))$, a contradiction. If $b > 0$ and $\beta > 1/3$, then $n^\beta/\psi(n) \rightarrow \infty$, so $\mathcal{D}_j(n)/\psi(n) \rightarrow \infty$, contradicting the upper bound. Britt and Beaton's case is $\beta = 3/8$, $b > 0$: there $\mathcal{D}_j(n)/n^{3/8}$ would tend to $b > 0$, while $\mathcal{D}_j(n)/n^{3/8} \leq C_j n^{-1/24}(\log n)^{2/3} \rightarrow 0$. $\square$

Only the upper half of Theorem 1.1's deficit bound, that is the lower counting bound (5.3), is needed for $\beta = 3/8$, and the simpler ladder of Section 5.4 already suffices for it, since $n^{1/3} \log n = o(n^{3/8})$. Section 9.1 gives the same argument for $\beta = 3/8$ and for a pure $n^{1/3}$ deficit; Part II, Section 13.1, repeats it. Nothing here disputes Britt and Beaton's exact trees or finite enumerations, on which every Part relies, and nothing is said about how well (1.4) fits finite data (Section 39, question 5).

2 Decorated commitments and exact counting

2.1 Auxiliary objects and terminal extraction

The growth operation increments every positive entry, may increment certain zeros, and prepends a new zero. For class 759, the auxiliary class permits zero-valued instances of 010 while excluding the other forbidden patterns. For class 247, the auxiliary class additionally permits zero-valued instances of 000. Let p be the initial-zero-run length. Zeros lying after that initial run may have to be incremented in later stages before growth within the initial run is again allowed. A commitment word records their future nonempty batches of increments. Its letters $1, \dots, b$ must all occur. The number c in the label (p, c) counts remaining batches, rather than the number of pending zeros.

For 759 the onto commitment words avoid 212, 112, 213; for 247 they also avoid 111. The exact word counts supplied by the source are $C_b \binom{k-1}{b-1}$ and $C_b \binom{b}{k-b}$ for length k , where $C_b = \binom{2b}{b}/(b+1)$ is the Catalan number. We use these published word-count identities as part of the combinatorial input.

Both label trees start at $(0, 0)$ and have edges

$$(p, c) \longrightarrow (p+1, c), \quad (2.1)$$

$$(p, c) \longrightarrow (p+1, c-1) \quad (c > 0), \quad (2.2)$$

$$(p, 0) \longrightarrow (p-\ell, b) \quad (0 \leq \ell \leq p-1), \quad (2.3)$$

where the third edge has multiplicity

$$M_{759}(\ell, b) = C_b \binom{\ell}{b}, \quad M_{247}(\ell, b) = C_b \binom{b+1}{\ell-b}. \quad (2.4)$$

Out-of-range binomial coefficients mean zero. Thus the 247 support is $b \leq \ell \leq 2b+1$. The admissible terminal labels are

$$\mathcal{T}_{759} = \{(p, 0) : p \geq 0\}, \quad \mathcal{T}_{247} = \{(p, 0) : 0 \leq p \leq 2\}. \quad (2.5)$$

Counting every auxiliary node would give the wrong sequences. In particular, a long all-zero initial access path is legal in the auxiliary 247 tree, although it is not itself a target avoider. Only the final extraction uses (2.5).

Here is the multiplicity interpretation. Without a new commitment, increment a nonempty suffix of r initial zeros. The new prefix has length $p-r+1$, so $\ell = r-1$. For 759 every $r \geq 1$ is allowed; for 247 only $r = 1, 2$ is allowed. With a new commitment, increment a suffix of $r \geq 0$ zeros and one additional zero to its left, leaving $k \geq 1$ zeros between. Then $\ell = r+k$ and the new prefix length is $p-\ell$. Summing the word counts gives

$$\sum_{k=b}^{\ell} C_b \binom{k-1}{b-1} = C_b \binom{\ell}{b}$$

for 759. For 247, r is zero or one, and Pascal's identity gives

$$C_b \left[\binom{b}{\ell-b} + \binom{b}{\ell-b-1} \right] = C_b \binom{b+1}{\ell-b}.$$

For $b = 0$, use the suffix rules directly. They give $M_{759}(\ell, 0) = 1$ for every $\ell \geq 0$, and $M_{247}(\ell, 0) = 1$ for $\ell = 0, 1$, with zero otherwise, as in (2.4). At $p \geq 1$, both classes have a unit boundary self-loop. Class 247 also has a unit boundary edge from p to $p - 1$ for $p \geq 2$.

2.2 Why completed histories do not overcount

Given a physical growth step $S \rightarrow T$, recover S uniquely by removing the first zero of T and replacing each remaining entry t by $\max(t - 1, 0)$. The formerly zero positions incremented in that step are exactly those whose corresponding entries in T equal one. Iterating this operation recovers the entire physical history from a final sequence.

For a completed commitment, its word is also recoverable: order the subsequent nonempty batches in which the pending zeros are incremented, and give each pending position the index of its batch. Onto-ness ensures that no empty, hence invisible, commitment can be inserted. Idle steps are already determined by the physical history. It follows that two different fully completed decorated histories cannot have the same physical endpoint. At an unfinished endpoint, future commitment words need not yet be distinguishable physically; this is another reason to use only (2.5).

Conversely, the source construction gives every target avoider its completed decorated history. Its reverse-growth chain can encounter the allowed zero-valued exceptions in the auxiliary class, and the later increment order supplies the commitment words. Every legal completed history with the relevant terminal label is a target avoider. Thus the multiplicity-weighted paths used below count distinct objects exactly. When restricting paths to a fixed sequence of cycles, their deterministic cycle lengths fix all cut positions; no further ambiguity is introduced.

3 Critical tilts and completed cycles

Choose parameters

j	μ	x	y	α	r
759	9	3	1/2	3/2	2/3
247	8	2	1/3	4/3	5/8

(3.1)

and let $h(p, c) = x^p y^c$. Give an edge of multiplicity m the normalized weight $q(s, t) = mh(t)/(\mu h(s))$. A length- n path of multiplicity m then has weight

$$m\mu^{-n}h(\text{endpoint}), \tag{3.2}$$

since the root has weight one.

For $c > 0$, the outgoing row mass is $x/\mu + x/(\mu y) = 1$. For boundary rows, let $\mathcal{C}(v) = \sum_{b \geq 0} C_b v^b$ and $u = 1/x$. The nonnegative commitment sums extended to every $\ell \geq 0$ are

$$G_{759}(u, y) = \frac{1}{1-u} \mathcal{C}\left(\frac{yu}{1-u}\right), \tag{3.3}$$

$$G_{247}(u, y) = (1+u)\mathcal{C}(yu(1+u)). \tag{3.4}$$

Both Catalan arguments are $1/4$, where the series converges and $\mathcal{C}(1/4) = 2$. Both G values equal three. Adding the ordinary edge gives boundary row masses at most $2/3$ and $5/8$, respectively. The root has just the ordinary edge, of mass x/μ . Thus q is substochastic. Its total mass at any original integer time is at most one. At valid terminals $h = x^p \geq 1$; (3.2) gives

$$a_j(n) \leq \mu_j^n \quad (n \geq 0). \quad (3.5)$$

3.1 Exact completed cycle weights

From $(p, 0)$, take a commitment edge with $b \geq 1$ and $\ell \leq p - 1$. Follow it by $w \geq b$ bulk edges, exactly b of which fulfill a commitment, the last fulfillment occurring on the final edge. Choose the first $b - 1$ fulfillment positions among the first $w - 1$ edges. There are $\binom{w-1}{b-1}$ choices, and every proper bulk prefix has $c > 0$. The cycle duration is $w + 1$, displacement $w - \ell$, and total weight

$$Q_j(b, \ell, w) = \mu_j^{-w-1} x_j^{w-\ell} M_j(\ell, b) \binom{w-1}{b-1}. \quad (3.6)$$

Here “cycle” means completed commitment excursion, not necessarily a return to the same height. It is legal if the multiplicity is nonzero, $w \geq b$, and $\ell \leq p - 1$. During the bulk portion, p only increases.

Define the negative-binomial and binomial masses by

$$\text{NB}_{b,q}(r) = \binom{r-1}{b-1} q^b (1-q)^{r-b}, \quad \text{Bin}_{m,q}(r) = \binom{m}{r} q^r (1-q)^{m-r}.$$

Direct cancellation in (3.6) yields the exact identities

$$Q_{759}(b, \ell, w) = \frac{C_b}{4^b} \frac{\ell}{9b} \text{NB}_{b,2/3}(\ell) \text{NB}_{b,2/3}(w), \quad (3.7)$$

$$Q_{247}(b, \ell, w) = \frac{C_b}{4^b} \frac{3}{16} \text{Bin}_{b+1,1/3}(\ell - b) \text{NB}_{b,3/4}(w). \quad (3.8)$$

For 247 the exact center of ℓ is $(4b + 1)/3$, so its bounded shift from $\alpha b = 4b/3$ must be retained in local estimates.

4 A uniform moderate local lower bound

Lemma 4.1. *There exist constants $c, C > 0$, b_0 , and $\delta > 0$ such that, for both classes and integer $b \geq b_0, \ell, w$ satisfying $|\ell - \alpha b| \leq \delta b$ and $|w - \alpha b| \leq \delta b$, the probabilistic supports hold and*

$$Q_j(b, \ell, w) \geq cb^{-5/2} \exp \left\{ -\frac{C}{b} [(\ell - \alpha b)^2 + (w - \alpha b)^2] \right\}. \quad (4.1)$$

The geometric constraint $\ell \leq p - 1$ is separate.

Proof. Two-sided Stirling inequalities

$$\sqrt{2\pi m} (m/e)^m e^{1/(12m+1)} \leq m! \leq \sqrt{2\pi m} (m/e)^m e^{1/(12m)} \quad (m \geq 1)$$

imply, uniformly for r/N in a fixed compact interval $I \subset (0, 1)$,

$$\text{Bin}_{N,q}(r) \geq c_I N^{-1/2} e^{-ND(r/N\|q)}, \quad D(t\|q) = t \log(t/q) + (1-t) \log \frac{1-t}{1-q}.$$

Since $D(q\|q) = D'(q\|q) = 0$ and $D''(t\|q) = 1/[t(1-t)]$, Taylor's theorem on I gives

$$\text{Bin}_{N,q}(r) \geq cN^{-1/2} \exp\{-C(r - Nq)^2/N\} \quad (4.2)$$

when $|r - Nq| \leq \eta N$ for some fixed $\eta > 0$. The exact relation $\text{NB}_{b,q}(r) = (b/r) \text{Bin}_{r,q}(b)$ gives

$$\text{NB}_{b,q}(r) \geq cb^{-1/2} \exp\{-C(r - b/q)^2/b\}, \quad |r - b/q| \leq \delta b,$$

because $r \asymp b$ and b/r stays away from zero and one. For the 247 binomial factor,

$$(\ell - b) - (b + 1)/3 = (\ell - 4b/3) - 1/3,$$

and $(u - 1/3)^2 \leq 2u^2 + 2/9$ absorbs the bounded displacement. Also $C_b/4^b \geq cb^{-3/2}$ by Stirling, and ℓ/b in (3.7) is bounded below. Multiplication proves the result.

For explicit support margins one may take $\delta = 1/12$ and enlarge b_0 . The 759 ratios $b/\ell, b/w$ lie in $[12/19, 12/17]$; the 247 ratio b/w lies in $[12/17, 4/5]$ and $(\ell - b)/(b + 1)$ lies in $[1/8, 5/12]$. All are inside $[1/8, 7/8]$, which also contains the relevant success probabilities. Thus the compact-set constants are uniform. In particular this proof allows deviations of size $\sqrt{b \log b}$; it is not an application of a fixed-width central-limit window outside its range. $\square$

5 The logarithmic staircase counting lower bound

5.1 Rounded heights and legal upward and downward stages

For integers $k \geq k_0$, set

$$p_k = \lfloor k^2 \log(k+1) \rfloor, \quad b_k = \lfloor p_k/4 \rfloor, \quad \ell_k = \lfloor \alpha b_k + 1/2 \rfloor, \quad d_k = p_{k+1} - p_k. \quad (5.1)$$

Fix k_0 sufficiently large, independently of n . An upward stage from p_k uses $(b_k, \ell_k, \ell_k + d_k)$ and ends at p_{k+1} . A downward stage from p_{k+1} uses $(b_k, \ell_k, \ell_k - d_k)$ and ends at p_k . The mean-value theorem applied to $t^2 \log(t+1)$, with the floor error bounded by one, gives

$$p_k \asymp b_k \asymp k^2 \log k, \quad d_k = O(k \log k) > 0, \quad d_k/b_k = O(1/k), \quad d_k^2/b_k = O(\log k).$$

The limiting margins are strictly positive: $\ell_k/p_k \rightarrow \alpha/4 < 1$, $\ell_k/b_k \rightarrow \alpha > 1$, and, for 247, $2 - \alpha > 0$. Therefore a single fixed k_0 makes $\ell_k \leq p_k - 1$, $\ell_k - d_k \geq b_k$, and every commitment support condition hold for all subsequent k . The downward departure height is larger and imposes no stronger ℓ constraint. Nearest-integer rounding gives $|\ell_k - \alpha b_k| \leq 1/2$; the waiting-time deviations are at most $d_k + 1/2 = o(b_k)$. Lemma 4.1 consequently gives a stage weight at least k^{-A} for a fixed $A > 0$. Indeed its negative logarithm is bounded by

$$O(1) + \frac{5}{2} \log b_k + O((d_k + 1)^2/b_k) = O(\log k).$$

Ascending from p_{k_0} to p_K and descending in reverse order has exact length

$$T_K = \sum_{k=k_0}^{K-1} [(\ell_k + d_k + 1) + (\ell_k - d_k + 1)] = \sum_{k=k_0}^{K-1} (2\ell_k + 2) = O(K^3 \log K) \quad (5.2)$$

and normalized weight at least $\exp\{-CK \log K\}$. The signed increments cancel exactly in the duration, not just asymptotically.

5.2 Every integer length at the plateau

At height p_K put $b = b_K$, $L = \ell_K + 1$, and $R = \lfloor K \log(K + 1) \rfloor$. For every integer $s \in [-R, R]$, choose $\ell = w = L - 1 + s$. This is a same-height return cycle of length $L + s$. Since $R/b = O(1/K)$ and $(R + 1)^2/b = O(\log K)$, the same strict support margins and Lemma 4.1 give a uniform weight at least $K^{-A'}$ for every such cycle. In particular, the available durations contain the entire integer interval $[L - R, L + R]$, where $0 < R < L$, $L \asymp K^2 \log K$, and $R \asymp K \log K$.

Lemma 5.1 (Integer filling). *If L, R are positive integers with $R < L$, every integer $N \geq (L^2 - R^2)/(2R)$ is a sum of $m = \lceil N/(L + R) \rceil$ integers in $[L - R, L + R]$.*

Proof. The upper endpoint satisfies $N \leq m(L + R)$. The lower endpoint obeys

$$m(L - R) \leq \left(\frac{N}{L + R} + 1 \right) (L - R) \leq N,$$

where the last inequality is precisely the assumed bound on N . Start all summands at $L - R$ and distribute the integer excess with capacity $2R$ per summand. This realizes every integer between the two extreme sums; no divisibility or parity assumption is needed. $\square$

Here the sufficient threshold is $O(K^3 \log K)$, and $m = O(N/(K^2 \log K) + 1)$. We fix one composition of N once and for all. Counting different compositions is unnecessary and is not used.

5.3 Assembly at every sufficiently large integer time

Let $p_0 = p_{k_0}$ be fixed. Reach $(p_0, 0)$ by p_0 ordinary edges. For 759, return to this valid endpoint after the descent. For 247, append $p_0 - 2$ unit downward boundary edges and end at $(2, 0)$. The total fixed access and termination lengths are $F_{759} = p_0$, $F_{247} = 2p_0 - 2$, and their normalized weights are fixed positive constants.

Choose a sufficiently small fixed $\varepsilon > 0$ and set

$$K = \left\lfloor \varepsilon(n/\log n)^{1/3} \right\rfloor.$$

There are fixed constants M_1, M_2 bounding T_K and the filling threshold by $M_i K^3 \log K$. Choose $\max(M_1, M_2)\varepsilon^3 \leq 1/4$. Eventually $K \geq k_0$, $\log K \leq \log n$, and $F_j \leq n/4$, so

$$N := n - F_j - T_K \geq n/2 \quad \text{and} \quad (L^2 - R^2)/(2R) \leq n/4.$$

Insert the exact m -cycle plateau filling between ascent and descent. The total original length is exactly n . Its normalized weight is at least

$$\begin{aligned} \exp \left\{ -C \left[K \log K + \left(\frac{n}{K^2 \log K} + 1 \right) \log K \right] \right\} &\geq \exp \{ -C'(K \log K + n/K^2) \} \\ &\geq \exp \{ -C''\psi(n) \}. \end{aligned}$$

The floor causes no denominator problem: eventually $K \geq (\varepsilon/2)(n/\log n)^{1/3}$. The final tilt is 3^{p_0} for 759 and 4 for 247. These are fixed constants, so division in (3.2) only changes the constant in the bound. We have proved

$$a_j(n) \geq \mu_j^n e^{-C_j \psi(n)} \tag{5.3}$$

for every sufficiently large integer n . Ending at p_K instead would incur the much larger endpoint factor x^{p_K} ; the descent is essential.

5.4 A simpler bound already excludes the old exponent

For perspective, one can use $p_k = k^2$, $b_k = \lfloor k^2/4 \rfloor$, and the same nearest-integer ℓ_k . Then $d_k = 2k + 1$ is $O(\sqrt{b_k})$, and the fixed-window consequence of Lemma 4.1 gives stage weights at least ck^{-5} . The total ladder length is $O(K^3)$. At the top, choose return-cycle lengths in $[L - K, L + K]$, with $L \asymp K^2$; the filling threshold is $O(K^3)$ and the number of plateau cycles is $O(n/K^2 + 1)$. Taking $K = \lfloor \varepsilon n^{1/3} \rfloor$ and using the same fixed endpoints proves

$$a_j(n) \geq \mu_j^n \exp\{-Cn^{1/3} \log n\}.$$

This weaker independent construction alone already rules out a negative nonzero $n^{3/8}$ term. The logarithmic staircase is needed for the matching order in Theorem 1.1.

6 The boundary renewal kernel and its joint transforms

For the matching upper counting bound, add a cemetery state to q to obtain a killed Markov chain. Killing occurs only on departure from $c = 0$, because bulk rows have mass one. Put $a = x/\mu$; the two bulk probabilities are a and $1 - a = x/(\mu y)$. An excursion with b commitments therefore completes almost surely, with a negative-binomial duration. A terminal path satisfies

$$\frac{a_j(n)}{\mu_j^n} \leq \text{total } q\text{-mass of length-}n \text{ paths ending at } c = 0, \quad (6.1)$$

because its contribution to the count divided by μ^n is its q -weight times $x^{-p} \leq 1$. Allowing all boundary terminal heights in 247 enlarges the sum and is valid for this upper bound.

Embed the chain at successive visits to $c = 0$, counting a one-edge boundary-to-boundary move as its own renewal step. Such a step is an ordinary edge (duration 1, displacement 1, weight a), a $b = 0$ commitment edge (duration 1, displacement $-\ell$), or a completed cycle (3.6) with duration $L = w + 1$ and displacement $D = w - \ell$. Let $K(p, p')$ be the total weight of all legal such steps, with durations summed. For a fixed $b \geq 1$, extend the ℓ -sum to all nonnegative integers. Set $d_b = C_b/4^b$. Direct summation gives

$$\sum_{\ell, w} Q_j(b, \ell, w) e^{\lambda(w-\ell) + \eta(w+1)} = A_j(\lambda, \eta) d_b R_j(\lambda, \eta)^b, \quad (6.2)$$

where

$$A_{759}(\lambda, \eta) = \frac{e^\eta}{9(1 - e^{-\lambda/3})}, \quad R_{759}(\lambda, \eta) = \frac{4e^\eta}{9(1 - e^{-\lambda/3})(1 - e^{\lambda+\eta/3})}, \quad (6.3)$$

$$A_{247}(\lambda, \eta) = \frac{e^\eta(1 + e^{-\lambda/2})}{8}, \quad R_{247}(\lambda, \eta) = \frac{e^\eta(1 + e^{-\lambda/2})}{2(1 - e^{\lambda+\eta/4})}. \quad (6.4)$$

The same $A_j d_b R_j^b$ formula at $b = 0$ is the sum of all zero-commitment edges, with $L = 1, D = -\ell$. It inserts no fictitious bulk steps. To verify the calculation, use

$$\begin{aligned} \sum_{\ell \geq b} \binom{\ell}{b} z^\ell &= \frac{z^b}{(1 - z)^{b+1}}, & \sum_{\ell} \binom{b+1}{\ell - b} z^\ell &= z^b (1 + z)^{b+1}, \\ \sum_{w \geq b} \binom{w-1}{b-1} t^w &= \left(\frac{t}{1-t} \right)^b & (b \geq 1). \end{aligned}$$

For each fixed b , the sums over ℓ, w converge in a fixed neighborhood of $(0, 0)$, which can be chosen independently of b . *The unrestricted sum over all b is not asserted to converge at positive tilt.* Every positive-tilt use below has the finite legal cutoff $b \leq p - 1$ or the finite cutoff $b \leq H$.

The critical values and derivatives are

j	$A_j(0, 0)$	$R_j(0, 0)$	$\partial_\lambda \log R_j(0, 0)$	$\partial_\lambda^2 \log R_j(0, 0)$	$\partial_\eta \log R_j(0, 0)$
759	1/6	1	0	3/2	3/2
247	3/16	1	0	2/3	4/3

(6.5)

In particular there are fixed $B, \epsilon_0 > 0$ such that

$$\log R_j(\lambda, 0) \leq B\lambda^2, \quad \log R_j(0, \eta) \leq B\eta \quad (0 \leq \lambda, \eta \leq \epsilon_0). \quad (6.6)$$

The prefactors are bounded and Lipschitz there. These are analytic Taylor bounds for explicit functions, not assumptions about the counting sequences. At zero tilt, $\sum_{b \geq 0} d_b = 2$ gives the unrestricted boundary row mass $a + 2A_j(0, 0) = r_j < 1$, in agreement with (3.1).

[write] **5 October 2026.** Part II, Section 17, restates the transforms (6.2)–(6.4) with the legal drop cutoff and identifies the tilted drop and duration laws; Part III uses them from there.

7 A uniform Catalan estimate and a global spatial barrier

Lemma 7.1. *There is an absolute $B_0 < \infty$ such that for every integer $m \geq 0$ and $0 \leq \gamma \leq 1/4$,*

$$\sum_{b=0}^m d_b \exp \left\{ \gamma b \frac{\log(m+2)}{m+2} \right\} \leq 2 + B_0 \gamma. \quad (7.1)$$

Proof. The central-binomial bound gives $d_b \leq B_1(b+1)^{-3/2}$ and hence $\sum_{b=1}^m b d_b \leq B_2 \sqrt{m}$. Put $\theta = \gamma \log(m+2)/(m+2)$. For $m \geq 1$, use $e^u - 1 \leq u e^u$ to obtain

$$\begin{aligned} \sum_{b=0}^m d_b (e^{\theta b} - 1) &\leq \theta e^{\theta m} \sum_{b=1}^m b d_b \\ &\leq B_2 \gamma \log(m+2) (m+2)^{-1/2+\gamma} \leq B_3 \gamma. \end{aligned}$$

The last bound is uniform since $\gamma \leq 1/4$ and $\sup_{t \geq 2} (\log t) t^{-1/4} < \infty$. Add $\sum_{b=0}^m d_b \leq 2$. The case $m = 0$ is immediate. $\square$

Define for real $p \geq 0$

$$f(p) = \sqrt{(p+2) \log(p+2)}, \quad \phi(p) = \kappa[f(p) - f(0)], \quad V(p) = e^{\phi(p)}. \quad (7.2)$$

The constant $\kappa > 0$ will be fixed sufficiently small. Writing $t = p + 2$, direct differentiation gives

$$f'(p) = \frac{\log t + 1}{2\sqrt{t \log t}}, \quad f''(p) = -\frac{(\log t)^2 + 1}{4t^{3/2}(\log t)^{3/2}} < 0.$$

Thus f is globally increasing and concave, with no patch at small heights. If $\lambda_p = \phi'(p)$, then

$$0 \leq \lambda_p \leq B_4 \kappa, \quad \lambda_p^2 \leq B_4 \kappa^2 \frac{\log(p+2)}{p+2}, \quad \frac{V(p')}{V(p)} \leq e^{\lambda_p(p'-p)}. \quad (7.3)$$

For $p \geq 1$, apply this last inequality to legal transitions, then extend the ℓ -sum while retaining $b \leq p - 1$. Equations (6.2) and (6.6) give

$$\frac{(KV)(p)}{V(p)} \leq ae^{\lambda_p} + A_j(\lambda_p, 0) \sum_{b=0}^{p-1} d_b R_j(\lambda_p, 0)^b.$$

Choose κ small enough that every λ_p is in the fixed Taylor neighborhood. Increasing the cutoff to p if convenient, Lemma 7.1 with $\gamma = O(\kappa^2)$ shows that this sum is at most $2 + O(\kappa^2)$ uniformly in p . The Lipschitz prefactor then yields $(KV)(p)/V(p) \leq r_j + O(\kappa)$. At the root the only edge gives $aV(1) \leq ae^{\lambda_0}$, so the same uniform bound applies. Fix κ so small that a common $\rho < 1$ satisfies

$$KV \leq \rho V \leq V \quad \text{at every boundary state,} \quad V(0) = 1. \quad (7.4)$$

Let P_m be the embedded killed chain, with $V(\text{cemetery}) = 0$, and let τ_H be the first renewal index with live height $p \geq H$. The integrability needed at each step follows from the preceding row bounds. Bounded optional stopping for the nonnegative supermartingale gives, for every m ,

$$V(H)\mathbb{P}_0(\tau_H \leq m) \leq \mathbb{E}_0[V(P_{\tau_H})\mathbf{1}_{\{\tau_H \leq m\}}] \leq \mathbb{E}_0[V(P_{\tau_H \wedge m})] \leq 1.$$

Taking the increasing limit of the hit probabilities proves

$$\mathbb{P}_0(\tau_H < \infty) \leq V(H)^{-1} \leq C_1 e^{-\kappa \sqrt{(H+2) \log(H+2)}}. \quad (7.5)$$

There is no unbounded-stopping-time equality, nor any assertion that the stopped V -values themselves increase. Overshoots are harmless because V is increasing. Equivalently, one can sum the superharmonic inequality over the first-hit stopping line.

[write] 5 October 2026. Part II, Lemma 24.2, uses the same barrier function $\sqrt{(p+2) \log(p+2)}$ and the same supermartingale argument, and adds the bound for the coefficient contribution of length- n paths; that form is Part III's Proposition 30.1(3).

8 A time tilt and exact original time extraction

Fix an integer $H \geq 2$ and put

$$\eta_H = \delta \frac{\log(H+2)}{H+2}, \quad (8.1)$$

where $\delta > 0$ is a sufficiently small fixed constant. Let $K_H(z)$ be the boundary renewal kernel restricted to starting and ending heights in $\{0, \dots, H-1\}$, weighting a step of duration L by z^L . The restriction is on boundary heights only; interior bulk heights are not artificially constrained.

A legal departure below H has $b \leq H$. Drop the endpoint restriction and extend ℓ , retaining that finite cutoff. The exact transforms, Taylor bounds and Lemma 7.1 imply

$$\begin{aligned} \|K_H(e^{\eta_H})\|_\infty &\leq ae^{\eta_H} + A_j(0, \eta_H) \sum_{b=0}^H d_b R_j(0, \eta_H)^b \\ &\leq r_j + O(\delta), \end{aligned} \quad (8.2)$$

uniformly in H . Indeed use $\gamma = B\delta \leq 1/4$ in the Catalan lemma and $\eta_H = O(\delta)$. Choose δ so small that (8.2) is bounded by a fixed $\rho_1 < 1$ for both classes. The root row is smaller as well.

The nonnegative generating function for all finite paths from the root ending at a boundary height below H and having every boundary visit below H is

$$F_H(z) = \sum_{m \geq 0} [K_H(z)^m \mathbf{1}](0). \quad (8.3)$$

Every renewal step has strictly positive original duration. Every original path ending at $c = 0$ decomposes uniquely into these steps. Consequently $[z^n]F_H(z) = B_H(n)$ is exactly the confined length- n q -mass. In particular this coefficient uses original time, not renewal index. At the positive evaluation point,

$$F_H(e^{\eta_H}) \leq \sum_{m \geq 0} \rho_1^m = \frac{1}{1 - \rho_1}.$$

Positivity of coefficients now gives

$$B_H(n) \leq \frac{1}{1 - \rho_1} e^{-\eta_H n}. \quad (8.4)$$

The full random excursion durations, and their dependence on spatial displacements, are retained throughout. Replacing a duration by its mean would not justify this coefficient bound.

8.1 Splitting at the first high boundary visit

Partition length- n paths ending at $c = 0$ by whether any boundary visit has height at least H . The confined paths contribute at most (8.4). For each remaining path, split at its first such boundary visit, at original time $s \leq n$. Given its prefix, all suffixes of length exactly $n - s$ have total q -mass at most one, by substochasticity of the original step chain. Therefore the mass of these paths is at most the sum of their first-hit prefix weights. Those prefixes belong to the first-hit stopping line of the embedded chain. Including prefixes whose hit occurs after time n only enlarges the bound, so (7.5) controls this sum. In combination with (6.1),

$$\frac{a_j(n)}{\mu_j^n} \leq C_1 e^{-\kappa \sqrt{(H+2) \log(H+2)}} + C_2 e^{-\delta n \log(H+2)/(H+2)}. \quad (8.5)$$

Take

$$H = \lfloor n^{2/3} (\log n)^{1/3} \rfloor.$$

Then

$$\sqrt{(H+2) \log(H+2)} \sim \sqrt{2/3} \psi(n), \quad \frac{n \log(H+2)}{H+2} \sim \frac{2}{3} \psi(n).$$

Both contributions in (8.5) are bounded by $C e^{-c\psi(n)}$ for suitable fixed positive constants. Absorbing the prefactor by reducing c for sufficiently large n proves the upper inequality in (1.1). The lower inequality was established in (5.3). Since $\psi(n) = o(n)$, the two growth limits in (1.2) follow. This proves Theorem 1.1.

9 Consequences and the inverse counting threshold

9.1 The exponent and the literal numerical proposal

For every fixed $\beta > 1/3$, Theorem 1.1 gives

$$\frac{\log a_j(n) - n \log \mu_j}{n^\beta} \longrightarrow 0. \quad (9.1)$$

In particular $\psi(n)/n^{3/8} = n^{-1/24}(\log n)^{2/3} \rightarrow 0$. If (1.3) were a full equivalent, taking logarithms would instead give the limit $-b < 0$ at $\beta = 3/8$, a contradiction. The issue is the fixed negative nonzero coefficient of $n^{3/8}$ in the logarithm, not the numerical accuracy of an approximation over a finite range. A polynomial factor n^g or fixed amplitude cannot repair the conflict since their logarithms are $O(\log n)$.

The positive lower deficit bound also gives $D_j(n)/n^{1/3} \rightarrow \infty$. Hence a pure $n^{1/3}$ deficit with a fixed finite positive coefficient is ruled out as well. One precise sense of the stretched exponent is

$$\frac{\log D_j(n)}{\log n} \longrightarrow \frac{1}{3},$$

while the matching logarithmic multiplier is $(\log n)^{2/3}$ in the two-sided order estimate. None of this supplies a limit for $D_j(n)/\psi(n)$, nor does it justify differentiating an unspecified Θ term to obtain ratio asymptotics.

[write] **5 October 2026.** Part II, Theorem 13.1, now supplies the limit $\mathcal{D}_j(n)/\psi(n) \rightarrow \sigma_{D_j}$; the remark that the two-sided bound supplies none stays true of this Part. No ratio asymptotic is proved in any Part (Section 39, question 3). Remark 1.2 extends the argument of this subsection to every stretched form $Cn^g\mu_j^n \exp\{-bn^\beta\}$.

9.2 A positive inverse correction with the correct ceilings

For $T > 1$ define

$$N_j(T) = \min\{n \geq 0 : a_j(n) \geq T\}, \quad s = \frac{\log T}{\log \mu_j}.$$

The set is nonempty by exponential growth. The sequence is in fact strictly increasing for $n \geq 1$: appending the new largest entry n to a valid sequence of length n is an injection. A forbidden triple using that last entry would require an earlier entry at least n , which is impossible. The valid sequence $(0, 1, \dots, n-1, n-1)$ lies outside this injection's image, proving strictness. The only initial tie is $a_j(0) = a_j(1) = 1$. The following threshold proof does not need that monotonicity.

[write] **5 October 2026.** This strict monotonicity is proved only in this Part; Parts II and III neither restate nor use it. It gives $a_j(n+1)/a_j(n) > 1$ for $n \geq 1$, nothing like a ratio limit.

Corollary 9.1. *For each class there are constants $0 < d_j < d_j^+ < \infty$ such that, for all sufficiently large real T ,*

$$\lceil s + d_j\psi(s) \rceil \leq N_j(T) \leq \lceil s + d_j^+\psi(s) \rceil. \quad (9.2)$$

Consequently

$$N_j(T) - \frac{\log T}{\log \mu_j} = \Theta((\log T)^{1/3}(\log \log T)^{2/3}),$$

with a positive correction.

Proof. Write $L = \log \mu_j$ and use eventual constants c, C in $c\psi(n) \leq nL - \log a_j(n) \leq C\psi(n)$. Choose $0 < d < c/L$ and $D > C/L$. For large T , no bounded n reaches T . If $n < s$, the global bound $a_j(n) \leq \mu_j^n < T$ rules it out. If $s \leq n < s + d\psi(s)$, monotonicity of the real function ψ gives

$$\log a_j(n) \leq nL - c\psi(n) < \log T + (dL - c)\psi(s) < \log T.$$

Thus every integer below $\lceil s + d\psi(s) \rceil$ is excluded, proving the lower ceiling direction, including when its argument is an integer. For the upper bound take $n = \lceil s + D\psi(s) \rceil$. Since $\psi(s) = o(s)$, we have $n/s \rightarrow 1$ and $\psi(n)/\psi(s) \rightarrow 1$. Therefore

$$\log a_j(n) \geq nL - C\psi(n) \geq \log T + DL\psi(s) - C\psi(n) > \log T$$

eventually. Finally $\psi(s) \sim L^{-1/3}(\log T)^{1/3}(\log \log T)^{2/3}$, and the bounded ceiling errors are negligible on this diverging scale. $\square$

The constants here are existential. The report supplies neither calibrated finite thresholds nor a rounding-sharp one-term inverse equivalent. For all $T > 1$, the simpler global lower bound $\lceil \log T / \log \mu_j \rceil \leq N_j(T)$ remains valid.

[write] 5 October 2026. Refined twice. Part II, Corollary 26.1, gives $N_j(T) = s + (\sigma_{D_j} / \log \mu_j) \psi(s) + o(\psi(s))$, so any constants $d_j < \sigma_{D_j} / \log \mu_j < d_j^+$ serve in (9.2) for large T ; Part III, Corollary 36.1, adds the term $\frac{7\sigma}{3\lambda} \psi(s) \log \log s / \log s$ with error $O(\psi(s) / \log s)$. Neither supplies calibrated finite thresholds.

10 Boundary regularity and differential equations

For either class write $a_n = a_j(n)$, $\mu = \mu_j$, $\rho = 1/\mu$, $b_n = a_n \mu^{-n}$, and

$$F(z) = \sum_{n \geq 0} a_n z^n, \quad f(\theta) = F(\rho e^{i\theta}) = \sum_{n \geq 0} b_n e^{in\theta}.$$

Theorem 1.1 implies $e^{-C\psi(n)} \leq b_n \leq e^{-c\psi(n)}$ eventually. The radius of F is exactly ρ .

Theorem 10.1. *Every derivative of F extends continuously from its open convergence disk to the closed disk $|z| \leq \rho$. The boundary function f is smooth and periodic. As the integer $k \rightarrow \infty$,*

$$\log \|f^{(k)}\|_\infty = 3k \log k - 2k \log \log(k+2) + O(k), \quad (10.1)$$

$$\log F^{(k)}(\rho) = 3k \log k - 2k \log \log(k+2) + O(k). \quad (10.2)$$

The boundary function has sharp Gevrey order 3. The point ρ is a genuine singularity of F , and F is not D -finite over $\mathbb{C}(z)$; in particular it is not algebraic.

Here Gevrey order s means that $\|f^{(k)}\|_\infty \leq AB^k(k!)^s$ for some $A, B > 0$ and all k . “Sharp order 3” means the estimate holds for $s = 3$ and fails for every $s < 3$. It is not a claim of a convergent boundary Taylor series.

[write] 5 October 2026. The $O(k)$ in (10.1)–(10.2) is identified as $\lceil \log(2D/\pi^2) - 3 \rceil k + o(k)$, plus $k \log \mu$ for the radial derivatives, by Part II, Corollary 26.2, and refined to $\lceil \log(2D/\pi^2) - 3 \rceil k - k \log \log k / \log k + O(k / \log k)$ by Part III, Corollary 37.1; those Parts write $\log \log k$, which differs from $\log \log(k+2)$ here by $O(1/(k \log k))$, so by $O(1/\log k)$ after multiplication by $2k$. The genuine singularity at ρ , the non- D -finiteness over $\mathbb{C}(z)$ and the non-algebraicity are proved only in this Part.

10.1 All boundary derivatives and their growth

For every fixed k , the sum $\sum n^k b_n$ is finite because $\psi(n)/\log n \rightarrow \infty$. Thus each derivative series for F converges absolutely and uniformly on $|z| \leq \rho$. The same argument for the Fourier series of f

gives smoothness and, for $k \geq 1$,

$$f^{(k)}(0) = i^k \sum_{n \geq 1} n^k b_n, \quad \|f^{(k)}\|_\infty = \sum_{n \geq 1} n^k b_n =: M_k. \quad (10.3)$$

The upper norm bound follows from the triangle inequality, and equality holds at $\theta = 0$ by nonnegativity of b_n .

Put $N_k = k^3 / [\log(k+2)]^2$. Then $N_k/k \rightarrow \infty$ and $\psi(N_k)/k \rightarrow 3^{2/3}$. In particular, there exist fixed constants $d, D > 0$ such that

$$dk \leq \psi(N_k) \leq Dk$$

for all large k . The single term $n_k = \lfloor N_k \rfloor$ and the lower coefficient bound give

$$M_k \geq \exp\{k \log N_k - O(k)\}. \quad (10.4)$$

For a matching upper bound, for any fixed $a > 0$ we have

$$\sup_{n \geq 2} n^k e^{-a\psi(n)} \leq \exp\{k \log N_k + O_a(k)\}. \quad (10.5)$$

For $n \leq N_k$ this is immediate. For $n \geq N_k$, monotonicity of $\log$ yields $\psi(n) \geq \psi(N_k)(n/N_k)^{1/3} \geq dk(n/N_k)^{1/3}$. Writing $u = n/N_k$, the remaining maximization is of $\log u - adu^{1/3}$ over $u \geq 1$, whose supremum is a finite constant independent of k . This proves (10.5).

Absorb finitely many exceptional coefficients into a fixed multiplicative factor. Apply (10.5) with $a = c/2$, leaving the summable factor $e^{-c\psi(n)/2}$ outside the supremum. The $n = 1$ term is harmless. Consequently $M_k \leq \exp\{k \log N_k + O(k)\}$. Together with (10.4) and $\log N_k = 3 \log k - 2 \log \log(k+2)$ this proves (10.1).

For the radial derivatives, with $(n)_k = n(n-1) \cdots (n-k+1)$,

$$F^{(k)}(\rho) = \mu^k \sum_{n \geq k} (n)_k b_n.$$

The upper bound follows from $(n)_k \leq n^k$. For the lower bound use $n = n_k \geq 2k$, so $(n_k)_k \geq (n_k/2)^k$. The extra factors μ^k and 2^{-k} change the logarithm only by $O(k)$. This proves (10.2). Stirling's formula now proves Gevrey order 3 and excludes every smaller order.

Moreover,

$$\left(\frac{F^{(k)}(\rho)}{k!} \right)^{1/k} = \frac{k^2}{[\log(k+2)]^2} \exp\{O(1)\} \rightarrow \infty.$$

Thus the Taylor series formed from the radial derivatives at ρ has radius zero. A holomorphic continuation through ρ would have a positive-radius Taylor series with those derivatives, which is impossible. This proves genuine singularity despite all finite boundary derivatives. It says nothing about analytic continuation through other points of the convergence circle.

10.2 The arithmetic obstruction to a differential equation

Suppose first that F were D-finite over $\mathbb{Q}(z)$, meaning it satisfies a nonzero linear differential equation with rational-function coefficients. Its coefficients are integers, hence have denominator one and only trivial algebraic conjugates; $a_n \leq \mu^n$ gives the required exponential bound. Therefore it is a G-function.

We use the André–Chudnovsky–Katz theorem in the form stated by Fischler and Rivoal [2, Theorem 6, pp. 328–329]: a minimal differential operator of a G-function is Fuchsian, with rational local exponents. Choose a small sector about ρ containing the interval immediately to its left and avoiding other operator singularities; this selects the original branch of F . If ρ is an ordinary point of the operator, F is already holomorphic there. Otherwise, writing $w = \rho - z$, its solutions on that sector are finite sums

$$\sum_{q \in E} \sum_{j=0}^{J_q} w^q (\log w)^j h_{qj}(w), \quad (10.6)$$

where $E \subset \mathbb{Q}$ is finite and the h_{qj} are holomorphic at zero. This is the only arithmetic regular-singularity theorem imported into the argument.

A convergent representation (10.6) that is smooth to every order along $w > 0$ must actually be holomorphic at $w = 0$. To see this, combine exponent classes modulo integers and expand the finitely many holomorphic factors. The supports form a finite union of discrete sets $q + \mathbb{N}$, bounded below. Separate the nonnegative integral powers without logarithms; they form a convergent holomorphic function. If any nonholomorphic term remains, choose its least exponent r and at that exponent its largest nonzero logarithmic degree j . Its coefficient is a nonzero polynomial $P(\log w)$ of degree j . For some $\epsilon > 0$ and finite J , the remaining analytic tails satisfy

$$R^{(m)}(w) = O(w^{r+\epsilon-m}(1 + |\log w|)^J) \quad (w \downarrow 0)$$

after every fixed number m of derivatives. This follows by factoring the first surviving power from each of the finitely many convergent tails and bounding its analytic factor and derivatives on a smaller disk. For the leading term, the exact identity

$$\left(\frac{d}{dw}\right)^m [w^r P(\log w)] = w^{r-m} \prod_{\ell=0}^{m-1} \left(r - \ell + \frac{d}{dt}\right) P(t) \Big|_{t=\log w}$$

controls all possible logarithmic cancellation.

- If $r < 0$, this residual function is unbounded as $w \downarrow 0$.
- If $r \geq 0$ is nonintegral, differentiate $\lfloor r \rfloor + 1$ times. Its falling-factorial multiplier is nonzero, and the resulting negative power makes that derivative unbounded.
- If $r \geq 0$ is integral, then $j \geq 1$. Differentiating $r + 1$ times gives a nonzero leading multiple of $w^{-1}(\log w)^{j-1}$, again unbounded.

Higher exponents and lower logarithmic degrees cannot cancel the stated leading terms. Termwise differentiation is justified by the convergent local representation, so no uncontrolled asymptotic remainder is being differentiated. Each case contradicts the finite derivatives established above. The solution must therefore be holomorphic at ρ , contradicting its genuine singularity. Thus F is not D-finite over $\mathbb{Q}(z)$.

Finally, D-finiteness over $\mathbb{C}(z)$ descends to $\mathbb{Q}(z)$ for a rational-coefficient series. Clear denominators in a proposed complex differential equation and fix its finite order and coefficient degree bounds. Its finitely many unknown coefficients satisfy a homogeneous linear system with rational entries, one equation for each series coefficient. The rational row space has a finite basis and the same rank over $\mathbb{Q}$ and $\mathbb{C}$. A nonzero complex null vector would therefore give a nonzero rational one and a rational differential equation, already excluded. This completes the proof of Theorem 10.1. The familiar implication algebraic $\Rightarrow$ D-finite gives nonalgebraicity.

These conclusions do not identify a unique dominant singularity or a complex slit-domain expansion, and do not supply a leading stretched constant or further terms of a transseries.

11 What remains to be proved

The two bounds identify a scale, leaving several distinct problems open.

1. **A leading logarithmic constant.** Does $D_j(n)/\psi(n)$ converge to a positive constant? If so, one needs a sharper principle for the optimal height profile or the killed renewal kernel. Optimizing the present nonsharp inequalities cannot identify it.
2. **An equivalent for the sequence.** Even a leading logarithmic constant would leave lower-order exponent terms, polynomial factors, and an amplitude unresolved. Their existence and normalization would require separate arguments.
3. **Ratios and effective error rates.** A two-sided deficit bound does not establish $a_j(n+1)/a_j(n) \rightarrow \mu_j$ or a correction rate for this ratio. Effective constants and usable onset bounds are also not supplied here.
4. **All orders and transseries.** No complete asymptotic expansion, all-orders remainder theorem, or transseries follows from the matching order estimate. Such claims would need an identified analytic structure and uniform error control.
5. **Numerical crossover.** The logarithmic multiplier changes slowly, so finite-range pure-power fits merit fresh investigation against the proven scale. Determining a practical crossover or an effective fitting window is a separate numerical project; no new digits or crossover prediction are claimed in this report.

11.1 Further questions and research

[write] (5 October 2026.) Under Vladimir’s standing rule of 4 October 2026, this subsection records the status of every claim of this Part stated without proof, with what the later Parts settle. Nothing in this Part was found to be wrong; the external numerical forms that it refutes are on record in Remark 1.2.

1. *A leading logarithmic constant* (item 1 above). Answered: Part II, Theorem 13.1, proves $\mathcal{D}_j(n)/\psi(n) \rightarrow \sigma_{D_j}$, by the “sharper principle for the optimal height profile” that the item asks for (the singular variational problem of Section 19).
2. *An equivalent for the sequence* (item 2). Open; Section 39, question 2. Part III determines one more exponent term, but not the next one, the polynomial factor or the amplitude.
3. *Ratios and effective error rates* (item 3). Ratios open (question 3). Part III gives a quantitative remainder $O(\psi(n)/\log n)$, with constants that are not effective; effective constants and onsets remain open (question 4).
4. *All orders and transseries* (item 4). Open (question 4).
5. *Numerical crossover* (item 5). Open (question 5).

6. *Other singularities on the circle* (the last paragraph of Section 10: no unique dominant singularity, no slit-domain expansion). Open (question 6).

12 Verification and reproducibility

The analytic proof above is the basis for the infinite statements. The reader package also contains an independently implemented exact finite companion. It checks the original pattern-avoidance definitions, decorated physical histories, state recurrences, exact cycle factors, joint transforms including $b = 0$, critical derivatives, and finite instances of the rounded construction. In particular it distinguishes completed endpoints from all auxiliary nodes. Rational upper and lower bounds for logarithms certify the floors used in staircase tests; floating-point rounding is not used for those floors.

The authoritative entry points are `checks/verify.py` and `checks/negative_tests.py`. Their recorded outputs are `data/verification_results.json` and `data/negative_results.json`. The companion’s README records all exact ranges and provenance. The tests have explicit exceptions, so their guards remain active under optimized Python. Mathematical, strict-schema, and inventory mutations are run in fresh copies and must fail with their expected diagnostics. These tests are regression certificates for finite identities; they are not finite proofs of the uniform analytic estimates, asymptotic theorem, boundary derivative growth, Gevrey conclusion, or non-D-finiteness.

The complete offline replay is

```
python3 -B reproduce.py -output ../report126-replay.json
```

It checks closed inventories, compares normal and optimized exact-check results byte-wise, exercises mutations, builds the PDF twice in clean external directories, and requires identical PDF bytes. It then creates a deterministic ZIP, extracts a fresh copy, repeats exact checks and PDF reconstruction, and requires identical repacking. All outputs are outside the reader directory, whose bytes are checked again at the end. The optional `-skip-pdf` switch records that PDF reconstruction was omitted; it does not claim a full replay. The build environment is recorded because different TeX or compression versions can change bytes without changing mathematical content.

No downloaded source paper, development scratch notes, numerical fit, or externally published artifact is required for replay. Integrity inventories are tamper-detection records relative to the supplied verifier, not cryptographic signatures or a proof of authenticity against simultaneous replacement of that verifier and its inventory.

[write] 5 October 2026. In this directory the companion’s files carry the prefix `126-scale-`: `checks/verify.py` and `checks/negative_tests.py` are `code/126-scale-checks-verify.py` and `code/126-scale-checks-negative_tests.py`, the recorded outputs are `data/126-scale-verification_results.json` and `data/126-scale-negative_results.json`, and the companion’s README is `126-scale-checks-README.md`. The checkers enforce the closed inventory of the delivered layout (`checks/manifest.sha256`, `CHECKSUMS.sha256`), which is not shipped, so they do not run here; the report’s README gives the rerun route from the arrival commit.

Part II

Sharp logarithmic deficit constants for two inversion sequence classes

Bundle Report 127, archive A279551_A279556_Sharp_Deficit_Constants_and_Inversion_Source.zip, delivered as `report127.tex` (24 pages), dated 2 October 2026. Delivered title block “Report 127. Sharp logarithmic deficit constants for two inversion sequence classes. A279556 and A279551. Standalone proof and reproducible source companion”; no author line. Printed as delivered, with `[write]` notes, except its Sections 2–4, which are Report 126’s Sections 2–4 word for word and are printed in Part I. Delivered Section k is Section $k + 12$ here, and its statement $k.j$ is $(k + 12).j$; references to the displays of its Sections 2–4 and to its Lemma 4.1 point to Part I, where they keep their numbers. Section 27.1 is added in the write.

Reading conventions of Part II	Elsewhere in the report
$D = D_j \in \{1, \frac{1}{2}\}$, diffusion coefficient, “not a deficit”	Part I: $D_j(n)$ is the deficit
$a_n^* = E_n/n$; $[a, b]$ in Theorem 21.1; $b = 1 - \delta$	$a = x/\mu$; b the number of commitments
$\mu_i = \mathbb{E}[L_i \mid \text{past}]$ (proof of Theorem 21.1)	$\mu = \mu_j$ the growth constant
M_k martingale (Section 21); M_k moments (Section 26)	Part I: M_k moments only
$k = S/n$ (Sections 24–25); $\lambda = \log \mu$ (Section 26)	k stair index or derivative order; λ the tilt
$[l, r]$ a strip (Lemma 20.2); slack e	r the row mass; e Euler’s number

Abstract of this Part (as delivered). We prove the sharp leading logarithmic asymptotics for two classes of inversion sequences enumerated by commitment trees. For A279556 the growth constant is 9 and the logarithmic deficit constant is $(3\pi^2/2)^{1/3}$; for A279551 they are 8 and $(3\pi^2)^{1/3}$. The common deficit scale is $n^{1/3}(\log n)^{2/3}$. The proof retains the exact legal initial-drop cutoff and every cycle’s random original duration. A uniform local critical-root theorem gives a state-dependent tilted law. Separate clock and reward martingales, an exact likelihood identity, and discrete all-integer endpoint repairs prove the coefficient lower bound. A positive space–time supersolution, with long-cycle and large-height tails controlled, proves the coefficient upper bound. We solve the associated singular variational problem and derive precise inverse-threshold and boundary-derivative consequences. The combinatorial input is the exact commitment construction of Britt and Beaton. This is a leading logarithmic asymptotic, not a coefficient equivalent or an all-orders expansion. No global priority claim is made.

13 Definitions and the sharp theorem

An inversion sequence of length n is an integer sequence $e = (e_1, \dots, e_n)$ with $0 \leq e_i < i$. Let $a_{247}(n)$ count those with no $i < j < k$ satisfying $e_i \leq e_j$ and $e_i \geq e_k$. Let $a_{759}(n)$ count those with no $i < j < k$ satisfying these two inequalities and $e_j \neq e_k$. In pattern notation the forbidden patterns are (000, 010, 110, 120) and (010, 110, 120), respectively. Thus

$$a_{247}(n) = A279551(n), \quad a_{759}(n) = A279556(n), \quad a_j(0) = 1.$$

All logarithms are natural. Throughout the proof, D denotes the positive diffusion coefficient in the following table, not a deficit:

j	μ_j	D_j	$\sigma_{D_j} = (3\pi^2/(2D_j))^{1/3}$
759	9	1	$(3\pi^2/2)^{1/3}$
247	8	1/2	$(3\pi^2)^{1/3}$

For $n > 1$ define

$$H = H_n = n^{2/3}(\log n)^{1/3}, \quad S = S_n = n^{1/3}(\log n)^{2/3}, \quad \theta_n = H/n = S/H. \quad (13.1)$$

In particular $H^2 = nS$, $H^3 = n^2 \log n$, $HS = n \log n$, and $\theta_n^2 = S/n$.

Theorem 13.1. *For $j \in \{759, 247\}$, as $n \rightarrow \infty$ through all integers,*

$$\log a_j(n) = n \log \mu_j - \sigma_{D_j} S_n + o(S_n). \quad (13.2)$$

Equivalently,

$$a_j(n) = \mu_j^n \exp\{-(\sigma_{D_j} + o(1))S_n\}.$$

Moreover $a_j(n) \leq \mu_j^n$ for every $n \geq 0$ and $\lim_{n \rightarrow \infty} a_j(n)^{1/n} = \mu_j$.

The constants are approximately 2.455445701568578 for A279556 and 3.093667726280136 for A279551. These are evaluations of the exact formula, not fits to sequence data. The error in (13.2) is only $o(S_n)$; exponentiating it does not give a multiplicative $1 + o(1)$ error. In particular the theorem identifies neither an exact multiplier nor lower-order exponent terms.

[write] **5 October 2026.** This answers the first question of Part I (Section 11, item 1) and implies Part I's Theorem 1.1. Part III, Theorem 29.1, identifies the next exponent term, $\frac{7\sigma_D}{3} S_n \log \log n / \log n$, with remainder $O(S_n / \log n)$; the multiplier and the polynomial factor are still unknown.

13.1 Combinatorial source and scope

The exact input is Britt and Beaton's commitment construction [1, Sections 3.8–3.9, Lemmas 5–6 and Appendix A]. We reproduce its model, weights, and terminal extraction below, and prove all analytic estimates needed for (13.2). Their Section 4.2 explicitly labels its asymptotic estimates as nonrigorous numerical analysis. The literal proposed forms $An^g \mu_j^n e^{-bn^{3/8}}$, with $A, b > 0$, cannot be full equivalents, since $S_n = o(n^{3/8})$. This observation concerns the asymptotic forms, not their finite enumerations or exact trees. No global literature priority is asserted.

[write] **5 October 2026.** Part I, Remark 1.2, records the two forms (Britt and Beaton's (4.12)–(4.13)) with their fitted values and excludes every stretched form $Cn^g \mu_j^n \exp\{-bn^\beta\}$. This Part does not cite Report 126.

The two global bounds are established directly. We do not invoke a path large-deviation principle for a fixed-law renewal process: the law here depends on height and time, and the original duration is correlated with displacement. The lower argument uses exact tilted laws with a genuine random clock; the upper argument uses a positive kernel on original space–time. Both retain the actual endpoint weights.

14 Decorated commitments and exact counting

[write] (5 October 2026.) Report 127’s Section 2 (subsections “Auxiliary objects and terminal extraction” and “Why completed histories do not overcount”) is word for word Report 126’s Section 2, and is printed once, as Section 2 of Part I, with its edge rules, the multiplicities (2.4) and the terminal sets (2.5).

15 Critical tilts and completed cycles

[write] (5 October 2026.) Report 127’s Section 3 is word for word Report 126’s Section 3, printed as Section 3: the parameter table (3.1), the telescoping weight (3.2), the bound (3.5), the cycle weight (3.6) and the factorizations (3.7)–(3.8).

16 A uniform moderate local lower bound

[write] (5 October 2026.) Report 127’s Section 4 is word for word Report 126’s Section 4, printed as Section 4; Report 127’s Lemma 4.1 is Lemma 4.1, with (4.1) and (4.2).

17 Exact legal renewal transforms

A boundary path has a unique decomposition into ordinary boundary edges, $b = 0$ commitment edges, and completed commitment excursions (3.6). In a completed excursion the last fulfillment is the first return to zero outstanding commitments. Write $\Delta = w - \ell$ and $L = w + 1$ for its displacement and original duration. For $b = 0$ use $\Delta = -\ell, L = 1$; the ordinary edge has $\Delta = L = 1$ and weight $a = x/\mu$.

Set $d_b = C_b/4^b$. Extending only the drop sum for a fixed b gives

$$\sum_{\ell, w} Q_j(b, \ell, w) e^{\lambda\Delta + \eta L} = A_j(\lambda, \eta) d_b R_j(\lambda, \eta)^b, \quad (17.1)$$

with the duration-one interpretation at $b = 0$, where

$$A_{759} = \frac{e^\eta}{9(1 - e^{-\lambda/3})}, \quad R_{759} = \frac{4e^\eta}{9(1 - e^{-\lambda/3})(1 - e^{\lambda+\eta/3})}, \quad (17.2)$$

$$A_{247} = \frac{e^\eta(1 + e^{-\lambda/2})}{8}, \quad R_{247} = \frac{e^\eta(1 + e^{-\lambda/2})}{2(1 - e^{\lambda+\eta/4})}. \quad (17.3)$$

Indeed, the elementary identities used in this summation are

$$\begin{aligned} \sum_{\ell \geq b} \binom{\ell}{b} z^\ell &= \frac{z^b}{(1 - z)^{b+1}}, & \sum_{\ell} \binom{b+1}{\ell - b} z^\ell &= z^b(1 + z)^{b+1}, \\ \sum_{w \geq b} \binom{w-1}{b-1} t^w &= \left(\frac{t}{1-t} \right)^b & (b \geq 1). \end{aligned}$$

At fixed b the sums converge in a fixed neighborhood of $(0, 0)$, independent of b . No convergence of the unrestricted positive-tilt sum over all b is asserted.

Under the fixed- b exponential tilt, drop and duration are independent. Using $\text{NB}^{\text{fail}}(m, q)$ for failures before m successes of probability q , the tilted drop laws are

$$759: \quad \ell = b + \text{NB}^{\text{fail}}(b + 1, 1 - e^{-\lambda}/3), \quad (17.4)$$

$$247: \quad \ell = b + \text{Bin}(b + 1, e^{-\lambda}/(2 + e^{-\lambda})). \quad (17.5)$$

They also hold at $b = 0$. For $b \geq 1$, independently,

$$w \sim \begin{cases} \text{NB}^{\text{trials}}(b, 1 - e^{\lambda+\eta}/3), & 759, \\ \text{NB}^{\text{trials}}(b, 1 - e^{\lambda+\eta}/4), & 247. \end{cases} \quad (17.6)$$

Here the superscript “trials” means that successes are counted too. Independence at fixed b survives the legal-drop restriction because that restriction concerns ℓ alone. Displacement and duration are not independent, since both involve w .

Let $F_{b,\lambda}(p-1)$ be the drop CDF in (17.5). The exact legal row is

$$T_p(\lambda, \eta) = ae^{\lambda+\eta} + A_j(\lambda, \eta) \sum_{b=0}^{p-1} d_b R_j(\lambda, \eta)^b F_{b,\lambda}(p-1). \quad (17.7)$$

At $p = 0$ the sum is empty. The cutoff is truly $\ell \leq p-1$; using merely $b \leq p-1$ would change the sharp result. The constants and derivatives at the origin are

j	a	A_0	α	v	$D = v/\alpha$	$r = a + 2A_0$	$1 - r$
759	1/3	1/6	3/2	3/2	1	2/3	1/3
247	1/4	3/16	4/3	2/3	1/2	5/8	3/8

(17.8)

Specifically $R(0, 0) = 1$, $(\log R)_\lambda(0, 0) = 0$, $(\log R)_\eta(0, 0) = \alpha$, and $(\log R)_{\lambda\lambda}(0, 0) = v$. Analyticity gives

$$\log R(\lambda, \eta) = \alpha\eta + \frac{v}{2}\lambda^2 + O(|\lambda|^3 + |\lambda\eta| + \eta^2). \quad (17.9)$$

The identity $\sum_{b \geq 0} d_b = 2$ shows that the unrestricted zero-tilt boundary row is $r < 1$, consistently with original-time substochasticity established earlier.

17.1 Uniform concentration estimates

For fixed b , normalize the unrestricted zero-tilt commitment mass by $A_0 d_b$. There are fixed $c, C, \theta_0 > 0$ such that

$$\mathbb{P}(|\Delta| > z) \leq C \exp\{-c \min(z^2/(b+1), z)\}, \quad (17.10)$$

$$\mathbb{P}(L > u) \leq C \exp\{-cu + C(b+1)\}. \quad (17.11)$$

Both include $b = 0$. To prove the first, (17.1) and (17.9) give, for $|\vartheta| \leq \theta_0$,

$$\mathbb{E}e^{\vartheta\Delta} \leq \exp\{C|\vartheta| + Cb\vartheta^2\}.$$

Use Chernoff’s bound with either sign and magnitude proportional to $\min(z/(b+1), 1)$; bounded z is absorbed in C . A fixed small positive η in (17.1) proves (17.11).

Uniformly for λ near zero, the tilted drop law satisfies

$$\mathbb{P}(|\ell - \mathbb{E}\ell| \geq z) \leq 2 \exp\{-c \min(z^2/(b+1), z)\}. \quad (17.12)$$

Each law in (17.5) is a sum of $b + 1$ independent geometric or Bernoulli variables, plus b . Their centered log MGFs are bounded by Cv^2 on a common neighborhood of zero. The same Chernoff argument applies. The means are

$$\mathbb{E}\ell = \alpha(\lambda)b + O(1), \quad \alpha(\lambda) = \begin{cases} (1 - e^{-\lambda}/3)^{-1}, & 759, \\ 1 + e^{-\lambda}/(2 + e^{-\lambda}), & 247, \end{cases} \quad (17.13)$$

so $\alpha(\lambda) = \alpha + O(\lambda)$ uniformly.

18 The uniform sharp local root

Theorem 18.1. *Fix $B < \infty$. Uniformly for $|\lambda| \leq B\sqrt{\log p/p}$ and c in a fixed compact interval, put*

$$s = \frac{1}{2} \log p + \log \log p + c, \quad \eta = s/p - D\lambda^2/2.$$

Then, as integer $p \rightarrow \infty$,

$$T_p(\lambda, \eta) = r + 2A_0\sqrt{\alpha/\pi}e^c + o(1). \quad (18.1)$$

The unique root $T_p(\lambda, \eta_p(\lambda)) = 1$ therefore satisfies

$$\eta_p(\lambda) = \frac{\frac{1}{2} \log p + \log \log p + c_* + o(1)}{p} - \frac{D}{2}\lambda^2, \quad c_* = \frac{1}{2} \log(\pi/\alpha), \quad (18.2)$$

uniformly on the same λ range.

Proof. Since $v = \alpha D$, (17.9) yields

$$\beta := \log R(\lambda, \eta) = \alpha s/p + o(1/p). \quad (18.3)$$

Indeed, p times the remainder is $O((\log p)^{3/2}/\sqrt{p} + (\log p)^2/p) = o(1)$ uniformly. Also $A = A_0 + o(1)$ and $ae^{\lambda+\eta} = a + o(1)$.

Set $h = \sqrt{p} \log p$. The tilt moves the mean drop cutoff by $O(p|\lambda| + 1) = O(\sqrt{p \log p}) = o(h)$. Equations (17.12)–(17.13) imply

$$b \leq p/\alpha - h \implies F_{b,\lambda}(p-1) \geq 1 - e^{-c_1(\log p)^2}, \quad (18.4)$$

$$b \geq p/\alpha + h \implies F_{b,\lambda}(p-1) \leq e^{-c_1(\log p)^2}. \quad (18.5)$$

At either endpoint the distance between threshold and mean is a positive fixed multiple of h ; use $b + 1 \leq p$ in the concentration denominator. Rounding changes the margin by $O(1)$ only. For $m_- = \lfloor p/\alpha - h \rfloor$ and $m_+ = \lceil p/\alpha + h \rceil$, the legal sum is squeezed between the untruncated sums through m_- and m_+ , with a multiplicative $1 - o(1)$ below and an additive error above bounded by

$$e^{-c_1(\log p)^2} \sum_{b=0}^{p-1} d_b R^b = o(1).$$

The last sum grows only polynomially in p by (18.3). This is the essential use of the exact drop cutoff.

For either comparison endpoint m , we have $m \sim p/\alpha$ and $m\beta = s + o(1)$, since $h \log p/p = o(1)$. Stirling's formula gives $d_b = \pi^{-1/2} b^{-3/2} (1 + O(1/b))$ for $b \geq 1$. We claim

$$\sum_{b=0}^m d_b e^{\beta b} = 2 + \frac{e^s}{\sqrt{\pi m} s} (1 + o(1)). \quad (18.6)$$

For $b \leq m/2$, the excess over the unweighted sum is at most

$$C\beta\sqrt{m} e^{\beta m/2} = O(p^{-1/4} (\log p)^{3/2}) = o(1),$$

using $\sum_{b \leq m} b d_b = O(\sqrt{m})$. The omitted unweighted tail is $O(m^{-1/2})$. For $b > m/2$, write $k = m - b$. After extracting $e^{\beta m}/(\sqrt{\pi m}^{3/2})$, the remaining sum is

$$\sum_{0 \leq k < m/2} e^{-\beta k} (1 - k/m)^{-3/2} (1 + O(1/m)) \sim \beta^{-1}.$$

For a direct proof, first restrict to $k \leq K/\beta$; there $k/m = O(K/s) \rightarrow 0$. The complementary geometric tail is at most Ce^{-K}/β . Let $p \rightarrow \infty$ and then $K \rightarrow \infty$. This proves (18.6), uniformly in the stated ranges. Now

$$\frac{e^s}{\sqrt{\pi m} s} \rightarrow 2\sqrt{\alpha/\pi} e^c,$$

which proves (18.1).

For $p \geq 2$ the legal $b = 1, \ell = 1$ term is positive; its bulk MGF diverges at the upper duration-MGF edge. The row is continuous and strictly increasing in η , tends to zero as $\eta \rightarrow -\infty$, and therefore has exactly one root. At $p = 0, 1$ the explicit rows are

$$T_0 = ae^{\lambda+\eta}, \quad T_1 = e^\eta (ae^\lambda + 1/\mu),$$

and also have unique roots. In (18.1) bracket the root with $c = c_* \pm \varepsilon$. Since $1 - r = 2A_0$ in both classes, $c_* = (1/2) \log(\pi/\alpha)$, and uniformity plus monotonicity give (18.2). $\square$

The local $\log \log p$ term in (18.2) is not a next-order expansion of the coefficient logarithm. On the global scales (13.1), this theorem implies

$$\frac{n}{S} \eta_{\lfloor Hx \rfloor}(\theta_n q) \rightarrow \frac{1}{3x} - \frac{D}{2} q^2 \quad (18.7)$$

locally uniformly for $x > 0$ and bounded q . The factor $1/3$ comes from $HS/n = \log n$ and $\log H \sim (2/3) \log n$.

[write] 5 October 2026. Part III, Lemma 31.1, makes this theorem quantitative (error $O(\log \log p / \log p)$ in the scalar root) and anchors the tilt dependence (errors $O(p^{-3/2}(\log p)^C)$ for the root and $O(p^{-3/4}(\log p)^C)$ for its derivative). It also shows how the local terms $\frac{1}{2} \log p + \log \log p$ do enter the global expansion: accumulated along the minimizing profile they give the potential $\frac{1}{3} + \frac{7}{6} \frac{J}{N}$ and the term $\frac{7\sigma p}{3} SJ/N$. The sentence above, that the local $\log \log p$ term is not by itself a next-order expansion of the coefficient logarithm, remains true; Part III supplies the additional estimates that item 2 of Section 27 asks for.

19 The singular variational minimum

For nonnegative $f \in H_0^1(0, 1)$ with $\int_0^1 1/f < \infty$, set

$$I_D(f) = \int_0^1 \left(\frac{f'(t)^2}{2D} + \frac{1}{3f(t)} \right) dt. \quad (19.1)$$

Finite potential means $f > 0$ almost everywhere; endpoint values are understood as Sobolev traces.

Theorem 19.1. *The unique minimizer of (19.1) is the interior positive symmetric function f_* given parametrically by*

$$f_*(t) = M \sin^2 \vartheta, \quad t = \frac{\vartheta - \sin \vartheta \cos \vartheta}{\pi}, \quad 0 \leq \vartheta \leq \pi, \quad M = \left(\frac{2D}{3\pi^2} \right)^{1/3}. \quad (19.2)$$

Its action is $I_D(f_*) = 1/M = \sigma_D$.

Proof. Direct differentiation, using $M^3\pi^2 = 2D/3$, gives

$$f'_* = M\pi \cot \vartheta, \quad f''_* = -D/(3f_*^2), \quad (19.3)$$

$$\frac{f_*'^2}{2D} = \frac{1}{3}(1/f_* - 1/M), \quad \int_0^1 \frac{dt}{f_*(t)} = 2/M. \quad (19.4)$$

Near zero, $f_*(t) \sim (3D/2)^{1/3}t^{2/3}$, with the reflected behavior at one. Thus $f_* \in H_0^1$, both terms of its action are integrable, and (19.4) gives $I_D(f_*) = 1/M$.

For any admissible g , the first variation vanishes:

$$\int_0^1 \left[\frac{f'_*(g' - f'_*)}{D} - \frac{g - f_*}{3f_*^2} \right] dt = 0. \quad (19.5)$$

To justify integration by parts, the Sobolev trace estimate is $|g(t)| \leq \sqrt{t} \|g'\|_{L^2(0,t)} = o(\sqrt{t})$. Since $f'_* = O(t^{-1/3})$, the boundary product vanishes. Also $|g|/f_*^2 = O(t^{-5/6})$ and $1/f_* = O(t^{-2/3})$ are integrable, and the kinetic cross term is integrable by Cauchy–Schwarz. The other endpoint is identical. Apply (19.3) on $[\varepsilon, 1 - \varepsilon]$ and pass to the limit to obtain (19.5). The scalar identity

$$\frac{1}{g} - \frac{1}{f_*} + \frac{g - f_*}{f_*^2} = \frac{(g - f_*)^2}{gf_*^2}$$

holds almost everywhere. Subtracting the two actions and using (19.5) yields the nonnegative remainder

$$I_D(g) - I_D(f_*) = \frac{1}{2D} \int_0^1 (g' - f'_*)^2 dt + \frac{1}{3} \int_0^1 \frac{(g - f_*)^2}{gf_*^2} dt \geq 0.$$

Equality forces $g = f_*$, proving uniqueness. $\square$

The variational calculation alone is not a coefficient theorem. The remaining proof sections establish the required global bounds for the actual legal kernel, including both singular endpoint layers.

[write] **5 October 2026.** Part III, Lemma 32.1, solves the same problem with $1/(3f)$ replaced by A/f for A near $\frac{1}{3}$: the minimum is $\sigma_D(3A)^{2/3}$, attained by $f_A = M_A \sin^2 \varphi$ with $M_A = (2DA/\pi^2)^{1/3}$, and at $A = \frac{1}{3}$ it is this theorem ($M_{1/3} = M$).

20 Exact tilted laws and their legal moments

At height p , define the row-one probability law on legal cycles by

$$\mathcal{P}_{p,\lambda}(\text{cycle}) = Q_p(\text{cycle})e^{\lambda\Delta + \eta_p(\lambda)L}. \quad (20.1)$$

The notation Q_p includes ordinary and $b = 0$ boundary edges. All expectations in this section use (20.1) unless a fixed- b unrestricted law is explicitly named.

Lemma 20.1. *For each finite B , uniformly as $p \rightarrow \infty$ and $|\lambda| \leq B\sqrt{\log p/p}$, there are constants $c, C > 0$ with*

$$cp \leq \mathbb{E}L \leq Cp, \quad \mathbb{E}L^2 \leq Cp^2, \quad \mathbb{E}\Delta^2 \leq Cp \log p. \quad (20.2)$$

Proof. At the root tilt, let $s_p = (1/2)\log p + \log \log p + c_* + o(1)$. Then (18.2) and (17.9) give

$$\log R = \alpha s_p/p + o(1/p), \quad A = A_0 + o(1). \quad (20.3)$$

The negative-binomial parameters in (17.6) stay in a compact subset of $(0, 1)$. Thus for $1 \leq b \leq p-1$, $\mathbb{E}[(w+1)^2 | b] \leq C(b+1)^2 \leq Cp^2$. Multiply by the legal masses $Ad_b R^b F_{b,\lambda}(p-1)$, which together with the ordinary edge sum exactly to one. The $b = 0$ and ordinary durations equal one. This proves $\mathbb{E}L^2 \leq Cp^2$ and, by Cauchy–Schwarz, $\mathbb{E}L \leq Cp$.

For the lower mean bound, restrict b to the integer interval

$$J_p = [p/\alpha - 2p/\log p, p/\alpha - p/\log p]. \quad (20.4)$$

It is contained in $[1, p-1]$ for large p . On J_p , the Catalan estimate, (20.3), and $e^{s_p} \asymp \sqrt{p} \log p$ show that $Ad_b R^b \geq c \log p/p$. There are order $p/\log p$ such integers. The tilted mean drop lies below p by order $p/\log p$; the $O(p|\lambda| + 1)$ shift is smaller, uniformly. Equation (17.12) implies $F_{b,\lambda}(p-1) = 1 - o(1)$ throughout J_p . Hence this interval has probability at least a positive constant. There $L \geq b+1 \geq cp$, proving $\mathbb{E}L \geq cp$.

For the displacement bound, put $h = \sqrt{p} \log p$ and $m_+ = \lceil p/\alpha + h \rceil$. The local-root proof gives

$$\sum_{b \leq m_+} d_b R^b = O(1), \quad F_{b,\lambda}(p-1) \leq e^{-c(\log p)^2} \quad (b > m_+). \quad (20.5)$$

At fixed b , before applying legality, drop and duration are independent, have variances $O(b+1)$, and their mean coefficients agree at zero tilt. Analyticity therefore gives

$$\mathbb{E}(w - \ell) = O(1 + (b+1)(|\lambda| + |\eta|)).$$

Since $\eta = O(\log p/p)$, for $b \leq p-1$,

$$\mathbb{E}[(w - \ell)^2 | b] \leq C[(b+1) + (b+1)^2(\lambda^2 + \eta^2)] \leq Cp \log p. \quad (20.6)$$

For $b \leq m_+$, drop the legality indicator only in this numerator; (20.5) makes the summed contribution $O(p \log p)$. For $b > m_+$ use legality itself: $0 \leq \ell \leq p-1$, and w is independent of that event. Consequently

$$\mathbb{E}[(w - \ell)^2 \mathbf{1}_{\{\ell \leq p-1\}} | b] \leq 2\mathbb{E}[w^2 | b]F_{b,\lambda}(p-1) + 2p^2 F_{b,\lambda}(p-1) \leq Cp^2 F_{b,\lambda}(p-1).$$

The unrestricted sum through $p-1$ is at most polynomial in p by (20.3); the exponentially small factor in (20.5) makes this far tail $o(1)$. The $b = 0$ drop second moment is uniformly bounded. The ordinary edge contributes at most one. This proves (20.2). Extending the unrestricted mixture all the way to $b = p$ in the central bound would not be justified; its mass need not be $O(1)$. $\square$

Lemma 20.2. *For fixed $0 < l < r < \infty$ and $Q < \infty$, uniformly for $p/H \in [l, r]$ and $|q| \leq Q$,*

$$\frac{\mathbb{E}_{p, \theta_n q} \Delta}{\mathbb{E}_{p, \theta_n q} L} = \theta_n [Dq + o(1)]. \quad (20.7)$$

Proof. The transform is analytic near its root and its η derivative is positive. Implicit differentiation gives exactly

$$\eta'_p(\lambda) = -\mathbb{E} \Delta / \mathbb{E} L. \quad (20.8)$$

The function $\log T_p$ is convex in (λ, η) and strictly increasing in η . Its sublevel set is the hypograph of the root function, so η_p is concave. Put $\psi_n(x, q) = (n/S)\eta_{\lfloor Hx \rfloor}(\theta_n q)$. Equation (18.7) gives locally uniform convergence to $1/(3x) - Dq^2/2$. For differentiable concave functions, the derivative at q lies between the forward and backward difference quotients with step $u > 0$. Uniform convergence bounds these quotients uniformly in x and on compact q intervals. Let first $n \rightarrow \infty$, then $u \downarrow 0$. The limiting quadratic has a uniform modulus of continuity of its derivative, hence $\partial_q \psi_n(x, q) \rightarrow -Dq$ uniformly. Because $(n/S)\theta_n = 1/\theta_n$, this says $\eta'_p(\theta_n q) = \theta_n [-Dq + o(1)]$. Use (20.8) to conclude. $\square$

No derivative of an unspecified asymptotic remainder has been taken. Concavity supplies the derivative control that is actually needed.

21 A positive strip path with its genuine random clock

Theorem 21.1. *Let $0 \leq a < b \leq 1$ and let f be C^2 and positive on a neighborhood of $[a, b]$. Start at boundary height p_n and integer original time s_0 with $p_n/H \rightarrow f(a)$ and $s_0/n \rightarrow a$. At departure time s and height p use (20.1) with*

$$\lambda_s = \theta_n q(s/n), \quad q = f'/D. \quad (21.1)$$

Stop at the first boundary time $s_N \geq nb$. With probability tending to one under this tilted law,

$$s_N/n \rightarrow b, \quad \max_{i \leq N} |p_i/H - f(s_i/n)| \rightarrow 0. \quad (21.2)$$

For every fixed positive action tolerance, the original normalized mass of a set of these paths is at least

$$\exp \left\{ -S \left[\int_a^b \left(\frac{f'^2}{2D} + \frac{1}{3f} \right) dt + o(1) \right] \right\}. \quad (21.3)$$

Here (21.2) is convergence in probability; equivalently in (21.3) any fixed positive tolerance may replace $o(1)$ for all sufficiently large n . The estimates are uniform for bounded C^2 families on common neighborhoods with a common positive strip margin and converging starting heights and times.

Proof. Choose a strip $[lH, rH]$ whose rescaled interior contains the values of f with a fixed positive margin, and extend f slightly past b . Stop additionally on leaving that strip and after

$$m_n = \lceil 2n/(cH) \rceil + 1$$

cycles, where c is a uniform lower-mean constant from Lemma 20.1 on the strip. All martingale increments below are set to zero after this auxiliary stopping time.

Write $\mu_i = \mathbb{E}[L_i \mid \text{past}]$. On active steps, $\mu_i \geq cH$ and $\mathbb{E}[L_i^2 \mid \text{past}] \leq CH^2$. The clock martingale

$$Z_k = \sum_{i < k} (L_i - \mu_i)$$

has expected quadratic variation at most $Cm_n H^2 = O(nH)$. Doob's L^2 inequality yields

$$\max_{k \leq m_n} |Z_k|/n \longrightarrow 0 \quad \text{in probability.} \quad (21.4)$$

Also

$$\mathbb{E} \sum_{i < m_n, \text{active}} (L_i/n)^2 = O(H/n) = o(1). \quad (21.5)$$

Thus $\max L_i/n \rightarrow 0$ in probability. If neither time nb nor the strip boundary has been reached by the cap, accumulated conditional mean duration exceeds $2n$ whereas actual elapsed time is at most n . Equation (21.4) excludes this with probability $1 - o(1)$.

At each active step let

$$v_i = \frac{\mathbb{E}[\Delta_i \mid \text{past}]}{\mathbb{E}[L_i \mid \text{past}]} = \theta_n[f'(s_i/n) + \varepsilon_{n,i}], \quad \sup_i |\varepsilon_{n,i}| \rightarrow 0,$$

by Lemma 20.2. The separate reward martingale is

$$M_k = \sum_{i < k} (\Delta_i - v_i L_i).$$

Its conditional second moments are at most $2\mathbb{E}\Delta_i^2 + 2v_i^2 \mathbb{E}L_i^2 \leq CH \log H$, since $\theta_n^2 H^2 = O(H \log H)$. Hence

$$\mathbb{E}M_{m_n}^2 = O(n \log n), \quad \max_{k \leq m_n} |M_k|/H \longrightarrow 0 \quad \text{in probability,} \quad (21.6)$$

because $n \log n/H^2 \rightarrow 0$. Up to the stop, elapsed time is at most $n + \max L_i = O(n)$ with high probability. Therefore, uniformly at stopped boundary times,

$$p_k/H = p_0/H + \sum_{i < k} f'(s_i/n) L_i/n + o_{\mathbb{P}}(1).$$

The difference between the sum and $f(s_k/n) - f(s_0/n)$ is at most a fixed multiple of $\sum_i (L_i/n)^2$, by the Lipschitz bound for f' . This tends to zero by (21.5). The resulting tube convergence precludes strip exit because f has a positive strip margin. The cap was already excluded. Thus the first stop is the crossing of nb , its overshoot is $o(n)$, and (21.2) follows. Boundary-to-boundary linear interpolation has the same limit because the rescaled time mesh tends to zero.

For each finite realized path the inverse change of measure is exact:

$$\prod Q_{p_i} = \prod \mathcal{P}_{p_i, \lambda_{s_i}} e^{-\mathcal{A}_n}, \quad \mathcal{A}_n = \sum_i [\lambda_{s_i} \Delta_i + \eta_{p_i}(\lambda_{s_i}) L_i]. \quad (21.7)$$

It remains exact for stopped paths because stopping is a deterministic function of their observed finite history. No individual duration has been conditioned on a global exact-time event. On the strip, (18.7) gives

$$\begin{aligned} \frac{\mathcal{A}_n}{S} &= \sum_i q(s_i/n) \frac{p_{i+1} - p_i}{H} \\ &\quad + \sum_i \left[\frac{1}{3(p_i/H)} - \frac{D}{2} q(s_i/n)^2 + o(1) \right] \frac{L_i}{n}. \end{aligned} \quad (21.8)$$

Discrete summation by parts treats the first sum: the endpoint terms converge, while its remaining sum converges to $-\int f dq$. Replacing the random boundary path by f incurs at most its uniform error times the bounded total variation of q , plus endpoint errors. No total-variation estimate on the random path is required. Its limit is $\int q f' dt$. The second sum converges by the tube estimate, the vanishing time mesh, and the $o(n)$ terminal overshoot to $\int [1/(3f) - Dq^2/2] dt$. Consequently

$$\mathcal{A}_n/S \longrightarrow \int_a^b [f'^2/(2D) + 1/(3f)] dt$$

in probability. Restrict to a high-probability event with a prescribed small action tolerance and integrate (21.7) over it. The logarithm of its probability is $o(S)$, proving (21.3). All constants used above are uniform for the families stated in the theorem. $\square$

This theorem ends at its actual random boundary time and height. It does not assert an exact-time bridge or a zero endpoint. Those requirements are supplied next by a completely discrete construction.

22 Uniform entrance and exact integer time repair

Lemma 22.1. *There are constants C, p_* such that for every integer $p \geq p_*$ there is a legal normalized path family from the root to $(p, 0)$, of a deterministic integer length $E(p)$, with*

$$E(p) \leq Cp^{3/2}/\sqrt{\log p}, \quad \text{mass} \geq e^{-C\sqrt{p \log p}}. \quad (22.1)$$

There are constants C_0, C_1, p_ such that for every $p \geq p_*$ and every integer $N \geq C_0 p^{3/2}/\sqrt{\log p}$, there is a legal path family from $(p, 0)$ to $(2, 0)$ of length exactly N and mass*

$$\text{mass} \geq \exp\{-C_1[\sqrt{p \log p} + N \log p/p]\}. \quad (22.2)$$

Both constructions retain all elementary multiplicity and internal bulk-path realizations of each selected macrocycle.

Proof. Fix a sufficiently large integer k_0 and define

$$p_k = \lfloor k^2 \log(k+1) \rfloor, \quad b_k = \lfloor p_k/4 \rfloor, \quad \ell_k = \lfloor \alpha b_k + 1/2 \rfloor, \quad d_k = p_{k+1} - p_k. \quad (22.3)$$

The mean-value theorem and floor errors bounded by one give

$$p_k \asymp b_k \asymp k^2 \log k, \quad d_k = O(k \log k) > 0, \quad d_k^2/b_k = O(\log k).$$

The upward stage $(b_k, \ell_k, \ell_k + d_k)$ takes p_k to p_{k+1} ; the downward stage $(b_k, \ell_k, \ell_k - d_k)$ takes p_{k+1} to p_k . For all $k \geq k_0$ both are supported and legal: $\ell_k/p_k \rightarrow \alpha/4 < 1$, while $\ell_k - b_k$ is a positive fixed fraction of b_k and dominates d_k . In class 247, ℓ_k lies strictly between b_k and $2b_k + 1$. Lemma 4.1 gives each stage mass at least $\exp(-C \log k)$ and length $O(p_k)$. Thus ascent or descent between p_{k_0} and p_K has

$$\text{length } O(K^3 \log K), \quad \text{mass at least } e^{-CK \log K}. \quad (22.4)$$

Let K be largest with $p_K \leq p$, and put $g = p - p_K$. Then $0 \leq g < d_K = O(K \log K)$ and $p \asymp p_K$. A bridge from p_K to p uses $(b_K, \ell_K, \ell_K + g)$; the reverse bridge uses $(b_K, \ell_K, \ell_K - g)$. The same

strict support margins prove legality for all large K , and the deviations from αb_K are at most $g + 1$. Lemma 4.1 gives mass $e^{-C \log K}$ and duration $O(p_K)$. This includes $g = 0$, when the bridge is a return cycle. Fixed ordinary access from the root to p_{k_0} , followed by ascent and the upward bridge, proves (22.1), since

$$K \log K \asymp \sqrt{p \log p}, \quad K^3 \log K \asymp p^{3/2} / \sqrt{\log p}. \quad (22.5)$$

For repair, take the downward bridge to p_K , later descend to the base, and finally use a fixed sequence of $b = 0$ unit-drop boundary edges to $(2, 0)$. Such edges have positive weight in both classes and are legal from every height at least three. Let $F(p)$ be the length of these pieces without top returns. Then

$$F(p) \leq CK^3 \log K, \quad \text{their mass} \geq e^{-CK \log K}.$$

At p_K , before descending, allow the return cycles $(b_K, \ell_K + s, \ell_K + s)$ for integers $|s| \leq R$, where

$$L = \ell_K + 1, \quad R = \lfloor K \log(K + 1) \rfloor.$$

Their lengths are every integer in $[L - R, L + R]$. They are supported and legal for large K , since $R/b_K = O(1/K)$, and each has mass at least $e^{-C \log K}$ by Lemma 4.1.

Every integer

$$U \geq (L^2 - R^2)/(2R) = O(K^3 \log K) \quad (22.6)$$

is a sum of $m = \lceil U/(L + R) \rceil = O(U/p_K + 1)$ such lengths. Indeed $m(L - R) \leq U \leq m(L + R)$ follows from (22.6). Starting with m copies of $L - R$, distribute the integer excess into increments from zero to $2R$. This covers every residue; no divisibility condition remains. Taking C_0 sufficiently large makes $U = N - F(p)$ satisfy (22.6), uniformly in p . Fill precisely U at p_K . The total time is exactly N and its logarithmic cost is at most

$$CK \log K + C(U/p_K + 1) \log K \leq C_1[\sqrt{p \log p} + N \log p/p],$$

which proves (22.2).

For example, distribute the excess greedily to make the selected macrocycle pattern a deterministic function of (p, N) . Entrance likewise has a fixed pattern and exact length once p is chosen. A macrocycle weight Q still counts every elementary edge multiplicity and each of the $\binom{w-1}{b-1}$ bulk realizations. Choosing a pattern does not choose only one such realization. $\square$

[write] **5 October 2026.** This lemma extends Part I's staircase (Section 5, Lemma 5.1), which enters and leaves the fixed plateau heights p_K , to an arbitrary integer height p , by one bridge cycle, and to an endpoint $(2, 0)$ reached at any sufficiently large exact length. Part III uses it as Proposition 30.1(1)–(2).

23 The sharp coefficient lower bound

We now show

$$\limsup_{n \rightarrow \infty} \frac{n \log \mu - \log a_j(n)}{S} \leq \sigma_D. \quad (23.1)$$

Take the minimizer f_* in (19.2). Fix a small height $\varepsilon \in (0, M)$ and let $t_\varepsilon \in (0, 1/2)$ satisfy $f_*(t_\varepsilon) = \varepsilon$. Then $t_\varepsilon = O(\varepsilon^{3/2})$. Choose a large constant A , independent of ε, n , as specified below, and put

$$\delta = A\varepsilon^{3/2}, \quad b = 1 - \delta, \quad P_n = \lfloor \varepsilon H \rfloor.$$

Use Lemma 22.1 to enter P_n at deterministic time E_n , and write $a_n^* = E_n/n$ (not to be confused with the counting sequence). For fixed ε and large n ,

$$a_n^* \leq C\varepsilon^{3/2}, \quad \text{entrance mass} \geq e^{-C\sqrt{\varepsilon}S}. \quad (23.2)$$

Increase A to ensure $a_n^* \leq \delta/2$, and take ε small enough that $\delta < 1/4$. On $[a_n^*, b]$ define the reparameterized central path

$$f_{n,\varepsilon}(t) = f_*(t_\varepsilon + c_{n,\varepsilon}(t - a_n^*)), \quad c_{n,\varepsilon} = \frac{1 - 2t_\varepsilon}{b - a_n^*}. \quad (23.3)$$

Both endpoint heights equal ε . For each fixed ε , these functions form a bounded C^2 family on a common neighborhood with a positive strip margin. Theorem 21.1 applies, uniformly, to paths starting from $P_n/H = \varepsilon + o(1)$ at time E_n and stopping at their first boundary time $s_N \geq bn$. With probability tending to one,

$$p_N/H = \varepsilon + o(1), \quad s_N/n = b + o(1).$$

On a high-probability subevent, for large n ,

$$\frac{1}{2}\varepsilon H \leq p_N \leq 2\varepsilon H, \quad \frac{1}{2}\delta n \leq n - s_N \leq \delta n. \quad (23.4)$$

The upper time bound follows from $s_N \geq bn$; the lower from the $o(n)$ overshoot.

Increase A once more so that every height in (23.4) satisfies

$$\delta n/2 \geq C_0 p_N^{3/2} / \sqrt{\log p_N}$$

for sufficiently large n . This is possible because $H^{3/2}/\sqrt{\log H} \asymp n$, with constants uniform when a fixed ε is followed by $n \rightarrow \infty$. The repair of Lemma 22.1 then has exactly the residual time $n - s_N$, ends at $(2, 0)$, and has logarithmic cost at most

$$C[\sqrt{\varepsilon}S + (\delta/\varepsilon)S] \leq C_A \sqrt{\varepsilon}S. \quad (23.5)$$

The constant A is fixed before ε tends to zero.

For the central action, set

$$K_\varepsilon = \int_{t_\varepsilon}^{1-t_\varepsilon} \frac{f_*'^2}{2D} dt, \quad V_\varepsilon = \int_{t_\varepsilon}^{1-t_\varepsilon} \frac{1}{3f_*} dt.$$

The exact reparameterization formula is

$$I_D(f_{n,\varepsilon}; [a_n^*, b]) = c_{n,\varepsilon} K_\varepsilon + c_{n,\varepsilon}^{-1} V_\varepsilon. \quad (23.6)$$

As $\varepsilon \downarrow 0$, uniformly for sufficiently large n at fixed ε , $c_{n,\varepsilon} \rightarrow 1$ and $K_\varepsilon + V_\varepsilon \rightarrow \sigma_D$. Endpoint integrability was proved in Theorem 19.1. Thus (23.6) is at most $\sigma_D + o_\varepsilon(1)$ uniformly.

We check multiplicities before multiplying the three masses. The entrance macrocycle pattern and its length E_n are fixed, while all elementary realizations counted by its weights are retained. From any completed decorated path, the entrance cut is recovered at the deterministic original time E_n . The interior endpoint is the first subsequent boundary visit at or after bn . Canonical boundary-cycle decomposition makes this rule unambiguous. Thus distinct interior prefixes, entrance realizations, or elementary multiplicity choices remain distinct full paths. The repair pattern is deterministic

in $(p_N, n - s_N)$ and still retains every internal realization. Shared repair suffixes never identify different prefixes. The completed-history injectivity proved earlier then ensures that these are distinct enumerated objects, rather than only labels.

Multiplying (23.2), the mass in Theorem 21.1 with (23.6), and the worst repair mass (23.5) gives total normalized mass at exact original time n and endpoint $(2, 0)$ at least

$$\exp\{-S[\sigma_D + o_\varepsilon(1) + C_A\sqrt{\varepsilon} + o_n(1)]\}.$$

This endpoint is valid for both classes. Its h -weight is the fixed x_j^2 , so (3.2) yields

$$a_j(n) \geq \mu_j^n x_j^{-2} \exp\{-S[\sigma_D + o_\varepsilon(1) + C_A\sqrt{\varepsilon} + o_n(1)]\}.$$

First let $n \rightarrow \infty$ at fixed ε , then $\varepsilon \downarrow 0$. This proves (23.1).

24 Uniform subcritical rows and a global spatial tail

For the upper coefficient bound, write

$$k = S/n = \log n/H, \quad \sqrt{k} = S/H = \theta_n.$$

The local-root theorem is not needed in the subcritical argument below; the exact transforms and concentration estimates suffice.

Lemma 24.1. *Fix bounds on real q, h , and fix $R, e > 0$. For integer $0 \leq p \leq RH$, write $x = p/H$ when $p > 0$. If*

$$h + Dq^2/2 \leq 1/(3x) - e, \tag{24.1}$$

then uniformly over these choices,

$$T_p(\sqrt{k}q, kh) \leq r + o(1). \tag{24.2}$$

At $p = 0$ the conclusion holds without (24.1). The fixed bounds may depend on R and on a chosen smooth function, but never on n .

Proof. Put $\lambda = \sqrt{k}q$, $\eta = kh$, and $\beta = \log R(\lambda, \eta)$. Analyticity gives

$$\beta = \alpha k(h + Dq^2/2) + O(k^{3/2}), \quad \beta \leq Ck. \tag{24.3}$$

Also $A = A_0 + o(1)$ and $ae^{\lambda+\eta} = a + o(1)$ uniformly. If $\beta \leq 0$, the sum in (17.7) is at most two. Otherwise use, for $m \geq 0$,

$$\sum_{b=0}^m b d_b \leq C\sqrt{m+1}, \quad \sum_{b=0}^m d_b(e^{\beta b} - 1) \leq C\beta\sqrt{m+1}e^{\beta m}. \tag{24.4}$$

Choose a sufficiently small fixed $l > 0$. For $p \leq lH$, discard F and use $b \leq p$. The excess over two is at most

$$Ck\sqrt{H}e^{CklH} = O(n^{-1/3+Cl}(\log n)^{5/6}) = o(1)$$

if $Cl < 1/3$. This case needs no condition involving $1/x$. At $p = 0$ there is only the ordinary edge.

Now let $lH \leq p \leq RH$ and set $z_p = \sqrt{p} \log p$. The mean-cutoff shift $O(p|\lambda| + 1) = o(z_p)$ uniformly. By (17.12)–(17.13), for $b \geq p/\alpha + z_p$ the CDF is at most $Ce^{-c(\log p)^2}$. Thus the legal Catalan sum is

bounded by the untruncated sum through $m = \lceil p/\alpha + z_p \rceil$ plus $o(1)$. This tail error is $o(1)$ because the full sum through p is at most $2e^{Ckp} = n^{O(1)}$. An endpoint $m > p - 1$, if it occurs, only enlarges the upper bound. Using (24.1)–(24.3), $kp = x \log n$, $pk^{3/2} = o(1)$, and $kz_p = o(1)$, we obtain

$$\beta m \leq (1/3 - ex) \log n + o(1) \leq (1/3 - el) \log n + o(1).$$

Equation (24.4) now bounds the excess by

$$Ck\sqrt{H} n^{1/3-el} e^{o(1)} = O((\log n)^{5/6} n^{-el}) = o(1).$$

Hence the legal sum is at most $2 + o(1)$ throughout the strip. This proves (24.2). $\square$

The coefficient $1/3$ in this lemma uses both the true cutoff p/α and the factor $H^{-1/2}$ in the Catalan tail.

Lemma 24.2. *There are fixed $\kappa > 0, C < \infty$ such that the total q -mass of all first-hit boundary prefixes reaching height at least P is at most*

$$C \exp\{-\kappa \sqrt{(P+2) \log(P+2)}\}. \quad (24.5)$$

No restriction on their original duration is imposed. The same bound holds for the normalized coefficient contribution of length- n paths whose boundary heights reach P .

Proof. For real $p \geq 0$, put

$$G(p) = \sqrt{(p+2) \log(p+2)}, \quad V(p) = e^{\kappa(G(p) - G(0))}.$$

Direct differentiation gives

$$G'(p) = \frac{\log(p+2) + 1}{2\sqrt{(p+2) \log(p+2)}}, \quad G''(p) = -\frac{(\log(p+2))^2 + 1}{4(p+2)^{3/2} (\log(p+2))^{3/2}} < 0.$$

Thus G is increasing and concave. With $\lambda_p = \kappa G'(p)$,

$$0 \leq \lambda_p \leq C\kappa, \quad \lambda_p^2 \leq C\kappa^2 \frac{\log(p+2)}{p+2}, \quad V(p+\Delta)/V(p) \leq e^{\lambda_p \Delta}$$

for actual legal endpoints. Extend the drop sum in this exponential majorant, retaining $b \leq p - 1$. For the legal renewal kernel K ,

$$\frac{KV(p)}{V(p)} \leq ae^{\lambda_p} + A(\lambda_p, 0) \sum_{b=0}^{p-1} d_b \exp\left\{C\kappa^2 b \frac{\log(p+2)}{p+2}\right\}. \quad (24.6)$$

Here $\log R(\lambda, 0) \leq C\lambda^2$ near zero. For every $m \geq 0$ and $0 \leq \gamma \leq 1/4$,

$$\sum_{b=0}^m d_b e^{\gamma b \log(m+2)/(m+2)} \leq 2 + C\gamma. \quad (24.7)$$

Indeed (24.4) bounds the excess by $C\gamma \log(m+2)(m+2)^{-1/2+\gamma}$, uniformly $O(\gamma)$. In (24.6) increase the cutoff to $m = p$ and choose $\gamma = C\kappa^2$ small enough. Analyticity of A then gives $KV/V \leq r + O(\kappa)$

uniformly. At $p = 0$ only the ordinary edge occurs and satisfies the same bound. Fix sufficiently small κ so that for some $\rho < 1$,

$$KV \leq \rho V \leq V. \quad (24.8)$$

After adjoining a cemetery state with value zero, this is a nonnegative supermartingale. Sum over its first-hit stopping line, or first use bounded optional stopping and pass by monotone convergence. Because V is increasing, overshoots only strengthen the bound. The total first-hit prefix mass is at most $V(0)/V(P)$, proving (24.5).

For a first-hit prefix of original length $s \leq n$, original-time substochasticity makes the total q -mass of all suffixes of exactly $n - s$ steps at most one. The true terminal multiplier x_j^{-p} is at most one too. First-hit prefixes partition the relevant paths. Summing their bounds proves the coefficient assertion. Prefixes whose first hit is after n may be included harmlessly in the upper bound. $\square$

In particular, for each fixed $R > 0$, the contribution of paths whose boundary heights exceed RH is at most

$$\exp\{-(\kappa\sqrt{2R/3} + o(1))S\}, \quad (24.9)$$

since $\sqrt{RH \log(RH)}/S \rightarrow \sqrt{2R/3}$.

[write] **5 October 2026.** The barrier of Lemma 24.2 is Part I's (7.2), and its first-hit bound is (7.5); the new clause is the coefficient contribution of length- n paths. Part III uses it as Proposition 30.1(3).

25 A positive space time supersolution

Fix $u, v \in C^2([0, 1])$ with $v(0) = 0$. Suppose that for some $e > 0$,

$$u'(\tau)x + v'(\tau) + Du(\tau)^2/2 \leq 1/(3x) - e \quad (0 \leq \tau \leq 1, x > 0). \quad (25.1)$$

Fix $R > 0$ and define on integer original times and boundary heights

$$\Phi_n(t, p) = S[u(t/n)p/H + v(t/n)], \quad 0 \leq t \leq n, \quad 0 \leq p \leq RH. \quad (25.2)$$

Keep only legal complete cycles with $t + L \leq n$ and $0 \leq p + \Delta \leq RH$, and multiply their weights by

$$\exp\{\Phi_n(t + L, p + \Delta) - \Phi_n(t, p)\}. \quad (25.3)$$

We claim that every row of this finite space-time kernel is at most a fixed $\rho < 1$ for sufficiently large n .

Put $\delta_n = n^{-1/12}$. Call a cycle good if $L \leq \delta_n n$ and $|\Delta| \leq \delta_n H$. For admitted good cycles Taylor's theorem gives

$$\Phi_n(t + L, p + \Delta) - \Phi_n(t, p) \leq \lambda_t \Delta + \eta_{t,p} L + C_R \delta_n (S/n) L, \quad (25.4)$$

where

$$\lambda_t = (S/H)u(t/n), \quad \eta_{t,p} = (S/n)[u'(t/n)p/H + v'(t/n)].$$

For clarity, write $\tau = t/n$, $z = p/H$, $l = L/n$, $d = \Delta/H$. After division by S , the exact remainder is

$$\begin{aligned} &[u(\tau + l) - u(\tau)]d + z[u(\tau + l) - u(\tau) - u'(\tau)l] \\ &+ [v(\tau + l) - v(\tau) - v'(\tau)l]. \end{aligned}$$

Its absolute value is at most $Cl|d| + C_R l^2$, proving (25.4). In particular the mixed space–time term has not been dropped. Every Taylor argument lies in $[0, 1]$ because crossing cycles were excluded. By (25.1), the effective time tilt on the right of (25.4) satisfies Lemma 24.1 with $q = u(t/n)$ and slack at least $e/2$ eventually. The required bounds on q, h are fixed on $p/H \leq R$. Summing good cycles is therefore at most $r + o(1)$, uniformly in t, p .

For bad cycles, $|\Phi_n| \leq C_R S$ everywhere in the confined region. Discard their legal drop restriction but retain $b \leq p - 1 \leq RH$. Equations (17.10)–(17.11), followed by $\sum d_b \leq 2$, bound their transformed row mass by

$$C e^{C_R S} \left\{ e^{-c \min(\delta_n^2 H^2 / (RH+1), \delta_n H)} + e^{-c \delta_n n + C(RH+1)} \right\}. \quad (25.5)$$

The ordinary edge is good for large n . The bound in (25.5) is $\exp\{-\omega(S)\}$ because

$$\delta_n^2 H / S \rightarrow \infty, \quad \delta_n n / H \rightarrow \infty, \quad \delta_n n / S \rightarrow \infty.$$

A union bound on the two bad events suffices; independence between L and Δ is neither asserted nor needed. Combining good and bad cycles proves the claimed row bound, for example eventually with $\rho = (1 + r)/2$. Rows at $t = n$ are zero.

25.1 Extraction at exact original time

Starting at $(0, 0)$ in space–time, total transformed mass of all finite paths is at most

$$\sum_{m \geq 0} \rho^m = (1 - \rho)^{-1}. \quad (25.6)$$

Every original length- n path ending at $c = 0$ and with all boundary heights at most RH occurs at exactly $t = n$ in this kernel. Its canonical complete cycles retain every original integer duration. A cycle with $t + L > n$ cannot be present in such a boundary-terminal path, so deleting it is exact, not a clock approximation. The factors (25.3) telescope and $\Phi_n(0, 0) = 0$. The genuine coefficient terminal multiplier gives

$$\begin{aligned} e^{-\Phi_n(n,p)} x_j^{-p} &= e^{-Sv(1)} \exp\{-p[\log x_j + (S/H)u(1)]\} \\ &\leq e^{-Sv(1)} \end{aligned} \quad (25.7)$$

for all $p \geq 0$ and sufficiently large n , since $u(1)$ is fixed and $S/H \rightarrow 0$. For class 247, keeping its actual terminal set or enlarging it to all boundary heights gives the same upper bound. Hence the confined contribution to $a_j(n)/\mu_j^n$ is at most

$$(1 - \rho)^{-1} e^{-Sv(1)}. \quad (25.8)$$

All low heights, including the root, were covered by the subcritical lemma; no unproved entrance approximation is needed for this direction.

25.2 Smooth dual functions approaching the sharp constant

For the minimizer of (19.2), equations (19.4) imply

$$\int_0^1 \left[\frac{2}{3f_*} - \frac{f_*'^2}{2D} \right] dt = 1/M = \sigma_D. \quad (25.9)$$

Fix $0 < \zeta < 1/2$, set $r_\zeta = 1 - 2\zeta$, and put

$$u_\zeta(t) = f'_*(\zeta + r_\zeta t)/D.$$

This is smooth with bounded derivatives on $[0, 1]$, and $u'_\zeta(t) = -r_\zeta/[3f'_*(\zeta + r_\zeta t)^2] < 0$. For a further fixed $e > 0$, define $v(0) = 0$ and

$$v'(t) = 2\sqrt{-u'_\zeta(t)/3} - Du_\zeta(t)^2/2 - e. \quad (25.10)$$

The negative derivative is bounded away from zero on this compact interval, so v has all the regularity used above. For all $x > 0$,

$$u'_\zeta(t)x + 2\sqrt{-u'_\zeta(t)/3} \leq 1/(3x),$$

by expanding $(\sqrt{(-u'_\zeta)x} - \sqrt{1/(3x)})^2 \geq 0$. Thus (25.1) holds with slack e . Moreover

$$v(1) = \frac{1}{r_\zeta} \int_\zeta^{1-\zeta} \left[\frac{2\sqrt{r_\zeta}}{3f'_*(s)} - \frac{f'_*(s)^2}{2D} \right] ds - e \longrightarrow \sigma_D - e \quad (25.11)$$

as $\zeta \downarrow 0$, by endpoint integrability and (25.9). Choosing first small ζ and then small e therefore produces fixed, bounded smooth functions with $v(1)$ exceeding any prescribed number below σ_D . We never insert the singular limiting derivative itself in a kernel.

25.3 Completion of the upper coefficient bound

Given $\varepsilon > 0$, fix such u, v with $v(1) > \sigma_D - \varepsilon/2$. Then choose R so large that $\kappa\sqrt{2R/3} > \sigma_D + 1$. These choices are fixed before n tends to infinity. Split paths according to whether their boundary heights stay at most RH . Equations (24.9) and (25.8) give

$$\frac{a_j(n)}{\mu_j^n} \leq (1 - \rho)^{-1} e^{-Sv(1)} + \exp\{-(\kappa\sqrt{2R/3} + o(1))S\}.$$

The fixed prefactor disappears on the S scale. Thus eventually $a_j(n) \leq \mu_j^n e^{-(\sigma_D - \varepsilon)S}$, proving

$$\liminf_{n \rightarrow \infty} \frac{n \log \mu - \log a_j(n)}{S} \geq \sigma_D. \quad (25.12)$$

Combining (23.1) and (25.12) proves Theorem 13.1. Original-time substochasticity already gave the all- n bound $a_j(n) \leq \mu_j^n$, and $S_n = o(n)$ gives the exponential growth constants.

26 Precise inverse thresholds and derivative growth

Write $\lambda = \log \mu > 0$ and $\psi(t) = t^{1/3}(\log t)^{2/3}$. In this section $\sigma = \sigma_D$ and $a(n)$ denotes either sequence.

Corollary 26.1. *For real $T \rightarrow \infty$, let $N(T) = \min\{n : a(n) \geq T\}$ and $s = (\log T)/\lambda$. Then*

$$N(T) = s + \frac{\sigma}{\lambda} \psi(s) + o(\psi(s)). \quad (26.1)$$

Equivalently the correction coefficient multiplying $(\log T)^{1/3}(\log \log T)^{2/3}$ is $\sigma/\lambda^{4/3}$. The correction is positive.

Proof. Fix sufficiently small $\varepsilon > 0$ and constants $0 < d < (\sigma - \varepsilon)/\lambda$ and $D_+ > (\sigma + \varepsilon)/\lambda$. The global upper bound $a(n) \leq \mu^n$ excludes every integer $n < s$. For $n \geq s$ and sufficiently large T , the sharp coefficient upper bound and the increasing function ψ give

$$\log a(n) \leq \lambda n - (\sigma - \varepsilon)\psi(n) < \lambda s \quad \text{if } n < s + d\psi(s).$$

At $n = \lceil s + D_+\psi(s) \rceil$, regular variation gives $\psi(n)/\psi(s) \rightarrow 1$, and the coefficient lower bound implies $\log a(n) \geq \lambda s$ eventually. This proves the corresponding two bounds on $N(T)$ without coefficient monotonicity or a ratio limit. Let $\varepsilon \downarrow 0$ and let d, D_+ approach σ/λ . Integer ceilings contribute only $O(1) = o(\psi(s))$. Finally $\psi(s) \sim \lambda^{-1/3}(\log T)^{1/3}(\log \log T)^{2/3}$. $\square$

[write] **5 October 2026.** This refines Part I, Corollary 9.1, and is refined by Part III, Corollary 36.1, which adds $\frac{7\sigma}{3\lambda}\psi(s) \log \log s / \log s$ and replaces $o(\psi(s))$ by $O(\psi(s)/\log s)$.

Let

$$b_n = a(n)\mu^{-n}, \quad F(z) = \sum_{n \geq 0} a(n)z^n, \quad \rho = 1/\mu, \quad f(t) = F(\rho e^{it}).$$

The radius of convergence is ρ by Theorem 13.1. For every $k \geq 0$, $\sum n^k b_n < \infty$. Termwise differentiation therefore gives continuous boundary derivatives of all orders, uniformly in t . Since all $b_n \geq 0$, their norm is attained at $t = 0$:

$$\|f^{(k)}\|_\infty = M_k := \sum_{n \geq 0} n^k b_n \quad (k \geq 1). \quad (26.2)$$

Radial derivatives at ρ are also finite, with $F^{(k)}(\rho) = \mu^k \sum_{n \geq k} \binom{n}{k} b_n$.

Corollary 26.2. *As integer $k \rightarrow \infty$,*

$$\log M_k = 3k \log k - 2k \log \log k + [\log(2D/\pi^2) - 3]k + o(k), \quad (26.3)$$

$$\log F^{(k)}(\rho) = 3k \log k - 2k \log \log k + [\log \mu + \log(2D/\pi^2) - 3]k + o(k). \quad (26.4)$$

These are derivative-growth statements, not further terms in the coefficient asymptotics.

Proof. For fixed $c > 0$, consider the continuous objective $k \log x - c\psi(x)$ for $x \geq 2$. Its maximum occurs at $x \rightarrow \infty$, and the derivative equation is

$$c\psi(x)(1 + 2/\log x) = 3k. \quad (26.5)$$

The left side is increasing eventually. Taking logarithms in (26.5) first gives $\log x \sim 3 \log k$, and then

$$x = (3/c^3 + o(1)) \frac{k^3}{(\log k)^2}, \quad c\psi(x) = 3k + o(k).$$

Substitution yields

$$\log \sup_{n \geq 2} n^k e^{-c\psi(n)} = 3k \log k - 2k \log \log k + [\log(3/c^3) - 3]k + o(k). \quad (26.6)$$

The same expression holds for integer n : rounding the continuous maximizer changes its objective by $o(k)$.

For each sufficiently small fixed $\varepsilon > 0$, eventually

$$e^{-(\sigma+\varepsilon)\psi(n)} \leq b_n \leq e^{-(\sigma-\varepsilon)\psi(n)}.$$

A single saddle integer with $c = \sigma + \varepsilon$ supplies the lower bound on M_k . For the upper bound reserve a summable factor $e^{-\varepsilon\psi(n)}$, obtaining a constant times the supremum in (26.6) with $c = \sigma - 2\varepsilon$. The finitely many earlier coefficients contribute $e^{O(k)}$. Their addition changes the logarithm of the rapidly growing saddle bound by $o(1)$, so in particular it does not alter its linear term. Let $\varepsilon \downarrow 0$. This proves

$$\log M_k = 3k \log k - 2k \log \log k + [\log(3/\sigma^3) - 3]k + o(k).$$

For the radial upper bound, $(n)_k \leq n^k$. For the lower bound use the same saddle integer $n \asymp k^3/(\log k)^2$, for which

$$\log \frac{(n)_k}{n^k} = \sum_{j=0}^{k-1} \log(1 - j/n) = O(k^2/n) = o(k).$$

Hence the moment linear term is unchanged, while the external factor μ^k adds $k \log \mu$. Finally $3/\sigma_D^3 = 2D/\pi^2$. No differentiation of the coefficient's $o(S_n)$ remainder is used. $\square$

Under the convention $\|f^{(k)}\|_\infty \leq CA^k(k!)^s$, these estimates in particular show that the boundary function has sharp Gevrey order 3: the bound holds for $s = 3$ and fails for every $s < 3$. Indeed $\log(k!) = k \log k - k + O(\log k)$, so the leading coefficient 3 dominates every smaller choice of s and the negative $2k \log \log k$ permits $s = 3$.

[write] 5 October 2026. Corollary 26.2 identifies the $O(k)$ of Part I, Theorem 10.1; Part III, Corollary 37.1, adds the term $-k \log \log k / \log k$ with error $O(k/\log k)$. The sharp Gevrey order 3 was already proved in Part I, which also proves the genuine singularity at ρ and non-D-finiteness; this Part does not restate them.

27 What is established and what remains open

The sharp constant closes the leading logarithmic problem:

$$\frac{n \log \mu_j - \log a_j(n)}{n^{1/3}(\log n)^{2/3}} \longrightarrow \left(\frac{3\pi^2}{2D_j} \right)^{1/3}.$$

The proof retains the full legal cycle mixture, correlated random duration, exact original time, and true terminal extraction. The lower argument controls the clock and reward separately, and its repair works at every sufficiently large residual integer duration. The upper argument controls all small heights, long cycles, large jumps, and escaping boundary heights before exact coefficient extraction. The singular minimizing profile is approached only through fixed smooth or positive-strip approximants.

Several stronger conclusions are not supplied:

1. An asymptotic equivalent for $a_j(n)$, including its exact multiplicative constant, any polynomial factor, and all lower-order exponent terms, remains open here. A subleading logarithm can still produce an unbounded multiplicative error.

2. The local $\log \log p$ term does not by itself give a global next-order coefficient term. Accumulating it through a varying kernel and both endpoint layers requires additional estimates.

[write] 5 October 2026. Answered by Part III, Theorem 29.1: the global next term is $\frac{7\sigma_D}{3} S_n \log \log n / \log n$, obtained by accumulating the local terms through the varying kernel with integrable height errors and treating both endpoint layers (Sections 32–34). Item 2 is kept as the scope statement of this Part.

3. A limit or expansion for $a_j(n+1)/a_j(n)$ is not proved by this report. The unspecified logarithmic remainder cannot simply be differenced to infer a ratio theorem.
4. No quantitative coefficient remainder, effective onset, usable finite- n calibration, all-orders expansion, or transseries is asserted. The theorem supplies neither a practical numerical crossover nor a new fit to finite data.
5. Smooth boundary values and their derivative growth do not identify a unique dominant singularity or an exterior analytic continuation region. No exterior Δ -domain expansion is claimed.

The inverse and derivative corollaries follow from the sharp leading logarithm using comparison and optimization. Their extra precision should not be mistaken for a stronger coefficient equivalent.

27.1 Further questions and research

[write] (5 October 2026.) Under Vladimir’s standing rule of 4 October 2026, the status of the five items above. Nothing in this Part was found to be wrong.

1. *An asymptotic equivalent* (item 1). Open; Section 39, question 2. Part III’s remainder $O(S_n / \log n)$ still allows an unbounded multiplicative error, as item 1 warns.
2. *The global next-order term* (item 2). Answered by Part III, Theorem 29.1; the following term, of order $S_n / \log n$, is open (question 1).
3. *A ratio limit or expansion* (item 3). Open (question 3).
4. *Quantitative remainder, onsets, calibration, all orders, transseries, crossover* (item 4). Part III gives the quantitative but ineffective remainder $O(S_n / \log n)$; the rest is open (questions 4 and 5).
5. *Dominant singularity and continuation* (item 5). Open (question 6). Part I, Theorem 10.1, shows at least that ρ is a genuine singularity and that the generating functions are not D-finite.

28 Finite verification and offline reproduction

The analytic estimates in this report prove the infinite statements. The accompanying exact finite checker is a supplementary regression companion. Its own README records its checked ranges, provenance, and limitations. Tests of finite identities, support, or integer-time constructions cannot certify uniform asymptotic estimates or replace any martingale, tail, variational, or limiting argument above.

The authoritative checker and negative-test entry points are `checks/verify.py` and `checks/negative_tests.py`. Their recorded reference outputs are `data/verification_results.json`

and `data/negative_results.json`. The source directory is a closed inventory: changed or missing bytes, additional files or directories, and symlinks are rejected. Runtime guards use explicit exceptions and remain active under optimized Python.

The complete offline replay command is

```
python3 -B reproduce.py --output ../report127-replay.json
```

The replay checks the outer inventory, compares exact checks in normal and optimized Python bitwise, exercises mathematical and inventory mutations, and builds the PDF twice in fresh external directories. It requires byte-identical PDFs, packages a deterministic ZIP, extracts a fresh copy, repeats the checks and double-clean rebuild, and requires byte-identical repacking. All work products stay outside the reader directory, whose original bytes are checked again at the end. The optional `--skip-pdf` switch records that reconstruction was omitted; it does not claim a complete PDF replay.

The TeX and compression versions are recorded in `build-environment.txt`; other versions may produce equivalent content with different bytes. No downloaded paper, development note, network service, or numerical fit is required for reproduction. The exact succession laws cited below are the mathematical input, not a software dependency. Integrity inventories are records relative to the supplied verifier, not signatures or authenticity proofs against simultaneous malicious replacement of the verifier and its inventory.

[write] **5 October 2026.** In this directory the companion's files carry the prefix `127-sharp-`: `checks/verify.py` and `checks/negative_tests.py` are `code/127-sharp-checks-verify.py` and `code/127-sharp-checks-negative_tests.py`, the recorded outputs are `data/127-sharp-verification_results.json` and `data/127-sharp-negative_results.json`, and the companion's README is `127-sharp-checks-README.md`. Report 127's `build-environment.txt`, `integrity.py` and `build.sh` are byte copies of Report 126's (the last also of a file of `a307316-leafless-multigraphs`) and are shipped once, under Report 126's prefix or not at all. The checkers need the delivered closed layout; the report's README gives the rerun route.

Part III

The next logarithmic correction for two inversion sequence classes

Bundle Report 129, archive A279551_A279556_Loglog_Correction_and_Refined_Inversion_Source.zip, delivered as `report129.tex` (17 pages), dated 2 October 2026, the base of the merge. Delivered title block “Report 129. The next logarithmic correction for two inversion sequence classes. A279556 and A279551. Full quantitative upgrade with a proved input companion”; no author line. Printed as delivered, with `[write]` notes. Delivered Section k is Section $k + 28$ here, and its statement $k.j$ is $(k + 28).j$. “Report 127” is Part II: its Section k is Section $k + 12$ here, except that its Sections 2–4 are Part I’s Sections 2–4. Remark 29.2 and Section 39 are added in the write.

Reading conventions of Part III	Elsewhere in the report
$N = \log n$, $J = \log \log n$; $N(T)$ in Section 36	Parts I, II: $N(T)$, N_k , cycle counts
$e = e_n = N^{-6}$ the strip height	Euler’s number in exponentials; Part II’s slack e
$P(n)$ a power of $\log n$; $P_n = \lfloor eH \rfloor$	Part II: $P_n = \lfloor \varepsilon H \rfloor$
$\kappa_p = \eta_p(0)$; κ the barrier constant (Proposition 30.1)	Parts I, II: κ the barrier constant
$a_n = E(P_n)/n$, $b_n = 1 - \delta$ (Section 34); $b_n = a_j(n)\mu^{-n}$ (Section 37)	Part II: a_n^* , $b = 1 - \delta$
$R(t) = \psi(t)/\log t$, $G(t)$ the main term; R also the strip bound	Part II: $G(p)$ the barrier function

Abstract of this Part (as delivered). For the two commitment-tree enumerations A279556 and A279551 we prove

$$n \log \mu - \log a(n) = \sigma_D n^{1/3} (\log n)^{2/3} \left(1 + \frac{7}{3} \frac{\log \log n}{\log n} + O\left(\frac{1}{\log n}\right) \right),$$

where $(\mu, D) = (9, 1)$ and $(8, 1/2)$, respectively, and $\sigma_D = (3\pi^2/(2D))^{1/3}$. The proof gives a quantitative local critical root, an anchored derivative estimate, a shrinking-strip random-clock construction, and an integrable dual penalty. Exact integer-time extraction and endpoint repair are preserved. We derive refined inverse thresholds and boundary-derivative growth by explicit comparison bounds, without differentiating an unknown remainder. The unchanged Report 127 is included as the proved reference for the exact combinatorial model, spatial barrier, and uniform discrete entrance and repair. The reader bundle therefore contains the proofs of the earlier inputs as well as all new estimates. This is a logarithmic expansion, not a coefficient equivalent, and it supplies no effective numerical onset.

29 The theorem and the reader bundle

An inversion sequence of length n is a sequence $e = (e_1, \dots, e_n)$ of integers with $0 \leq e_i < i$. Write $a_{247}(n)$ for the number with no $i < j < k$ satisfying $e_i \leq e_j$ and $e_i \geq e_k$, and $a_{759}(n)$ for the number with no such triple also satisfying $e_j \neq e_k$. Their forbidden pattern sets are (000, 010, 110, 120) and (010, 110, 120), respectively. In the usual indexing,

$$a_{247}(n) = \text{A279551}(n), \quad a_{759}(n) = \text{A279556}(n), \quad a_j(0) = 1.$$

All logarithms are natural. The diffusion parameter D and the critical exponential growth constant μ are

j	μ	D	$\sigma_D = (3\pi^2/(2D))^{1/3}$
759	9	1	$(3\pi^2/2)^{1/3}$
247	8	1/2	$(3\pi^2)^{1/3}$

Set, for sufficiently large n ,

$$\begin{aligned} N = \log n, \quad J = \log \log n, \quad H = n^{2/3} N^{1/3}, \quad S = n^{1/3} N^{2/3}, \\ \theta = H/n = S/H, \quad k_0 = S/n = N/H. \end{aligned} \quad (29.1)$$

Thus $H^2 = nS$, $HS = nN$, and $\theta^2 = k_0$. The symbol $P(n)$ below always denotes a fixed finite power of $\log n$, with its exponent allowed to increase between estimates. It never hides a positive power of n .

Theorem 29.1. *For either class, as n tends to infinity through all integers,*

$$n \log \mu - \log a_j(n) = \sigma_D S + \frac{7\sigma_D}{3} \frac{SJ}{N} + O\left(\frac{S}{N}\right). \quad (29.2)$$

Equivalently, there are constants C, n_0 such that for all $n \geq n_0$,

$$e^{-G(n)-CR(n)} \leq a_j(n) \mu^{-n} \leq e^{-G(n)+CR(n)}, \quad (29.3)$$

where

$$\psi(t) = t^{1/3}(\log t)^{2/3}, \quad R(t) = \frac{\psi(t)}{\log t}, \quad G(t) = \sigma_D \psi(t) + \frac{7\sigma_D}{3} \frac{\psi(t) \log \log t}{\log t}.$$

The constants are not asserted to be effective or numerically calibrated.

The coefficient $7/3$ comes from adding the $J/6$ contributed by $\frac{1}{2} \log H$ to the J contributed by $\log \log H$, and then evaluating the resulting $7J/(6N)$ potential correction along the singular minimizing curve. Retaining an integrable height error is necessary to distinguish this coefficient from the remainder.

[write] **5 October 2026.** Since $SJ/N = o(S)$, this theorem implies Part II's Theorem 13.1 and Part I's Theorem 1.1, and it answers item 2 of Part II's Section 27. The coefficient of S/N is not determined (Section 35); it is question 1 of Section 39.

29.1 How the earlier input is used

The bundle contains the unchanged PDF, TeX, checkers, and build source of Report 127 under `earlier-inputs/report127/`. That report proves the exact counting interpretation and the analytic inputs stated in Section 30; its proofs are part of this reader bundle. It is not necessary to download a previous development note. The present report gives every new quantitative estimate used to strengthen the leading logarithmic law.

Statements in the unchanged Report 127 describing the next logarithmic correction as open refer to the scope of that earlier report. They are superseded here by Theorem 29.1. Its earlier proof and files have not been silently revised. The underlying exact commitment construction is due to Britt and Beaton [1]; the arguments here make no global literature-priority claim.

[write] **5 October 2026.** In this report “Report 127” is Part II, printed in full, with its Sections 2–4 in Part I; the proofs of the inputs are therefore in this article. The directory `earlier-inputs/report127/` of the delivered package is not shipped: its 19 files are byte copies of Report 127’s own delivery, whose files are shipped with the prefix `127-sharp-` (except its PDF, TeX, delivery README and checksum manifests). Report 127 is printed with dated [write] notes, which add cross-references and do not change its text.

The finite checks are regression and algebra checks, not replacements for these proofs. Package and rendering checks have the narrower scopes recorded in the accompanying README and generated check results.

Remark 29.2 ([write] The same mechanism for A202061; 5 October 2026). The report `a202061-ascent-120-deficit`, in the same collection directory, treats a different sequence, the 120-avoiding ascent sequences (OEIS A202061), through a different combinatorial model (a positive height walk with binomial blocks, not commitment trees). Its results have the same shape as Parts I–III, with the scale $F = n^{1/3}L^{2/3}$, $L = \log n$, equal to S here:

- the order $\Theta(F)$ of the deficit, its Theorem `a61:fd:thm:main`, with the same exclusion of a pure stretched exponent $3/8$, there estimated numerically by Conway, Conway, Elvey Price and Guttmann (its Corollary `a61:fd:cor:consequences`), here by Britt and Beaton with Guttmann’s assistance (Remark 1.2);
- the sharp constant, its Theorem `a61:sh:thm:main`;
- the second term $\frac{7C_\star}{3}F \log L/L + O(F/L)$, its Theorem `a61:o2:thm:main`, the counterpart of Theorem 29.1. There, as here, the critical local root has the form $(\frac{1}{2} \log p + \log \log p + c)/p$, and the global potential is $\frac{\alpha}{3} + \frac{7\alpha}{6} \frac{\log L}{L} + O(1/L)$ in its normalization, against $A_n = \frac{1}{3} + \frac{7}{6} \frac{J}{N}$ in (32.1). In its upper bound the $\frac{7}{6}$ has the origin described after Theorem 29.1 here ($\frac{1}{6}$ from $\frac{1}{2} \log$ and 1 from $\log \log$ of the height); in its lower construction the 1 comes from the width of a usable length window.

Its constant $C_\star = (3\pi^2\alpha^2/(2v))^{1/3}$ is written with its own α , v and height normalization; here $\sigma_D = (3\pi^2\alpha/(2v))^{1/3}$ with $D = v/\alpha$ (Part II, (17.8)). The sequences and the constants are different, and no relation between them is asserted. Neither report cites the other: Parts I–III are an independent instance of the method, not re-proofs of anything in that report. That report goes further: its Part IV identifies the next constant, the term $C_\star\kappa F/L$ with $\kappa = -1.9401\dots$ (Theorem `a61:o3:thm:main`), and its Part V every fixed inverse-logarithmic order (Theorem `a61:ao:thm:main`). Those are the model for question 1 of Section 39.

30 Exact model and earlier proved inputs

30.1 Counting and normalization

The two auxiliary label trees start at $(0, 0)$ and have edges

$$\begin{aligned} (p, c) &\longrightarrow (p+1, c), \\ (p, c) &\longrightarrow (p+1, c-1) \quad (c > 0), \\ (p, 0) &\longrightarrow (p-\ell, b) \quad (0 \leq \ell \leq p-1). \end{aligned}$$

The last edge has multiplicity

$$M_{759}(\ell, b) = C_b \binom{\ell}{b}, \quad M_{247}(\ell, b) = C_b \binom{b+1}{\ell-b}, \quad C_b = \frac{1}{b+1} \binom{2b}{b}.$$

Out-of-range binomial coefficients are zero. The exact terminal sets are

$$\mathcal{T}_{759} = \{(p, 0) : p \geq 0\}, \quad \mathcal{T}_{247} = \{(p, 0) : 0 \leq p \leq 2\}.$$

Multiplicity-weighted paths ending in these sets count the respective sequences exactly. In particular the extra auxiliary nodes are not all valid terminal objects. The completed decorated-history injectivity proof is in Report 127, Section 2.

Use $(\mu, x_j, y_j) = (9, 3, 1/2)$ for 759 and $(8, 2, 1/3)$ for 247, and normalize an edge of multiplicity m by

$$q(s, t) = \frac{mh(t)}{\mu h(s)}, \quad h(p, c) = x_j^p y_j^c.$$

Every length- n path of multiplicity m has normalized mass $m\mu^{-n}h(\text{endpoint})$. Thus at a valid terminal its coefficient multiplier is x_j^{-p} . These elementary rows are substochastic, so the mass at every exact original integer time is at most one, and $a_j(n) \leq \mu^n$ for every n .

At a boundary $(p, 0)$, a completed commitment with $b \geq 1$ has drop $\ell \leq p - 1$, followed by $w \geq b$ bulk steps, with its last fulfillment at the final bulk step. Its total mass, duration, and displacement are

$$Q_j(b, \ell, w) = \mu^{-w-1} x_j^{w-\ell} M_j(\ell, b) \binom{w-1}{b-1}, \quad L = w + 1, \quad \Delta = w - \ell. \quad (30.1)$$

For $b = 0$ use $L = 1$, $\Delta = -\ell$, and mass $\mu^{-1} x_j^{-\ell} M_j(\ell, 0)$. The ordinary edge has $L = \Delta = 1$ and mass $a = x_j/\mu$. Every boundary path decomposes uniquely into these cycles, including the duration-one edges. A “cycle” here need not return to the same height.

30.2 Exact transforms and fixed batch laws

Put $d_b = C_b/4^b$. Summing unrestricted drops at a fixed b gives $A_j(\lambda, \eta) d_b R_j(\lambda, \eta)^b$, where

$$A_{759} = \frac{e^\eta}{9(1 - e^{-\lambda}/3)}, \quad R_{759} = \frac{4e^\eta}{9(1 - e^{-\lambda}/3)(1 - e^{\lambda+\eta}/3)}, \quad (30.2)$$

$$A_{247} = \frac{e^\eta(1 + e^{-\lambda}/2)}{8}, \quad R_{247} = \frac{e^\eta(1 + e^{-\lambda}/2)}{2(1 - e^{\lambda+\eta}/4)}. \quad (30.3)$$

These follow from geometric and binomial summations in Report 127, Section 5. They are analytic on a fixed neighborhood of $(0, 0)$, independently of b ; no unrestricted infinite positive-tilt sum over b is being asserted.

At a fixed b under the exponential tilt, the drop laws are

$$759 : \quad \ell = b + \text{NB}^{\text{fail}}(b+1, 1 - e^{-\lambda}/3), \quad (30.4)$$

$$247 : \quad \ell = b + \text{Bin}(b+1, e^{-\lambda}/(2 + e^{-\lambda})), \quad (30.5)$$

where the negative binomial counts failures before the stated number of successes. Independently, for $b \geq 1$, w is a negative-binomial trials variable with b successes and success probability $1 - e^{\lambda+\eta}/3$

or $1 - e^{\lambda+\eta}/4$, respectively. At $b = 0$ duration is one. The legal cutoff conditions only the drop, so fixed- b independence is preserved; Δ and L themselves are not independent.

Let $F_{b,\lambda}$ be the appropriate drop CDF. The exact legal row is

$$T_p(\lambda, \eta) = ae^{\lambda+\eta} + A_j(\lambda, \eta) \sum_{b=0}^{p-1} d_b R_j(\lambda, \eta)^b F_{b,\lambda}(p-1). \quad (30.6)$$

The sum is empty at $p = 0$. The constants are

j	a	A_0	α	D	$r = a + 2A_0$
759	1/3	1/6	3/2	1	2/3
247	1/4	3/16	4/3	1/2	5/8

In both cases $1 - r = 2A_0$ and

$$\log R(\lambda, \eta) = \alpha\eta + \frac{\alpha D}{2}\lambda^2 + O(|\lambda|^3 + |\lambda\eta| + \eta^2). \quad (30.7)$$

For $b \leq p$, the tilted drop mean is $\alpha b + O(p|\lambda| + 1)$. Uniformly for λ near zero,

$$\mathbb{P}(|\ell - \mathbb{E}\ell| \geq u) \leq C \exp\{-c \min(u^2/(b+1), u)\}. \quad (30.8)$$

This follows because the centered log moment generating functions of the geometric or Bernoulli summands are bounded by $C\vartheta^2$ on a fixed interval. At zero tilt, the fixed- b normalized laws also obey

$$\mathbb{P}(|\Delta| > z) \leq Ce^{-c \min(z^2/(b+1), z)}, \quad (30.9)$$

$$\mathbb{P}(L > u) \leq Ce^{-cu + C(b+1)}. \quad (30.10)$$

A Chernoff bound applied to (30.7) proves the first; a fixed positive duration tilt proves the second. Both include $b = 0$.

30.3 Uniform endpoint and spatial estimates

We state the discrete inputs with their uniformity explicitly. Their proofs, including integer-time filling, are in Report 127, Sections 4, 10, and 12.

Proposition 30.1 (Earlier proved inputs). *There are fixed positive constants with the following properties.*

1. *For every sufficiently large integer p there is a legal normalized path family from the root to $(p, 0)$, of deterministic integer length $E(p)$, satisfying*

$$E(p) \leq Cp^{3/2}/\sqrt{\log p}, \quad \text{mass} \geq e^{-C\sqrt{p \log p}}. \quad (30.11)$$

2. *For every sufficiently large integer p and every integer $m \geq C_0 p^{3/2}/\sqrt{\log p}$ there is a legal normalized family from $(p, 0)$ to $(2, 0)$ of length exactly m , with*

$$\text{mass} \geq \exp\{-C_1[\sqrt{p \log p} + m \log p/p]\}. \quad (30.12)$$

All elementary multiplicity and bulk-path realizations of its selected macrocycles are retained.

3. The total normalized mass of all first-hit boundary prefixes reaching height at least P is at most

$$C e^{-\kappa \sqrt{(P+2) \log(P+2)}}. \quad (30.13)$$

No restriction on their original duration is needed. The same estimate bounds the normalized coefficient contribution of length- n paths whose boundary heights reach P .

For precision about these references, the entrance and repair proof uses staircase heights $p_k = \lfloor k^2 \log(k+1) \rfloor$ and legal moderate-deviation cycles of mass at least $e^{-C \log k}$. Their accumulated time is $O(K^3 \log K)$ and logarithmic cost $O(K \log K)$. Return-cycle blocks and bounded terminal edges fill every remaining integer duration. Thus the constants do not depend on a fixed positive lower bound for p/H . The spatial proof uses the increasing concave function $\sqrt{(p+2) \log(p+2)}$ and a sufficiently small exponential multiple as a positive supermartingale. First-hit prefix extraction followed by original-time substochasticity gives the coefficient assertion. These proof summaries clarify the hypotheses; their full arguments remain in the supplied unchanged companion.

[write] **5 October 2026.** Where the cited proofs are in this report: Report 127's Section 2 (exact counting and the injectivity of completed decorated histories) and Section 3 (normalization) are Part I's Sections 2 and 3; its Section 4 (the moderate local bound used by the staircase) is Part I's Section 4; its Section 5 (transforms, tilted laws, concentration) is Section 17; its Section 10 is Section 22, where Lemma 22.1 proves items 1 and 2; its Section 12 is Section 24, where Lemma 24.2 proves item 3. Part I, Section 7, proves the first-hit part of item 3 by the same barrier.

31 A quantitative local critical root

For sufficiently large p and the tilt ranges below, write $\eta_p(\lambda)$ for the unique solution of $T_p(\lambda, \eta_p(\lambda)) = 1$ and $\kappa_p = \eta_p(0)$. The row is continuous and strictly increasing in η near the brackets constructed below, so those brackets prove existence and uniqueness of the relevant root.

Lemma 31.1. *For each fixed $B \geq 1$, uniformly for $|\lambda| \leq p^{-1/2}(\log p)^B$, as $p \rightarrow \infty$,*

$$\kappa_p = \frac{\frac{1}{2} \log p + \log \log p + c_* + O(\log \log p / \log p)}{p}, \quad c_* = \frac{1}{2} \log(\pi/\alpha), \quad (31.1)$$

$$\eta_p(\lambda) = \kappa_p - D\lambda^2/2 + O(p^{-3/2}(\log p)^{C_B}), \quad (31.2)$$

$$\eta'_p(\lambda) = -D\lambda + O(p^{-3/4}(\log p)^{C_B}). \quad (31.3)$$

For each fixed bounded real z , uniformly in this same range,

$$T_p(\lambda, \eta_p(\lambda) - z/p) = r + (1-r)e^{-z} + o(1). \quad (31.4)$$

The finite exponent C_B may be enlarged in each appearance.

Proof. First take $\lambda = 0$ and set

$$s = \frac{1}{2} \log p + \log \log p + c, \quad \beta = \log R(0, s/p), \quad m = p/\alpha + O(\sqrt{p}(\log p)^2),$$

with bounded c . Then $\beta = \alpha s/p + O(s^2/p^2)$ and $\beta m = s + O(p^{-1/2}(\log p)^3)$. By (30.8), the legal CDF is squeezed between hard cutoffs $p/\alpha \pm \sqrt{p}(\log p)^2$, with exponentially small tails. The endpoint Laplace sum has the quantitative estimate

$$\sum_{b=0}^m d_b e^{\beta b} = 2 + \frac{e^{\beta m}}{\sqrt{\pi} m^{3/2} \beta} \left[1 + O\left(\frac{1}{\beta m}\right) \right] + O(p^{-1/4}(\log p)^C). \quad (31.5)$$

Indeed $d_b \leq C(b+1)^{-3/2}$, so the excess from $b \leq m/2$ is at most $C\beta\sqrt{m}e^{\beta m/2}$, of the displayed polynomially small order. For $b > m/2$, put $b = m - h$ and use $d_b = (\sqrt{\pi}b^{3/2})^{-1}(1 + O(1/b))$. The geometric zeroth moment gives the displayed main term, while its first-moment correction divided by m is

$$\frac{C \sum_{h \geq 0} h e^{-\beta h}}{m \sum_{h \geq 0} e^{-\beta h}} = O((m\beta)^{-1}).$$

The geometric tail, Stirling error, unweighted missing tail $O(m^{-1/2})$, and integer rounding are smaller.

Near p/α , the individual weighted Catalan coefficients are $O(\log p/p)$. The CDF transition window therefore costs $O(p^{-1/2}(\log p)^C)$. Substitution in (30.6) yields

$$T_p(0, s/p) = r + 2A_0 \sqrt{\alpha/\pi} e^c [1 + O(\log \log p / \log p)]. \quad (31.6)$$

Since $1 - r = 2A_0$, monotonicity brackets the root at $c = c_* + O(\log \log p / \log p)$. A bounded subtraction of z from s similarly proves (31.4) at zero tilt.

For the anchored estimate put $\eta_c = \kappa_p - D\lambda^2/2$. Analyticity and (30.7) give

$$\begin{aligned} \log R(\lambda, \eta_c) - \log R(0, \kappa_p) &= O(p^{-3/2}(\log p)^C), \\ A(\lambda, \eta_c) - A(0, \kappa_p) &= O(p^{-1/2}(\log p)^C). \end{aligned} \quad (31.7)$$

For example, the unsimplified remainder is bounded by a constant times $|\lambda|^3 + |\lambda|\kappa_p + \kappa_p^2 + \lambda^4$. The cancellation of $D\lambda^2/2$ inside $\log R$ is retained even when η_c is negative.

Choose $h = \sqrt{p}(\log p)^{B+2}$ and $m_{\pm} = p/\alpha \pm h$. The actual mean shift $O(p|\lambda| + 1)$ is smaller than h . The two CDFs therefore differ from the hard cutoff only inside this window, apart from tails bounded by $e^{-c(\log p)^{2B+4}}$. The entire unrestricted coefficient sum through p is at most a fixed power of p : (31.1) and (31.7) give $\log R = O(\log p/p)$. Thus these tails remain negligible after multiplication by that sum. The truncated unrestricted sum through m_+ is $O(1)$ by (31.5), and each coefficient in the window is $O(\log p/p)$. The exponent comparison changes any coefficient with $b \leq p$ by relative $O(p^{-1/2}(\log p)^C)$. Consequently

$$|T_p(\lambda, \eta_c) - 1| \leq Cp^{-1/2}(\log p)^C. \quad (31.8)$$

In an η neighborhood of radius $p^{-3/2}(\log p)^{C+1}$, we have $\partial_{\eta} T_p \geq cp$. To see this, restrict b to

$$[p/\alpha - 2p/\log p, p/\alpha - p/\log p]. \quad (31.9)$$

There are order $p/\log p$ terms with individual legal mass at least $c \log p/p$. The margin $p/\log p$ dominates both the tilt mean shift and its fluctuations; changes of the exponent in the stated neighborhood are $o(1)$. Each of these cycles has $L \geq b+1 \geq cp$. Bracket (31.8) using this derivative lower bound to prove (31.2). The same comparison after a bounded shift $-z/p$ proves (31.4).

Finally, $\log T_p$ is convex and strictly increasing in η . Its zero sublevel set is the hypograph of the root, so η_p is concave; analyticity and $\partial_{\eta} T_p > 0$ also make it differentiable. A differentiable concave function has its derivative between its adjacent secant slopes. Apply these inequalities to (31.2) with increment $h_p = p^{-3/4}(\log p)^C$, taking C sufficiently large and first enlarging the tilt exponent to $B+1$. Both endpoints then lie in that enlarged range. The secant error is

$$O(h_p) + O(p^{-3/2}(\log p)^C/h_p),$$

which proves (31.3). The scalar κ_p cancels exactly. Concavity applied only to an unanchored $o(1/p)$ remainder would not supply the needed derivative rate. $\square$

[write] **5 October 2026.** This lemma is the quantitative and anchored form of Part II's Theorem 18.1, which has $o(1)$ in place of $O(\log \log p / \log p)$ in (31.1) and no rate for the tilt dependence; Lemma 31.2 below sharpens Lemmas 20.1 and 20.2 in the same way.

Lemma 31.2 (Genuine duration moments). *For the exact row-one probability law*

$$\mathbb{P}_{p,\lambda}(\text{cycle}) = Q_p(\text{cycle})e^{\lambda\Delta + \eta_p(\lambda)L}, \quad (31.10)$$

in the tilt range of Lemma 31.1,

$$cp \leq \mathbb{E}L \leq Cp, \quad \mathbb{E}L^2 \leq Cp^2, \quad \mathbb{E}\Delta^2 \leq Cp(\log p)^C, \quad (31.11)$$

$$\frac{\mathbb{E}\Delta}{\mathbb{E}L} = D\lambda + O(p^{-3/4}(\log p)^C). \quad (31.12)$$

Proof. At fixed b , the duration law remains negative binomial with success parameter in a fixed compact subset of $(0, 1)$. Its second moment is $O((b+1)^2)$. Sum against the actual legal b -masses, which total at most one, and use (31.9) for the lower mean. For displacement, below m_+ the unrestricted fixed- b second moment is

$$O((b+1) + (b+1)^2(\lambda^2 + \eta_p(\lambda)^2)) \leq Cp(\log p)^C.$$

Only the unrestricted sum through m_+ is used here, and it is $O(1)$. Above m_+ , retain legality: $\ell \leq p-1$, and w is independent of that event. The second-moment numerator is bounded by $Cp^2 F_{b,\lambda}(p-1)$, whose exponentially small CDF beats the polynomial unrestricted mass. The $b=0$ and ordinary terms have bounded moments. Implicit differentiation of the exact row gives $\eta'_p(\lambda) = -\mathbb{E}\Delta/\mathbb{E}L$; use (31.3). No individual duration is replaced by its conditional mean and no displacement-duration covariance is dropped. $\square$

32 The corrected potential and integrable height error

Define

$$A_n = \frac{1}{3} + \frac{7}{6} \frac{J}{N}, \quad e = e_n = N^{-6}. \quad (32.1)$$

On a shrinking strip $ce \leq x = p/H \leq R$, with fixed $c > 0$ and $R < \infty$, and for $|q| \leq P(n)$, Lemma 31.1 gives

$$\frac{n}{S} \eta_p(\theta q) = \frac{A_n}{x} - \frac{D}{2} q^2 + E_n(x, q), \quad (32.2)$$

$$|E_n(x, q)| \leq \frac{C(1 + |\log x|)}{Nx} + n^{-1/3} P(n). \quad (32.3)$$

Indeed $\lambda\sqrt{p} = \sqrt{Nx} q$, so some fixed B covers all such tilts. The scalar algebra is

$$\begin{aligned} \frac{1}{2} \log p &= \frac{1}{3} N + \frac{1}{6} J + \frac{1}{2} \log x, \\ \log \log p &= J + \log(2/3) + O((J + |\log x|)/N). \end{aligned}$$

Divide their sum by Nx . The constants c_* and $\log(2/3)$ enter the $1/N$ remainder only; the anchored error in (31.2) becomes $n^{-1/3} P(n)$ after multiplication by n/S on this strip.

Lemma 32.1 (Variational profile and height integrability). *For A in a compact neighborhood of $1/3$, the minimum of*

$$I_A(f) = \int_0^1 \left(\frac{f'^2}{2D} + \frac{A}{f} \right) dt, \quad f(0) = f(1) = 0,$$

over positive absolutely continuous paths of finite action is

$$\min I_A = \frac{3A}{M_A} = \sigma_D(3A)^{2/3}, \quad M_A = (2DA/\pi^2)^{1/3}. \quad (32.4)$$

The minimizer is parameterized by

$$f_A(t) = M_A \sin^2 \varphi, \quad t = \frac{\varphi - \sin \varphi \cos \varphi}{\pi}, \quad 0 \leq \varphi \leq \pi. \quad (32.5)$$

It satisfies $f_A'' = -DA/f_A^2$ and

$$\int_0^1 \frac{dt}{f_A} = \frac{2}{M_A}, \quad \int_0^1 \frac{1 + |\log f_A|}{f_A} dt = \frac{2}{M_A} (1 - \log M_A + 2 \log 2) \quad (32.6)$$

when $M_A < 1$, as holds here for all sufficiently large n . All these endpoint estimates are uniform for the indicated A .

Proof. Since $dt = (2/\pi) \sin^2 \varphi d\varphi$, direct differentiation proves the ODE and the first integral in (32.6). The ODE first integral is $f_A'^2/(2D) = A(1/f_A - 1/M_A)$; integrating it proves (32.4). The second integral follows from $\int_0^\pi \log(\sin \varphi) d\varphi = -\pi \log 2$. For completeness the latter identity follows by symmetry and the double-angle formula: if $I = \int_0^{\pi/2} \log \sin u du$, then $2I = \int_0^{\pi/2} \log(\sin u \cos u) du = I - (\pi/2) \log 2$.

To verify minimality, convexity of $v \mapsto v^2/(2D)$ and of $x \mapsto A/x$ gives, on $[\varepsilon, 1 - \varepsilon]$, the pointwise lower tangent inequality around f_A . The linear terms integrate by parts to a boundary term because $f_A'' = -DA/f_A^2$. For a competing finite-action path f , its zero endpoint and Cauchy–Schwarz imply $f(t) = o(t^{1/2})$ as $t \downarrow 0$ and similarly at one. Since $f_A(t) \asymp t^{2/3}$ and $f_A'(t) = O(t^{-1/3})$, the boundary products $f_A'(t)(f(t) - f_A(t))$ vanish. Letting $\varepsilon \downarrow 0$ proves $I_A(f) \geq I_A(f_A)$. Strict convexity gives uniqueness. These endpoint powers also show integrability directly and uniformity when A stays in a compact positive interval. $\square$

A strip supremum bound $|\log x| = O(J)$ would turn (32.3) into an artificial bulk term of size J/N . We instead use the finite integral (32.6) in both proof directions.

[write] **5 October 2026.** At $A = \frac{1}{3}$, Lemma 32.1 is Part II's Theorem 19.1, whose proof also gives the exact quadratic remainder. The second integral in (32.6) was rechecked by hand at the placement from $\int_0^\pi \log \sin \varphi d\varphi = -\pi \log 2$.

33 The coefficient upper bound

33.1 A quantitative dual with an integrated penalty

Take $A = A_n$, $\zeta = N^{-9}$, and $r_n = 1 - 2\zeta$. Set

$$u(t) = f_A'(\zeta + r_n t)/D, \quad g(t) = f_A(\zeta + r_n t)/\sqrt{r_n}.$$

Then $u' = -A/g^2$, $\min g \asymp N^{-6} = e$, and $\max g < 1$ for all sufficiently large n . Every fixed derivative needed below is bounded by $P(n)$. Choose a sufficiently large fixed K and define $v(0) = 0$ by

$$v'(t) = \frac{2A}{g(t)} - \frac{D}{2}u(t)^2 - \frac{K(1 + |\log g(t)|)}{Ng(t)}. \quad (33.1)$$

The absolute value is smooth because $g < 1$. From Lemma 32.1,

$$v(1) \geq \sigma_D(3A_n)^{2/3} - C/N - C\sqrt{e}. \quad (33.2)$$

Indeed the unpenalized integral is

$$\frac{1}{r_n} \int_{\zeta}^{1-\zeta} \left(\frac{2A\sqrt{r_n}}{f_A} - \frac{f_A'^2}{2D} \right) dt.$$

Its omitted endpoint pieces are $O(\zeta^{1/3}) = O(\sqrt{e})$; the time-rescaling factors cost $O(\zeta)$. The penalty integrates to $O(1/N)$ by (32.6).

For each t let $F_t(x) = -Ax/g(t)^2 + 2A/g(t)$. The exact gap is

$$\frac{A}{x} - F_t(x) = \frac{A}{x}(1 - x/g(t))^2. \quad (33.3)$$

For all $ce \leq x \leq R$, sufficiently large fixed K makes

$$F_t(x) - \frac{K(1 + |\log g(t)|)}{Ng(t)} \leq \frac{A}{x} - \frac{C(1 + |\log x|)}{Nx} - \frac{3}{Nx}, \quad (33.4)$$

where C is from (32.3). If $1/2 \leq x/g \leq 2$, both $1/x$ and $1 + |\log x|$ are bounded by fixed multiples of their counterparts at g , so the penalty alone suffices. Outside that range (33.3) is at least $A/(4x)$, which absorbs the displayed error since $|\log x| = O(J)$ and $J/N \rightarrow 0$. The polynomially small part of (32.3) is also less than $1/(Nx)$ uniformly. Consequently

$$u'(t)x + v'(t) \leq \frac{n}{S}\eta_p(\theta u(t)) - \frac{2}{Nx}. \quad (33.5)$$

After multiplying by $k_0 = S/n$, the slack is exactly $2/p$.

33.2 Uniform rows including very small heights

For integer original time $0 \leq t \leq n$ and $0 \leq p \leq RH$, put

$$\Phi(t, p) = S[u(t/n)p/H + v(t/n)].$$

Retain only legal cycles with $t + L \leq n$ and endpoints in this height region, multiplying their masses by $e^{\Phi(t+L, p+\Delta) - \Phi(t, p)}$. Call a cycle good if $L \leq d_n n$ and $|\Delta| \leq d_n H$, where $d_n = n^{-1/12}$. Taylor's formula gives

$$\Phi(t + L, p + \Delta) - \Phi(t, p) \leq \theta u(t/n)\Delta + k_0[u'(t/n)p/H + v'(t/n) + d_n P(n)]L. \quad (33.6)$$

To include the mixed term explicitly, after dividing by S and writing $l = L/n$, $d = \Delta/H$, $x = p/H$, its remainder is

$$\begin{aligned} & [u(t/n + l) - u(t/n)]d \\ & + x[u(t/n + l) - u(t/n) - u'(t/n)l] \\ & + [v(t/n + l) - v(t/n) - v'(t/n)l], \end{aligned}$$

bounded by $P(n)(l|d| + l^2)$. This proves (33.6).

At $p \geq ceH$, its extra η shift times p is at most $RNd_nP(n) = o(1)$. Equation (33.5) leaves at least $1/p$ root slack. The shifted-row limit (31.4) with $z = 1$ bounds every such good-cycle row by a fixed number below one.

Choose the fixed constant $c > 0$ sufficiently small for the remaining heights. At $p \leq ceH$, the effective tilts in (33.6) obey

$$\lambda^2 \leq Ck_0/e, \quad |\eta| \leq Ck_0/e, \quad \beta = \log R(\lambda, \eta) \leq Ck_0/e,$$

with constants independent of small $c \leq 1$. Indeed $|u'|p/H \leq C/e$ there, and the terms of v' satisfy the same bound. If $\beta \geq 0$, discarding only the CDF gives

$$\sum_{b=0}^p d_b(e^{\beta b} - 1) \leq C\beta\sqrt{p+1}e^{\beta p} \leq n^{-1/3+Cc}P(n). \quad (33.7)$$

This uses $e^z - 1 \leq ze^z$ and $\sum_{b \leq p} bd_b \leq C\sqrt{p+1}$. For negative β there is no excess. Take c so that $Cc < 1/6$. The prefactor and ordinary-edge perturbations vanish uniformly, so these rows are bounded above by $r + o(1)$. At $p = 0$ there is only the ordinary edge. Equality with $r + o(1)$ is not claimed at fixed finite height, and no local-root approximation is used there.

For bad cycles, $|\Phi| \leq SP(n)$ on the retained region. The zero-tilt fixed- b estimates (30.9)–(30.10), retaining $b \leq RH$ and using $\sum d_b \leq 2$, give transformed mass at most

$$Ce^{SP(n)} \left\{ e^{-c \min(d_n^2 H^2 / (RH+1), d_n H)} + e^{-cd_n n + C(RH+1)} \right\} = o(1). \quad (33.8)$$

The first negative exponent has order $n^{1/2}$ times a fixed logarithmic power, and the second has order $n^{11/12}$ minus $O(H)$. Both dominate $SP(n)$. Thus all transformed rows are bounded by a single fixed $\rho < 1$ eventually.

33.3 Exact original time and large heights

The transformed mass over every possible number of cycles is at most $1/(1 - \rho)$. Keeping only paths with actual final original time n can only decrease it. Telescoping the transform and then applying the genuine terminal multiplier gives

$$e^{-\Phi(n,p)} x_j^{-p} = e^{-Sv(1)} e^{-p(\log x_j + \theta u(1))} \leq e^{-Sv(1)},$$

since $\theta P(n) \rightarrow 0$. Hence the confined normalized coefficient is at most $Ce^{-Sv(1)}$.

For paths with a boundary height beyond RH , the spatial input (30.13) gives contribution

$$\exp\{-(\kappa\sqrt{2R/3} + o(1))S\}.$$

Fix R sufficiently large that the coefficient exceeds $\sigma_D + 1$. This is negligible even at the new logarithmic scale. Finally,

$$\sigma_D(3A_n)^{2/3} = \sigma_D + \frac{7\sigma_D}{3} \frac{J}{N} + O(J^2/N^2), \quad J^2/N^2 = O(1/N).$$

Together with (33.2), $\sqrt{e} = N^{-3}$, and the fixed geometric prefactor, this proves

$$n \log \mu - \log a_j(n) \geq \sigma_D S + \frac{7\sigma_D}{3} \frac{SJ}{N} - O(S/N). \quad (33.9)$$

34 The coefficient lower bound with a shrinking strip

34.1 Entrance and the proposed central path

Take $P_n = \lfloor eH \rfloor$ and apply (30.11). Write $a_n = E(P_n)/n$. Since $\log(eH)/N \rightarrow 2/3$,

$$a_n \leq Ce^{3/2}, \quad \text{entrance logarithmic cost} \leq C\sqrt{e}S. \quad (34.1)$$

Let $f = f_{1/3}$ and choose the left endpoint time t_e with $f(t_e) = e$. For a sufficiently large fixed A_{rep} , put

$$\delta = A_{\text{rep}}e^{3/2}, \quad b_n = 1 - \delta, \quad c_n = \frac{1 - 2t_e}{b_n - a_n}, \quad f_n(t) = f(t_e + c_n(t - a_n)). \quad (34.2)$$

Increase A_{rep} so that $a_n \leq \delta/2$ and the uniform repair threshold below holds. Because $t_e = O(e^{3/2})$, $c_n = 1 + O(e^{3/2})$. On $[a_n, b_n]$ the path has both endpoints e , lies in $[e, M_{1/3}]$, and has every required fixed derivative bounded by $P(n)$. Extend it beyond b_n by the same formula for time $n^{-1/12}$; this is valid eventually because the distance to its singular endpoint has only a negative logarithmic order.

34.2 Controlled law and the clock

At an active departure time s , use (31.10) with

$$\lambda_s = \theta q(s/n), \quad q = f'_n/D. \quad (34.3)$$

Stop at the first boundary time $s_K \geq nb_n$, on leaving $[eH/2, RH]$, or after

$$m_n = \left\lceil \frac{4n}{ceH} \right\rceil + 1 \quad (34.4)$$

cycles, where c is the lower-mean constant in (31.11). All increments below are set to zero after stopping. The chosen R is larger than the profile's maximum. All active departure states satisfy the quantitative local lemmas.

The clock martingale $Z_k = \sum_{i < k} (L_i - \mathbb{E}[L_i \mid \text{past}])$ satisfies

$$\mathbb{E}Z_{m_n}^2 \leq m_n CH^2 \leq nHP(n).$$

Doob's inequality therefore yields

$$\max_k |Z_k|/n \leq n^{-1/12} \quad (34.5)$$

with failure probability at most $n^{-1/6}P(n)$. Also

$$\mathbb{E} \sum_i (L_i/n)^2 \leq n^{-1/3}P(n), \quad \sum_i (L_i/n)^2 \leq n^{-1/6} \quad (34.6)$$

with failure probability at most $n^{-1/6}P(n)$. On this event the maximum time mesh is at most $n^{-1/12}$. The cycle cap cannot be the first stop without the desired time crossing or spatial exit: absent a prior stop, $\mathbb{E}[L_i \mid \text{past}] \geq ceH/2$, and the cap in (34.4) gives accumulated conditional mean above $2n$. Its actual elapsed time would still be below n , contradicting (34.5). The factor 4 accounts for the lower strip height $eH/2$.

Put $v_i = \mathbb{E}[\Delta_i \mid \text{past}]/\mathbb{E}[L_i \mid \text{past}]$. The anchored derivative estimate gives, uniformly,

$$v_i = \theta f'_n(s_i/n) + O(n^{-1/2}P(n)), \quad v_i/\theta = f'_n(s_i/n) + O(n^{-1/6}P(n)). \quad (34.7)$$

The reward martingale $M_k = \sum_{i < k} (\Delta_i - v_i L_i)$ obeys $\mathbb{E} M_{m_n}^2 \leq nP(n)$. To check the rate without independence, use $(\Delta - vL)^2 \leq 2\Delta^2 + 2v^2L^2$ in (31.11); each active increment has second moment at most $HP(n)$ and $m_n \leq (n/H)P(n)$. Doob's inequality gives

$$\max_k |M_k|/H \leq n^{-1/12} \quad (34.8)$$

with failure probability at most $n^{-1/6}P(n)$.

Since the total time is at most $n + \max_i L_i$, these estimates imply at every visited boundary

$$p_k/H = p_0/H + \sum_{i < k} f'_n(s_i/n) L_i/n + O(n^{-1/12}P(n)). \quad (34.9)$$

The Riemann error in this sum is at most $P(n) \sum_i (L_i/n)^2$. The entrance rounding is $O(1/H)$. It follows, on an event whose probability tends to one, that

$$\sup_k |p_k/H - f_n(s_k/n)| \leq \tau_n := n^{-1/14}, \quad 0 \leq s_K/n - b_n \leq n^{-1/12}. \quad (34.10)$$

Here $n^{-1/12}P(n) = o(n^{-1/14})$. Since $\tau_n = o(e)$, the tube excludes a spatial exit: the proposed curve has margin at least $e/2$ from the lower boundary and fixed upper margin. The stop is therefore the intended original-time crossing. This proves a quantitative shrinking-strip result rather than appealing to a fixed-strip convergence theorem.

34.3 Exact likelihood and an integrated error

For each realized stopped prefix its inverse change of measure is exact:

$$\prod_i Q_{p_i}(\text{cycle}_i) = \prod_i \mathbb{P}_{p_i, \lambda_{s_i}}(\text{cycle}_i) e^{-B_n}, \quad B_n = \sum_i [\lambda_{s_i} \Delta_i + \eta_{p_i}(\lambda_{s_i}) L_i]. \quad (34.11)$$

No duration is rounded or frozen. Equations (32.2)–(32.3), evaluated at the actual random heights on (34.10), give

$$\begin{aligned} \frac{B_n}{S} &= \sum_i q(s_i/n) \frac{p_{i+1} - p_i}{H} \\ &+ \sum_i \left(\frac{A_n}{p_i/H} - \frac{D}{2} q(s_i/n)^2 \right) \frac{L_i}{n} + O(1/N) + n^{-c'_*} P(n) \end{aligned} \quad (34.12)$$

for some positive c'_* . The $O(1/N)$ is obtained by integration, not by a strip supremum. Specifically,

$$\sum_i \frac{1 + |\log(p_i/H)|}{p_i/H} \frac{L_i}{n} = O(1). \quad (34.13)$$

On $[e/2, R]$ this height function is Lipschitz, with constant $P(n)$ (also across $x = 1$). Its path-tube and mesh errors are polynomially small; its integral along f_n is uniformly bounded by (32.6) and the time change in (34.2). The short overshoot contributes only $n^{-1/12}P(n)$. This proves (34.13). The anchored-root remainder contributes $n^{-1/3}P(n)$.

For the first sum in (34.12), discrete summation by parts bounds the replacement error by

$$\tau_n(|q(\text{start})| + |q(\text{end})| + \text{TV}(q)) \leq \tau_n P(n).$$

Thus no bounded-total-variation assertion for the random boundary path is needed. Riemann errors for the deterministic smooth path are polynomially small. The second sum is treated by its tube, mesh, and lower-height bounds. Hence

$$\frac{B_n}{S} = \int_{a_n}^{b_n} \left(\frac{f_n'^2}{2D} + \frac{A_n}{f_n} \right) dt + O(1/N) + n^{-c_*} P(n). \quad (34.14)$$

The overshoot again has time at most $n^{-1/12}$ and every relevant integrand there is at most $P(n)$. The controlled event has probability tending to one, and is eventually at least $1/2$. Integrating (34.11) over it costs only an additional factor $1/2$ in the lower mass bound.

34.4 Every integer time and evaluation of the action

On the tube, for sufficiently large n ,

$$eH/2 \leq p_K \leq 2eH, \quad \delta n/2 \leq n - s_K \leq \delta n.$$

Choose the fixed A_{rep} large enough that the repair input (30.12) applies throughout this state and time range. It appends a legal family of exactly $n - s_K$ steps ending at $(2, 0)$, with logarithmic cost at most

$$C(\sqrt{e}S + (\delta/e)S) \leq C\sqrt{e}S. \quad (34.15)$$

The entrance cost has the same order. The terminal multiplier is exactly the constant x_j^{-2} . Concatenation is injective by the completed-history argument in Report 127: the entrance duration is deterministic, the interior stop is the first boundary visit at or after nb_n , and the repair macrocycle pattern is a deterministic function of its start state and residual time. Distinct decorated prefixes remain distinct completed histories. Every internal multiplicity realization counted by Q is retained.

Set

$$K_e = \int_{t_e}^{1-t_e} \frac{f'^2}{2D} dt, \quad W_e = \int_{t_e}^{1-t_e} \frac{dt}{f}.$$

The central action in (34.14) is exactly $c_n K_e + A_n c_n^{-1} W_e$. The endpoint estimates give

$$K_e + \frac{1}{3}W_e = \sigma_D + O(\sqrt{e}), \quad W_e = 2\sigma_D + O(\sqrt{e}), \quad c_n = 1 + O(e^{3/2}).$$

Therefore

$$\int_{a_n}^{b_n} \left(\frac{f_n'^2}{2D} + \frac{A_n}{f_n} \right) dt = \sigma_D + \frac{7\sigma_D}{3} \frac{J}{N} + O(\sqrt{e}). \quad (34.16)$$

The unperturbed profile suffices because the action is linear in A_n at that path. Combine (34.1), (34.14), (34.15), and (34.16). Since $\sqrt{e} = N^{-3} = O(1/N)$ and every $n^{-c}P(n) = o(1/N)$, this proves

$$n \log \mu - \log a_j(n) \leq \sigma_D S + \frac{7\sigma_D}{3} \frac{SJ}{N} + O(S/N). \quad (34.17)$$

Together with (33.9), this proves Theorem 29.1 at every sufficiently large integer n .

35 Error ledger and the scope of the new term

The following ledger lists the errors that must stay below SJ/N . Unless indicated otherwise, the stated errors are relative to S .

Source	Bound	Reason it is harmless
Scalar local root	$O(1/N)$ after integration	The height error is $(1+ \log x)/(Nx)$ and is integrated along the profile
Anchored tilt remainder	$n^{-1/3}P(n)$	Uniform on the strip down to eH
Controlled drift	$n^{-1/6}P(n)$	Obtained from anchored concavity, not differentiation of an unspecified error
Clock fluctuation	variance $\leq nHP(n)$	Threshold $n^{11/12}$ has failure probability $n^{-1/6}P(n)$
Reward fluctuation	variance $\leq nP(n)$	Threshold $Hen^{-1/12}$ has failure probability $n^{-1/6}P(n)$
Squared time mesh	expectation $n^{-1/3}P(n)$	Threshold $n^{-1/6}$ controls Riemann errors and overshoot
Path tube	radius $n^{-1/14}$	Smaller than every fixed negative logarithmic power and therefore than the strip margin
Upper Taylor remainder	$n^{-1/12}P(n)$ per unit time	Its root shift times p is $o(1)$, preserving $1/p$ slack
Very small heights	row excess $n^{-1/3+Cc}P(n)$	Choose fixed c with $Cc < 1/6$; the row is bounded above by $r + o(1)$
Entrance and repair	$O(N^{-3})$ cost; $O(nN^{-9})$ time	Uniform discrete constructions retain every residual integer duration
Likelihood height error	$O(1/N)$	Lipschitz tube and mesh comparison with an integrable deterministic profile
Large jumps and durations	negative exponents of orders $n^{1/2}$ and $n^{11/12}$	Both dominate $SP(n)$ even for the growing dual
Boundary heights above RH	exponent greater than $(\sigma_D + 1)S$	One fixed sufficiently large R suffices
Terminal and counting factors	fixed multiplier and $O(1)$ logarithmic cost	Exact terminal $(2, 0)$ and injective concatenation
Expansion of minimum	$O(J^2/N^2)$	This is $O(1/N)$

The estimate leaves an unknown coefficient at scale $S/N = n^{1/3}(\log n)^{-1/3}$. Exponentiating such a remainder does not give a multiplicative $1 + o(1)$ error. In particular neither an exact multiplier nor a polynomial prefactor is determined. The presence of c_* in the scalar root does not isolate the next global coefficient: other accumulated effects at the same scale have not been evaluated. No ratio theorem, full transseries, effective onset, or finite-data calibration is asserted.

[write] **5 October 2026.** These non-claims are collected, with the six open problems of the three Parts, in Section 39. The S/N coefficient is question 1 there.

36 Refined inverse thresholds

In this and the next section write $\sigma = \sigma_D$ and $\lambda = \log \mu$. Let ψ , R , and G be as in Theorem 29.1.

Corollary 36.1. *For real $T \rightarrow \infty$, define $N(T) = \min\{n \geq 0 : a_j(n) \geq T\}$ and $s = (\log T)/\lambda$. Then*

$$N(T) = s + \frac{\sigma}{\lambda} \psi(s) + \frac{7\sigma}{3\lambda} \frac{\psi(s) \log \log s}{\log s} + O\left(\frac{\psi(s)}{\log s}\right). \quad (36.1)$$

This conclusion requires neither monotonicity of the coefficients nor a ratio asymptotic.

Proof. Put $m(s) = s + G(s)/\lambda$ and consider integers within $O(\psi(s))$ of s . Direct differentiation of the explicit smooth functions gives $G'(t) = O(\psi(t)/t)$ and $R'(t) = O(R(t)/t)$. Uniformly in this range,

$$G(n) = G(s) + O(\psi(s)^2/s) = G(s) + o(R(s)), \quad R(n) = R(s)(1 + o(1)).$$

The small-error assertion follows from $\psi(s) \log s/s \rightarrow 0$. For a fixed sufficiently large K , let $n_+ = \lceil m(s) + KR(s) \rceil$. The coefficient lower envelope in (29.3) implies $\log a_j(n_+) \geq \lambda s$ eventually. Integer rounding costs $O(1) = o(R(s))$.

For the other direction, $a_j(n) \leq \mu^n$ excludes every integer $n < s$. The explicit function $t \mapsto \lambda t - G(t) + CR(t)$ is strictly increasing eventually, since its derivative tends to $\lambda > 0$. At $m(s) - KR(s)$ it is less than λs for large enough fixed K . The upper envelope therefore excludes every integer between s and this threshold. These two brackets prove $N(T) = m(s) + O(R(s))$, which is (36.1). $\square$

[write] **5 October 2026.** This refines Part I, Corollary 9.1, and Part II, Corollary 26.1. Part I also proves, in Section 9.2, that $a_j(n)$ is strictly increasing for $n \geq 1$, which this corollary does not need.

37 Refined boundary derivative growth

Let $b_n = a_j(n)\mu^{-n}$, $F(z) = \sum_{n \geq 0} a_j(n)z^n$, $\rho = 1/\mu$, and $f(t) = F(\rho e^{it})$. Theorem 29.1 gives radius ρ and $\sum n^k b_n < \infty$ for every fixed integer k . Thus f is smooth, all its derivatives are represented by uniformly convergent series, and positivity gives

$$\|f^{(k)}\|_\infty = M_k := \sum_{n \geq 0} n^k b_n \quad (k \geq 1). \quad (37.1)$$

Its maximum is attained at $t = 0$. Radial boundary derivatives are $F^{(k)}(\rho) = \mu^k \sum_{n \geq k} (n)_k b_n$, with the falling factorial $(n)_k = n(n-1) \cdots (n-k+1)$.

Corollary 37.1. *As $k \rightarrow \infty$ through integers,*

$$\begin{aligned} \log \|f^{(k)}\|_\infty &= 3k \log k - 2k \log \log k + [\log(2D/\pi^2) - 3]k \\ &\quad - \frac{k \log \log k}{\log k} + O\left(\frac{k}{\log k}\right), \end{aligned} \quad (37.2)$$

$$\begin{aligned} \log F^{(k)}(\rho) &= 3k \log k - 2k \log \log k + [\log \mu + \log(2D/\pi^2) - 3]k \\ &\quad - \frac{k \log \log k}{\log k} + O\left(\frac{k}{\log k}\right). \end{aligned} \quad (37.3)$$

These are consequences of the explicit envelopes (29.3); they assert no extra coefficient term.

Proof. We first optimize explicit smooth bounds. For any fixed real c , put $V_c(x) = G(x) + cR(x)$ for sufficiently large real x . With $y = \log x$, $a = 7/3$, and $b = c/\sigma$,

$$V_c(e^y) = \sigma e^{y/3} y^{2/3} \left(1 + \frac{a \log y + b}{y} \right). \quad (37.4)$$

Its first and second y derivatives are eventually positive, with second derivative asymptotic to $(\sigma/9)e^{y/3}y^{2/3}$. Thus $ky - V_c(e^y)$ is strictly concave eventually, and its unique eventual maximum tends to infinity. Finitely many smaller x have objective only $O(k)$. We differentiate only this explicit potential.

Write $B(y) = a \log y + b$. The stationary equation is

$$k = \frac{\sigma}{3} e^{y/3} y^{2/3} \left(1 + \frac{B(y) + 2}{y} + O\left(\frac{\log y}{y^2}\right) \right). \quad (37.5)$$

Taking logarithms and multiplying by three gives, with $L = \log k$,

$$y + 2 \log y + \frac{7 \log y + 3b + 6}{y} = 3L + 3 \log(3/\sigma) + O\left(\frac{(\log y)^2}{y^2}\right). \quad (37.6)$$

First this shows $y \sim 3L$; then it shows $y = 3L - 2 \log L + O(1)$. Substituting this back and expanding yields

$$y = 3L - 2\ell + c_0 - \frac{\ell}{L} + O(1/L), \quad \ell = \log L, \quad c_0 = \log(3/\sigma^3). \quad (37.7)$$

Here is the coefficient calculation explicitly. If $y = 3L - 2\ell + c_0 + d\ell/L + O(1/L)$, then $2 \log y$ contributes $-4\ell/(3L)$ at this order, and $7 \log y/y$ contributes $7\ell/(3L)$. Their combined ℓ/L coefficient in (37.6) is $d + 1$, so $d = -1$. The remaining residual is $O(1/L)$ and the left derivative tends to one, justifying the stated remainder by the mean-value theorem.

Divide (37.4) by (37.5). At the saddle,

$$V_c(e^y) = 3k \left(1 - \frac{2}{y} + O\left(\frac{\log y}{y^2}\right) \right) = 3k - \frac{2k}{L} + O(k\ell/L^2). \quad (37.8)$$

Consequently, for every fixed c ,

$$\sup_{x \geq x_0} \{k \log x - V_c(x)\} = 3kL - 2k\ell + (c_0 - 3)k - \frac{k\ell}{L} + O(k/L). \quad (37.9)$$

The saddle has $x \asymp k^3/(\log k)^2$. Rounding to a nearest integer changes the objective by $o(1)$: in a fixed neighborhood of that saddle, the absolute x derivative is $O(k/x)$, and $k/x \rightarrow 0$. Thus (37.9) also holds for the integer supremum.

Now use (29.3). A single saddle integer for V_C supplies the lower bound for M_k . For the upper bound reserve the summable positive factor $e^{-R(n)}$:

$$n^k b_n \leq e^{-R(n)} \exp\{k \log n - G(n) + (C + 1)R(n)\} \quad (n \geq n_0).$$

The series $\sum_{n \geq n_0} e^{-R(n)}$ is finite, and the remaining supremum is (37.9) with $c = -(C + 1)$. The earlier coefficients contribute $e^{O(k)}$, negligible beside the saddle on the logarithmic scale k/L . Therefore (37.9) proves (37.2), since $3/\sigma_D^3 = 2D/\pi^2$.

For radial derivatives, $(n)_k \leq n^k$ gives the upper bound. At the same lower-bound saddle $n \asymp k^3/(\log k)^2$,

$$\log \frac{(n)_k}{n^k} = \sum_{j=0}^{k-1} \log(1 - j/n) = O(k^2/n) = o(k/\log k).$$

The falling factorial therefore preserves the displayed refinement; the factor μ^k adds $k \log \mu$. $\square$

The unknown coefficient of $R(n)$ affects the moment only at $O(k/\log k)$, which explains why the additional term in (37.2) is nevertheless determined. In the convention $\|f^{(k)}\|_\infty \leq CA^k(k!)^s$, the sharp Gevrey order remains 3, as already established in Report 127.

[write] **5 October 2026.** “Report 127” is Part II (after Corollary 26.2). Sharp Gevrey order 3 was proved first in Part I, Theorem 10.1, which alone proves that ρ is a genuine singularity and that the generating functions are not D-finite. The dossier of the placement checked (37.9) numerically: for $k = 10^4, \dots, 10^{14}$, $c \in \{0, \pm 3\}$ and both values of D , the difference between $\sup_y \{ky - V_c(e^y)\}$ and the right side of (37.9) without its $O(k/L)$, divided by k/L , stays in $[-4.2, 2.5]$, while omitting the term $-k\ell/L$ leaves about $-4.8 k/L$ at $k = 10^{14}$. This is a consistency check of the saddle algebra, not part of the proof.

38 Verification and reproducibility

The report supplies every new quantitative estimate, and the unchanged earlier companion supplies the complete proofs of the stated inputs. A mathematical argument, a finite regression check, a byte-integrity check, and a rendered-page review have distinct scopes; none is a substitute for another.

The reader bundle is local and offline. It contains this report’s PDF and editable TeX, the unchanged Report 127 package, and finite-check and reproduction tools. The outer closed inventory rejects missing or changed files, extra files or directories, unsafe inventory paths, and symbolic links. The earlier package retains its own original inventory. The README gives the precise checker commands, their scope, and the build environment. Full replay verifies checks in normal and optimized Python, rebuilds the report in clean external directories, extracts a fresh deterministic ZIP, repeats the checks, and compares the repack. All runtime products stay outside the immutable package.

The finite companion tests explicit identities and bounded examples. It does not certify asymptotic uniformity, the singular variational argument, martingale convergence, or infinite tail estimates. A source inventory is a record relative to its supplied verifier, not an authenticity signature against simultaneous malicious replacement. Different TeX or compression versions may produce equivalent content with different bytes; the recorded environment is used for the bitwise replay.

[write] **5 October 2026.** In this directory the companion’s files carry the prefix `129-loglog-`: `checks/verify.py` and `checks/negative_tests.py` are `code/129-loglog-checks-verify.py` and `code/129-loglog-checks-negative_tests.py`, the recorded outputs are `data/129-loglog-verification_results.json` and `data/129-loglog-negative_results.json`, and the companion’s README is `129-loglog-checks-README.md`. The “unchanged Report 127 package” is not shipped as such: Report 127’s text is Part II, and its files are shipped with the prefix `127-sharp-`. Report 129’s `integrity.py` is a byte copy of Report 126’s, shipped once as `code/126-scale-integrity.py`. The outer closed inventory (`CHECKSUMS.sha256`) is not shipped, so the checkers do not run in this directory; the report’s README gives the rerun route from the arrival commit.

39 Further questions and research

[write] (5 October 2026.) This section was added in the write, under Vladimir’s standing rule of 4 October 2026. It collects every claim of the three manuscripts that is stated without proof or explicitly left open, with its sources, the sketch offered and what is missing; Sections 11.1 and 27.1 record which questions of Parts I and II a later Part answers. No demonstrably wrong claim was found in any of the three manuscripts. The one wrong claim on record is external: Britt and Beaton’s numerical forms with a stretched term $n^{3/8}$, which their Section 4.2 labels non-rigorous; Remark 1.2 proves that they fail, with every other stretched exponent.

Answered inside this report: the leading constant (Part I’s first question) by Theorem 13.1; the global next term (Part II’s second item) by Theorem 29.1; Part I’s “no quantitative coefficient remainder” by the remainder $O(S/\log n)$ of Theorem 29.1, whose constants are not effective.

1. *The coefficient of $S/\log n$.* Sources: Report 129, Section 35, and its delivered README (replaced in this directory by the report’s own). Is there a constant κ_j with $\mathcal{D}_j(n) = \sigma_D S + \frac{7\sigma_D}{3} \frac{SJ}{N} + \kappa_j \frac{S}{N} + o(\frac{S}{N})$, and what is it? Sketch from the source: the constant $c_* = \frac{1}{2} \log(\pi/\alpha)$ of the scalar root enters at this scale, but its presence “does not isolate the next global coefficient”, because “other accumulated effects at the same scale have not been evaluated”. In the ledger of Section 35 the terms of relative order $1/N$ are the scalar-root error (which contains c_* and the constant $\log(2/3)$ of the expansion after (32.3)) and the height error $(1 + |\log x|)/(Nx)$ integrated along the profile, with the matching dual penalty in the upper bound; all are bounded, none is evaluated. The entrance and repair costs ($O(N^{-3})$) and the expansion of $\sigma_D(3A_n)^{2/3}$ ($O(J^2/N^2)$) are of smaller order. Missing: a two-sided analysis to precision $o(S/N)$ that evaluates these terms. A model: `a202061-ascent-120-deficit`, Part IV, determined the analogous constant for A202061 (its Theorem `a61:o3:thm:main`), and its Part V every fixed inverse-logarithmic order by reduction to a classical action (Remark 29.2).

Numerical hint, not a fit. A floating-point run of the normalized commitment-tree recursion (the placement dossier’s, rerun at the write; at $n = 40$ it agrees to all printed digits with an exact big-integer run of the recursion, which reproduces the companions’ counts 1, 1, 2, 4, 10, 27, 79, 247, 816 and 1, 1, 2, 5, 15, 51, 190, 759, 3206) gives the residual $\rho_n = [\mathcal{D}_j(n) - \sigma_D S(1 + \frac{7}{3} \frac{J}{N})]/(\sigma_D S/N)$:

n	40	100	200	400
A279551 (247): $\mathcal{D}/(\sigma_D S)$	1.083	1.134	1.165	1.189
A279551 (247): ρ_n	−2.74	−2.95	−3.02	−3.05
A279556 (759): $\mathcal{D}/(\sigma_D S)$	1.184	1.202	1.216	1.228
A279556 (759): ρ_n	−2.37	−2.63	−2.75	−2.81

The residuals are bounded and drift slowly, as an $O(S/N)$ remainder allows, and would suggest κ_j near $-3\sigma_D$. They prove nothing: further terms of relative order J/N or J^2/N^2 cannot be excluded at these sizes, where $J/N \approx 0.30$ at $n = 400$, and the drift has not stopped. (For A202061 the analogous normalized constant is $\kappa = -1.94\dots$; nothing suggests that the constants of different sequences agree.)

2. *A coefficient equivalent.* Sources: Report 126, Section 11, item 2; Report 127, Section 27, item 1; Report 129, Theorem 29.1 and Section 35. An asymptotic $a_j(n) \sim C_j n^{g_j} \mu_j^n \exp\{-\Phi_j(n)\}$ needs every exponent term down to $o(1)$, then the power g_j and the amplitude C_j . Missing: everything below $O(S/\log n)$; exponentiating that remainder leaves an unbounded multiplicative error.

3. *Ratio asymptotics*. Sources: Report 126, item 3; Report 127, item 3; Report 129, Section 35. Does $a_j(n+1)/a_j(n) \rightarrow \mu_j$, and at what rate? Known: $a_j(n+1) > a_j(n)$ for $n \geq 1$ (Section 9.2) and $a_j(n)^{1/n} \rightarrow \mu_j$. Missing: the remainders are never differenced, and differencing an unspecified $o(S)$ or $O(S/N)$ is not valid (Report 127's proof of Corollary 26.2, Report 129's of Corollary 37.1).
4. *All orders, transseries, effective constants and onsets*. Sources: Report 126, items 3–4; Report 127, item 4; Report 129, Theorem 29.1 (“not asserted to be effective or numerically calibrated”) and Section 35. Missing: a complete expansion in inverse powers of $\log n$ with its $\log \log n$ structure, uniform error control at each order, and explicit constants with a usable onset. The A202061 report, Part V, is again the model for the inverse-logarithmic orders.
5. *Numerical crossover for finite-range fits*. Sources: Report 126, item 5; Report 127, item 4. Why do pure-power fits to the available terms suggest $n^{3/8}$, and from which n on does the proved scale dominate? The table under question 1 shows that the two proved terms are not small at the sizes where fits are made: $\mathcal{D}_j(400)/(\sigma_D S)$ is still 1.19 and 1.23, and $\frac{7}{3}J/N \approx 0.70$ at $n = 400$. This is an evaluation of the formulas, not an effective statement. Missing: a quantitative crossover analysis; no new digits are claimed.
6. *Singularities on the circle $|z| = \rho$* . Sources: Report 126, Theorem 10.1 and the paragraph closing Section 10; Report 127, item 5. Is ρ the only singularity of F on $|z| = \rho$; does F continue analytically through the other points of that circle, or into a slit domain, and with what expansion there? Known: every derivative of F extends continuously to the closed disk, ρ is a genuine singularity, and F is not D-finite, so no linear differential equation describes it. Missing: any information off the positive axis.

References

- [1] Nathan Britt and Nicholas Beaton, *Completing the enumeration of inversion sequences avoiding triples of relations*, arXiv:2512.21943v3, 29 September 2026. Exact commitment trees: Sections 3.8–3.9, equations (3.40)–(3.45), Lemmas 5–6 and Appendix A. Numerical estimates: Section 4.2, equations (4.12)–(4.13), and Appendix B.2. <https://arxiv.org/html/2512.21943v3>.
- [2] Stéphane Fischler and Tanguy Rivoal, *On the values of G -functions*, *Commentarii Mathematici Helvetici* **89** (2014), 313–341. Theorem 6, pp. 328–329. DOI: [10.4171/CMH/321](https://doi.org/10.4171/CMH/321). <https://ems.press/content/serial-article-files/43350>.
- [3] OEIS Foundation Inc., *A279551*. <https://oeis.org/A279551>. Sequence identifier. Report 126: the finite companion recomputes its own internal counts and does not claim a fresh OEIS data-file retrieval. Report 127: the companion's README specifies the provenance and scope of any finite coefficient comparisons.
- [4] OEIS Foundation Inc., *A279556*. <https://oeis.org/A279556>. Sequence identifier with the same data-provenance convention.
- [5] *Report 127. Sharp logarithmic deficit constants for two inversion sequence classes*, 2 October 2026. Unchanged proof and source companion included under `earlier-inputs/report127/`. Exact counting: Sections 2–3. Exact transforms and concentration: Section 5. Uniform entrance and every-integer-time repair: Section 10. Spatial barrier: Section 12. [**write**] Printed as Part II

of this report (Sections 2–3 as Part I's Sections 2–3; Sections 5, 10 and 12 as Sections 17, 22 and 24); the directory `earlier-inputs/report127/` is not shipped.